PHYS 325

Classical Mechanics I

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1 Newton's Laws of Motion

1.1 An introduction to classical mechanics

Classical mechanics is the study of the motion of objects, barring objects that do not move near the speed of light and objects at an atomic scale. There are three main formulations in classical mechanics: Newtonian, Lagrangian, and Hamiltonian.

Classical mechanics involves solving for the equation of motion of an object in a system. In essence, you solve for the position of the object as a function of time to see how the object moves based on a given force.

1.2 Space and time

1.2.1 Space

We represent the spatial position of an object using a position vector $\mathbf{r}$ which specifies the distance and direction to the point from the origin O of a chosen reference frame. Using the standard basis vectors (which are orthonormal basis vectors that span all of $\mathbb{R}^3$) $\{\mathbf{e_x}, \mathbf{e_y}, \mathbf{e_z}\}$, a point $P(x, y, z)$ is represented as

$$\mathbf{r} = x\mathbf{e_x} + y\mathbf{e_y} + z\mathbf{e_z} \tag{1.1}$$

An object may move with time, we specify its position using a vector-valued function

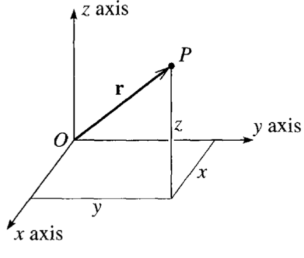


Figure 1.1 The point P is identified by its position vector $\mathbf{r}$, which gives the position of P relative to a chosen origin O. The vector $\mathbf{r}$ can be specified by its components (x, y, z) relative to chosen axes $Oxyz$.

$\mathbf{r}(t)$ which in cartesian coordinates is given by,

$$\mathbf{r}(t) = x(t)\mathbf{e_x} + y(t)\mathbf{e_y} + z(t)\mathbf{e_z} \tag{1.2}$$

For a general 3D coordinate system, the basis vectors are $\{\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3}\}$. Then, the position vector is,

$$\mathbf{r}(t) = r_1(t)\mathbf{e_1} + r_2(t)\mathbf{e_2} + r_3(t)\mathbf{e_3} \tag{1.3}$$

Going back to cartesian coordinates, the object's velocity and acceleration are given by,

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{e}_x + \frac{dy}{dt}\mathbf{e}_y + \frac{dz}{dt}ez \tag{1.4}$$

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2} = \frac{d^2x}{dt^2}\mathbf{e}_x + \frac{d^2y}{dt^2}\mathbf{e}_y + \frac{d^2z}{dt^2}\mathbf{e}_z \tag{1.5}$$

In classical mechanics, we use an overhead dot to represent derivatives with respect to time. Thus,

$$\mathbf{v}(t) = \dot{\mathbf{r}}(t) = \dot{x}(t)\mathbf{e}_x + \dot{y}(t)\mathbf{e}_y + \dot{z}(t)\mathbf{e}_z \tag{1.6}$$

$$\mathbf{a}(t) = \ddot{\mathbf{r}}(t) = \ddot{x}\mathbf{e}_x + \ddot{y}\mathbf{e}_y + \ddot{z}\mathbf{e}_z \tag{1.7}$$

Position has units of metres: m

1.2.2 Time

At non-relativistic speeds, time is a universal parameter t that all observers agree upon. In classical mechanics, We assume all observers are equipped with accurate clocks, all properly synchronised. This assumes information travels at infinite speeds. We know from relativity that information travels at most the speed of light. However, at non-relativistic speeds, this is a valid assumption that leads to accurate answers.

Time has units of seconds: s

1.2.3 Reference frames

A reference frame is a set of axes, with a fixed origin, used to label positions. An inertial reference frame is a frame that is not accelerating or rotating. Newton's laws hold in inertial reference frames, but they do not in non-inertial frames.

1.2.4 Mass

Mass is an inherent property of an object that characterises its inertia, or resistance to acceleration by an external force. Thus, objects with larger mass are harder to accelerate. There are two types of mass: inertial mass (m_I) and gravitational mass (m_G).

Inertial mass characterises an object's resistance against acceleration while gravitational mass characterises an object's resistance to accelerate due to the gravitational force.

A priori, m_I and m_G need not be equal but experiments show that they agree to 10^{-15} i.e.,

$$\left|\frac{m_I}{m_G} - 1\right| < 10^{-15} \tag{1.8}$$

Experiments have led to the weak-equivalence principle which states that all uncharged, freely-falling test particles follow the same trajectories in a uniform gravitational field, regardless of their composition. Thus, in classical mechanics we assume $m_I = m_G$.

In classical mechanics, we tend to simplify masses to point masses as they are easier to analyse. A point mass is an object with mass but no size; it has no internal degrees of freedom and cannot have energy due to internal vibrations or deformations. A point mass is a realistic approximation for many objects. However, they may be unrealistic depending on the system: if we analyse the orbital motion of the Earth around the Sun, the Earth can be treated as a point mass. However, when analysing the Earth's spin, it cannot be treated as a point mass. In classical mechanics, the word "particle" refers to a point mass.

Mass has units of kilograms: kg

1.2.5 Force

Force $\mathbf{F}$ is a vector quantity that can be classified as contact or no-contact forces.
Contact forces arise from contact between two bodies like a push or pull.
Non-contact forces like the electric, magnetic, and gravitational force arise from a "field".
Force is measured in newtons: N

1.3 Newton's First Law of Motion (N1L)

N1L (also known as the law of inertia) states that in the absence of external forces, an object moves at a constant velocity $\mathbf{v}$. In other words, in the absence of a force, a stationary body remains at rest, and a moving body moves at constant velocity.

1.4 Newton's Second Law of Motion (N2L)

N2L states that the net force (the vector sum of all the forces) $\mathbf{F}$ on a particle of mass m is given by

$$\mathbf{F} = m\mathbf{a} \tag{1.9}$$

We can concretely write this in vector notation as

$$F_x\mathbf{e}_x + F_y\mathbf{e}_y + F_z\mathbf{e}_z = m(a_x\mathbf{e}_x + a_y\mathbf{e}_y + a_z\mathbf{e}_z) = m(\ddot{x}\mathbf{e}_x + \ddot{y}\mathbf{e}_y + \ddot{z}\mathbf{e}_z) \tag{1.10}$$

Thus when $\mathbf{F} = 0$ and $m \neq 0$, $\mathbf{a} = 0 \implies \dot{\mathbf{v}} = 0$ so we recover N1L.
N2L can be restated in terms of momentum. The momentum of a particle of mass m travelling at velocity $\mathbf{v}$ is given by

$$\mathbf{p} = m\mathbf{v} \tag{1.11}$$

Thus, we can restate N2L as

$$\mathbf{F} = \dot{\mathbf{p}} \tag{1.12}$$

This states that the net force on a particle is equal to the rate of change of particle's momentum. This version of N2L includes the case where mass does change with time.

The goal of classical mechanics is to solve for $\mathbf{r}(t)$ when an object is subject to an external force. Thus, the general problem-solving step is to identify the external forces on an object, compute the vector sum, and solve the ODE $\mathbf{F} = m\ddot{\mathbf{r}}$ for $\mathbf{r}$. N2L in reality is a set of second-order ODEs as we have an ODE for each component of the vector in whatever coordinate system we use.

In cartesian coordinates, $\mathbf{F} = m\ddot{\mathbf{r}}$ is the following set of ODEs,

$$F_x = m\ddot{x}(t)\ (F_x = \mathbf{F}\cdot\mathbf{e}_x \text{ and } x(t) = \mathbf{r}\cdot\mathbf{e}_x) \tag{1.13}$$

$$F_y = m\ddot{y}(t)\ (F_y = \mathbf{F}\cdot\mathbf{e}_y \text{ and } y(t) = \mathbf{r}\cdot\mathbf{e}_y) \tag{1.14}$$

$$F_z = m\ddot{z}(t)\ (F_z = \mathbf{F}\cdot\mathbf{e}_z \text{ and } z(t) = \mathbf{r}\cdot\mathbf{e}_z) \tag{1.15}$$

Since this is a second-order ODE, we need two initial conditions to create a valid initial-value problem. This is the initial position and velocity of the particle.

$$\mathbf{r}(0) = \mathbf{r}_0 \tag{1.16}$$

$$\mathbf{v}(0) = \mathbf{v}_0 \tag{1.17}$$

1.5 Newton's Third Law of Motion (N3L) and the Conservation of Momentum

While N1L and N2L are concerned with the response of a single object to applied forces, N3L involves a relationship between two objects.

N3L states that if object 1 exerts a force $\mathbf{F_{1\to 2}}$ (force exerted on object 2 by 1), then object 2 exerts a reaction force equal in magnitude and opposite in direction on object 1,

$$\mathbf{F_{1\to2}} = -\mathbf{F_{2\to1}} \tag{1.18}$$

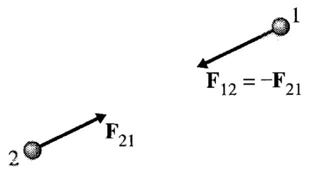


Figure 1.5 Newton's third law asserts that the reaction force exerted on object 1 by object 2 is equal and opposite to the force exerted on 2 by 1, that is, $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

N3L implies the conservation of momentum. Consider two objects 1 and 2, that have external forces $\mathbf{F_1^{ext}}$ and $\mathbf{F_2^{ext}}$ on them respectively. The net force on each object (including the reaction forces) is given by

$$\mathbf{F_1} = \mathbf{F_{1\to2}} + \mathbf{F_1^{ext}} \tag{1.19}$$

$$\mathbf{F_2} = \mathbf{F_{2\to1}} + \mathbf{F_2^{ext}} \tag{1.20}$$

By N2L, the net force on each system is given by their respective change in momenta

$$\dot{\mathbf{p_1}} = \mathbf{F_1} = \mathbf{F_{1\to2}} + \mathbf{F_1^{ext}} \tag{1.21}$$

$$\dot{\mathbf{p_2}} = \mathbf{F_2} = \mathbf{F_{2\to1}} + \mathbf{F_2^{ext}} \tag{1.22}$$

The total momenta of our system is given by $\mathbf{P} = \mathbf{p_1} + \mathbf{p_2}$. Thus, the change in momentum of our system is

$$\begin{aligned}
\dot{\mathbf{P}} &= \dot{\mathbf{p_1}} + \dot{\mathbf{p_2}} \\
&= \mathbf{F_1} + \mathbf{F_2} \\
&= \mathbf{F_{1\to 2}} + \mathbf{F_1^{ext}} + \mathbf{F_{2\to 1}} + \mathbf{F_2^{ext}} \\
&= \mathbf{F_{1\to 2}} + \mathbf{F_1^{ext}} - \mathbf{F_{1\to 2}} + \mathbf{F_2^{ext}} \\
\dot{\mathbf{P}} &= \mathbf{F_1^{ext}} + \mathbf{F_2^{ext}}
\end{aligned}$$

Defining the net external force $\mathbf{F^{ext}} = \mathbf{F_1^{ext}} + \mathbf{F_2^{ext}}$ yields,

$$\dot{\mathbf{P}} = \mathbf{F^{ext}} \tag{1.23}$$

This shows that the internal forces have no effect on the total momentum of the system. Additionally, if there is no external force on the system, then the change in momentum is zero, and momentum is conserved. A similar proof follows for an N-particle system (see Taylor chapter 1). In other words, when $\mathbf{F^{ext}} = 0, \dot{\mathbf{P}} = 0$

1.6 Newton's Laws and Reference Frames

Newton's laws are only valid as written above in inertial frames (non-accelerating and non-rotating reference frames)

In fact, N1L is only a special case of N2L in inertial frames.

Imagine we are accelerating to the right past a tree. In our RF, we see the tree acclerate to the left which violates N1L as there is no net external force acting on the tree. To explain this behaviour, we need to add fictitious forces in non-inertial frames like centrifugal force, coriolis force, or Euler's force. These forces do not arise because of physical interactions, but only purely from the acceleration of the reference frame.

Thus, we define inertial reference frames as reference frames where N1L holds.

N3L fails in relativistic phenomena as the action and reaction forces have to be measured at the same time t. However, there is no universal time near the speed of light due to time dilation.

From an inertial RF S, there are three ways to obtain another inertial RF

- Translation by a constant vector

  When we translate an RF by a constant vector $\mathbf{k}$, we are creating a new inertial RF by translating the origin of S by $\mathbf{k}$.

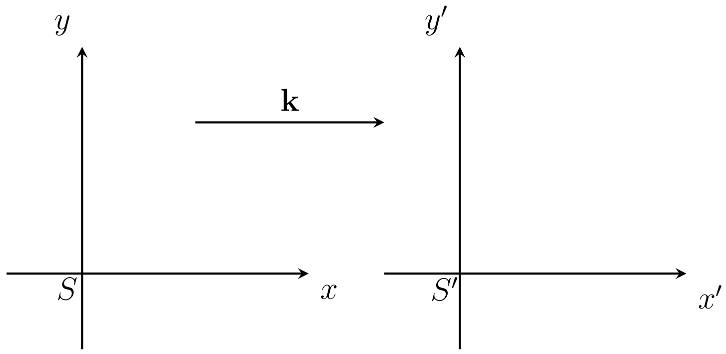
  

  Figure 1: Two coordinate systems S and S' with S' shifted to the right by $\mathbf{k}$.

  The position $\mathbf{r}'$ in S' is given by

  $$\mathbf{r}' = \mathbf{r} + \mathbf{k} \tag{1.24}$$

  Then differentiating to get acceleration,

  $$\ddot{\mathbf{r}}' = \ddot{\mathbf{r}} \tag{1.25}$$

  Thus, since S is inertial, $\mathbf{F} = 0 \implies \mathbf{r} = 0$. Since $\ddot{\mathbf{r}}' = \ddot{\mathbf{r}}$, $\mathbf{F} = 0 \implies \ddot{\mathbf{r}}' = 0$. Thus S' is also an inertial reference frame. One example is when we are looking at a ball falling from a height. We can either choose the origin to be the ground or translate it so that the origin is the original height of the ball.

- Boost by a constant velocity

  When we boost S by a constant velocity $\mathbf{v}$, S' moves at a constant velocity such that a position vector in S' is given by,

$$\mathbf{r}' = \mathbf{r} + \mathbf{v}t \tag{1.26}$$

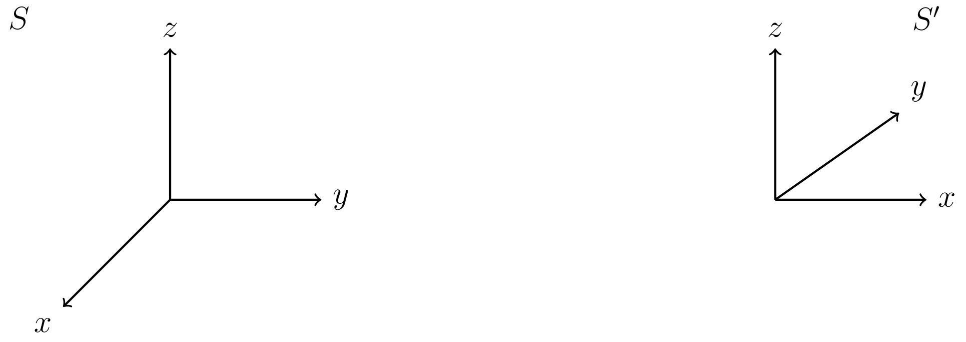
  Differentiating to get acceleration

$$\dot{\mathbf{r}'} = \dot{\mathbf{r}} + \mathbf{v} \tag{1.27}$$

$$\ddot{\mathbf{r}'} = \dot{\mathbf{r}} \tag{1.28}$$

  Thus, since S is inertial, $\mathbf{F} = 0 \implies \mathbf{r} = 0$. Since $\ddot{\mathbf{r}'} = \ddot{\mathbf{r}}$, $\mathbf{F} = 0 \implies \ddot{\mathbf{r}'} = 0$. Thus S' is also an inertial reference frame.

- Rotation by a constant angle We can rotate a RF by a constant angle by multiplying our original position vector $\mathbf{r}$ by a rotation matrix R. Rotating around the positive z-axis by an angle ϕ is given by



$$\mathbf{r}' = R\mathbf{r} \tag{1.29}$$

$$R = \begin{pmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.30}$$

Rotating reference frames is commonly done in box on incline problems, where we rotate the RF such that the x axis of S' is along the incline.

2 Equation of Motion for Different Forces

The general strategy to find the equation of motion for a system is to

1. Choose a reference frame that is well adapted to the problem (may have to translate, boost or rotate)
2. Draw a free body diagram by identifying all the external forces
3. Solve $\mathbf{F} = \dot{\mathbf{p}}$ ($\mathbf{F} = m\ddot{\mathbf{r}}$ at constant mass)

Assuming constant mass, we have several possible types of forces

2.1 Constant forces

If we have a constant force $\mathbf{F} = \mathbf{F}_0$ acting on a particle of mass m, then N2L gives

$$m\ddot{\mathbf{r}} = \mathbf{F}_0 \tag{2.1}$$

$$\implies \ddot{r} = \frac{\mathbf{F}_0}{m} \tag{2.2}$$

We integrate once to get the velocity of the particle,

$$\dot{\mathbf{r}} = \int \frac{\mathbf{F}_0}{m} dt = \frac{\mathbf{F}_0}{m} t + \mathbf{c_1} \tag{2.3}$$

We typically set the first constant of integration to be the initial velocity $\dot{\mathbf{r}}(0) = \mathbf{v}_0$. Then,

$$\dot{\mathbf{r}} = \frac{\mathbf{F}_0}{m} t + \mathbf{v}_0 \tag{2.4}$$

Integrate once again to get the position where we now specify the constant of integration to be the initial position $\mathbf{r}(0) = \mathbf{r}_0$.

$$\mathbf{r} = \int \frac{\mathbf{F}_0}{m} t + \mathbf{v}_0 dt = \frac{\mathbf{F}_0}{2m} t^2 + \mathbf{v}_0 t + \mathbf{r}_0 \tag{2.5}$$

Since force is constant, the above reduces to

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}t^2 \tag{2.6}$$

This is the familiar constant kinematics equation. We said it only holds for constant acceleration because constant force implies constant acceleration.

2.2 Time-dependent forces

If our force is now a function of time we have $\mathbf{F} = \mathbf{F}(t)$. Then by N2L,

$$m\ddot{\mathbf{r}} = \mathbf{F}(t) \tag{2.7}$$

$$\implies \ddot{\mathbf{r}} = \frac{\mathbf{F}(t)}{m} \tag{2.8}$$

Then,

$$\frac{dv}{dt} = \frac{\mathbf{F}(t)}{m} \tag{2.9}$$

By separation of variables we have,

$$dv = \frac{\mathbf{F}(t)}{m}dt \tag{2.10}$$

$$\implies \int dv = \int \frac{\mathbf{F}(t)}{m}dt \tag{2.11}$$

$$\implies \mathbf{v}(t) = \frac{\mathbf{f}(t)}{m} + \mathbf{c_1} \tag{2.12}$$

where $\mathbf{f}$ is the antiderivative of $\mathbf{F}$. Integrating again gives

$$\mathbf{r}(t) = \int \frac{\mathbf{f}(t)}{m}dt + \mathbf{c_1}t + \mathbf{c_2} \tag{2.13}$$

The constants of integration are linked to the initial velocity and position, but depending on the force as a function of time, is not the initial velocity and position itself.

Example 0

A particle of mass m moves along the line $-\infty < x < \infty$. It is subjected to a force $\mathbf{F}(t) = F_0 \cos(\alpha t)\mathbf{e}_x$. It starts at time $t_0 = 0$ at $x_0 = 0$ at rest. Find its position $x(t)$ for later times.

Solution

From N2L we have,

$$\ddot{x}(t) = \frac{F_0 \cos(\alpha t)}{m} \tag{2.14}$$

Integrating,

$$\dot{x}(t) = \int \frac{F_0 \cos(\alpha t)}{m} dt = \frac{F_0}{m\alpha} \sin(\alpha t) + c_1 \tag{2.15}$$

Since it starts at rest, $\dot{x}(0) = 0$. Then,

$$\dot{x}(0) = c_1 = 0 \tag{2.16}$$

Integrating to get position,

$$x(t) = -\frac{F_0}{m\alpha^2} \cos(\alpha t) + c_2 \tag{2.17}$$

Substitute the initial condition $x(0) = 0$ to get,

$$x(0) = -\frac{-F_0}{m\alpha^2} + c_2 = 0 \tag{2.18}$$

Then,

$$c_2 = \frac{F_0}{m\alpha^2} \tag{2.19}$$

Thus,

$$\mathbf{x}(t) = \frac{F_0}{m\alpha^2}(1 - \cos(\alpha t))\mathbf{e}_x \tag{2.20}$$

2.3 Position-dependent forces

If force is a function of position we have $\mathbf{F} = \mathbf{F}(\mathbf{r})$. Then by N2L,

$$\ddot{\mathbf{r}} = \frac{\mathbf{F}(\mathbf{r})}{m} \tag{2.21}$$

$$\implies \frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}(\mathbf{r})}{m} \tag{2.22}$$

From the chain rule we have,

$$\frac{d\mathbf{v}}{dt} = \frac{d\mathbf{v}}{d\mathbf{r}}\frac{d\mathbf{r}}{dt} \tag{2.23}$$

Since $\frac{d\mathbf{r}}{dt} = \mathbf{v}$,

$$\frac{d\mathbf{v}}{dt} = \mathbf{v}\frac{d\mathbf{v}}{d\mathbf{r}} \tag{2.24}$$

Then,

$$\mathbf{v} \cdot d\mathbf{v} = \frac{\mathbf{F}(\mathbf{r})}{m} \cdot d\mathbf{r} \tag{2.25}$$

Replace with the dummy variables and integrate $\mathbf{v}_0$ to v and $\mathbf{r}_0$ to $\mathbf{r}$. Then,

$$\int_{\mathbf{v}_0}^{\mathbf{v}} m\mathbf{v}' \cdot d\mathbf{v}' = \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}' \tag{2.26}$$

Then, we get

$$\frac{1}{2}m(\mathbf{v}^2 - \mathbf{v}_0^2) = \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}' \tag{2.27}$$

where $\mathbf{v}^2$ is the magnitude of $\mathbf{v}$. We define the kinetic energy as

$$T = \frac{1}{2}m\mathbf{v}^2 \tag{2.28}$$

Define the initial kinetic energy when it was at $\mathbf{r}_0$ as

$$T_0 = \frac{1}{2}m\mathbf{v}_0^2 \tag{2.29}$$

Then, the above integral becomes the work-energy theorem

$$T - T_0 = \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}' \tag{2.30}$$

If $\mathbf{F}$ is a conservative force, it is the gradient of a function called the potential-energy function (see section on conservative forces)

$$\mathbf{F} = -\nabla U \tag{2.31}$$

Substitute this into eq. (2.30),

$$T - T_0 = U(\mathbf{r}_0) - U(\mathbf{r}) \tag{2.32}$$

Then we have the conservation of energy,

$$U(\mathbf{r}_0) + T_0 = T + U(\mathbf{r}) = E_0 \tag{2.33}$$

By the conservation of energy we have,

$$\frac{1}{2} m\mathbf{v}^2 = E_0 - U(\mathbf{r}) \tag{2.34}$$

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \pm\sqrt{\frac{2}{m}\left(E_0 - U(\mathbf{r})\right)} \tag{2.35}$$

It is easier to solve for dt

$$\int_{t_0}^{t} dt = \int \pm \frac{1}{\sqrt{\frac{2}{m}(E_0 - U(\mathbf{r}))}} d\mathbf{r} \tag{2.36}$$

Then,

$$t(\mathbf{r}) = \int \pm \frac{1}{\sqrt{\frac{2}{m}(E_0 - U(\mathbf{r}))}} d\mathbf{r} + t_0 \tag{2.37}$$

We can invert this to get $\mathbf{r}(t)$. However, the integral may not have a closed-form solution. Thus, it is prudent to plot the potential energy to qualitatively understand the motion of the particle.

2.4 Energy Diagrams

Consider the plot of potential energy as a function of position

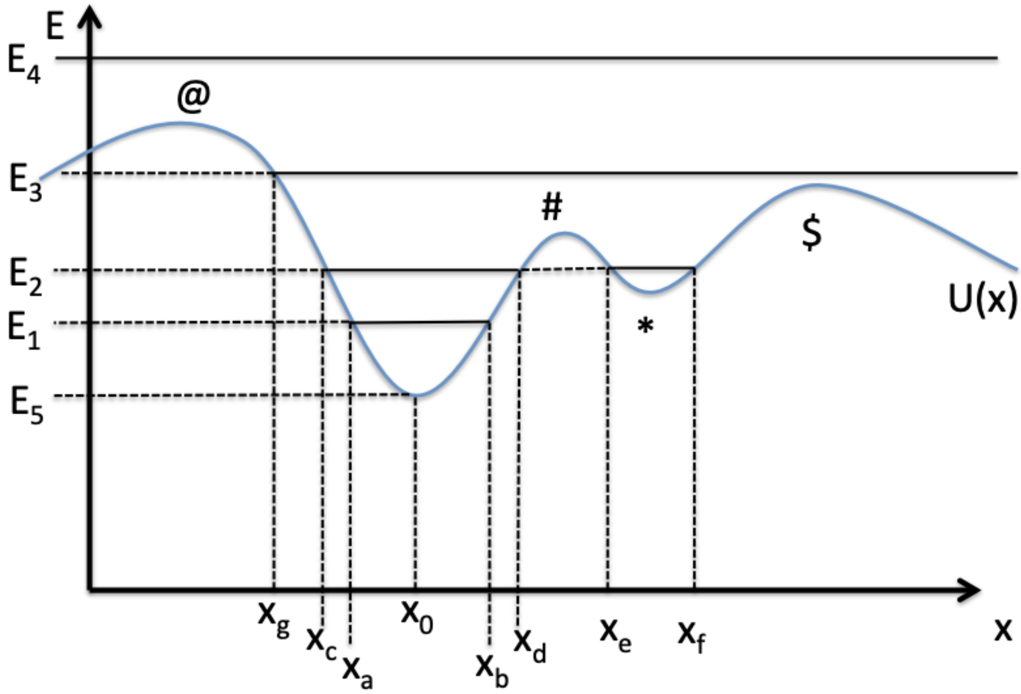


Figure 2: Potential energy of a particle

- The particle cannot have energy less than E_5. Then, $T = E_5 - U(x) < 0$ and the particle cannot have negative energy.

- If $U(x) = E_5$, the particle cannot move anywhere as it would need more energy which cannot magically be created. Here, $T = 0$

- If $U(x) = E_1$, the particle can move in the interval $[x_a, x_b]$. At x_a and x_b, the particle will have its maximum potential energy and so $T = 0$, x_a and x_b are known as turning points. At x_0, the particle has maximum kinetic energy.

- If $U(x) = E_2$, the particle can move between $[x_c, x_d]$ or $[x_e, x_f]$. It cannot move from the first region to the second region without additional energy

- If $U(x) = E_3$, then the particle will have position $x \geq x_3$. At $x = x_g$, the particle will move even though it has no kinetic energy since $F = -U'(x_g) \neq 0$ so the particle will accelerate
- If $U(x) = E_4$, the particle can be at any position

Definition 2.1 (Turning Point)**.** A turning point is a point in a potential where a particle has zero kinetic energy. At the turning point, it must have maximum potential energy.

When a particle reaches its turning point, it stops and reverses its direction of motion. Otherwise, it would result in negative kinetic energy. On a potential energy diagram, the turning points occur at the intersection of the maximum potential energy the particle can have with $U(\mathbf{r})$. We can analyse the motion near the equilibrium points. Consider the 1D case where a particle has potential energy $U = U(x)$. An equilibrium point occurs at the extrema of the curve. If x_p is an equilibrium point then

$$\frac{dU}{dx}|_{x_p} = 0 \tag{2.38}$$

Definition 2.2 (Stable Equilibrium Point)**.** An equilibrium point is stable if the potential is a minimum at that point. Formally, if x_0 is the equilibrium point, then it is stable if

$$\frac{d^2U}{dx^2}|x_0 > 0 \tag{2.39}$$

At a stable point, a small perturbation or push will cause the particle to return to the minimum. To the left of a minimum, $\frac{dU}{dx} < 0$ and to the right $\frac{dU}{dx} > 0$. A small push will cause a particle to experience a force in the opposite direction (since $F = -\frac{dU}{dx}$, causing the particle to return to the minimum.

Definition 2.3 (Unstable Equilibrium Point)**.** An equilibrium point is unstable if the potential is a maximum at that point. Formally, if x_0 is the equilibrium point, then it is unstable if

$$\frac{d^2U}{dx^2}|x_0 < 0 \tag{2.40}$$

At an unstable point, a small perturbation will cause the particle to accelerate away from the equilibrium point. To the left of a maximum we have $\frac{dU}{dx} > 0$ and to the right we have $\frac{dU}{dx} < 0$. Thus, a particle will see a force in the same direction as the perturbation, causing it to accelerate asway. The particle will accelerate until a turning point, falls into a stable equilibrium point, or escapes to infinity depending on the potential.

Definition 2.4 (Marginally Stable Equilibrium Point)**.** An equilibrium point is marginally unstable if the potential is a saddle point at that point. Formally, if x_0 is the equilibrium point, then it is marginally stable if

$$\frac{d^2U}{dx^2}|x_0 = 0 \tag{2.41}$$

If the particle is near a stable equilibrium point, we can get a more analytic understanding of its motion. Near the minimum, we can approximate the potential energy with a parabola. Let x_0 be the equilibrium point. We can taylor expand $U(x)$ about $x = x_0$ to

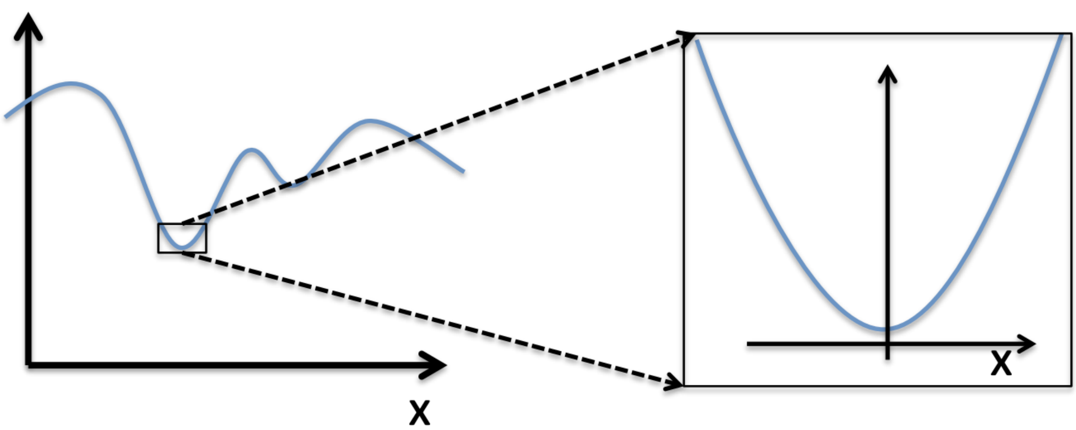


Figure 3: Near a stable equilibrium point

get.

$$U(x) \approx U(x_0) + \frac{dU}{dx}|_{x_0}|(x - x_0) + \frac{1}{2!}\frac{d^2U}{dx^2}|_{x_0}(x - x_0)^2 + O(x^3) \tag{2.42}$$

Since $U(x_0)$ is a constant, it involves a vertical translation of $U(x)$. We can set $U(x_0) = 0$. Further, since x_0 is an equilibrium we have $\frac{dU}{dx}|x_0 = 0$. Neglecting higher order terms,

$$U(x) = \frac{1}{2}\frac{dU}{dx}|_{x_0}(x - x_0)^2 \tag{2.43}$$

This is reminiscent of the potential energy of a simple harmonic oscillator $U(x) = \frac{1}{2}k(x - x_0)^2$ where $k = \frac{d^2U}{dx^2}|_{x_0}$. We can Taylor expand our force to get,

$$F(x) = -\frac{dU}{dx} \approx F(x_0) + \frac{dF}{dx}|_{x_0}(x - x_0) + O(x^2) \tag{2.44}$$

Since $F(x_0) = -\frac{dU}{dx}|_{x_0} = 0$ and $\frac{dF}{dx}|_{x_0} = -\frac{d^2U}{dx^2}|_{x_0} = -k$,

$$F(x) = -k(x - x_0) \tag{2.45}$$

Since we wanted to analyse the motion of the particle near the minimum, we can use N2L to get

$$m\ddot{x} = -kx + kx_0 \tag{2.46}$$

Rearrange,

$$m\ddot{x} + kx = kx_0 \tag{2.47}$$

We can divide by m and define $\omega = \sqrt{\frac{k}{m}}$ to get,

$$\ddot{x} + \omega^2 x = \omega^2 x_0 \tag{2.48}$$

This is the equation of a simple harmonic oscillator oscillating about $x = x_0$. The corresponding homogenous ODE is $\ddot{x} + \omega^2 x = 0$ which has roots of the characteristic polynomial as $r = \pm i\omega$. Then, the complimentary solution to the ODE is,

$$x_c = A\cos(\omega t) + B\sin(\omega t) \tag{2.49}$$

The particular solution is then,

$$x(t) = A\cos(\omega t) + B\sin(\omega t) + x_0 \tag{2.50}$$

The frequency and period of the oscillation is,

$$\omega = \sqrt{\frac{k}{m}} \tag{2.51}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \tag{2.52}$$

Thus, near a stable equilibrium point, a particle's motion can be described by a simple harmonic oscillator.

2.5 Velocity-dependent forces

If force is a function of velocity we have $\mathbf{F} = \mathbf{F}(\mathbf{v})$. Then by N2L,

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}(\mathbf{v})}{m} \tag{2.53}$$

We can use separation of variables to get,

$$\int \frac{1}{\mathbf{F}(\mathbf{v})} \cdot d\mathbf{v} = \int \frac{1}{m} dt \tag{2.54}$$

If we can solve this analytically, we get $\mathbf{v}(t)$. We can then integrate this to get $\mathbf{r}(t)$. The main class of velocity dependent forces is the drag force.

Definition 2.5 (Drag Force)**.** Drag force is friction due to motion through a medium (typically a fluid).

The most famous drag force is air-resistance. Drag forces always act in a direction opposite to the direction of motion. If a particle is travelling with a velocity $\mathbf{v}$, we can define a unit vector $\mathbf{e}_v$ that is in the direction of velocity.

$$\mathbf{e}_v = \frac{\mathbf{v}}{|\mathbf{v}|} \tag{2.55}$$

Then, the particle has velocity,

$$\mathbf{v} = v\mathbf{e}_v \tag{2.56}$$

A drag force will always act in the opposite direction to motion,

$$\mathbf{F}(\mathbf{v}) = -f(v)\mathbf{e}_v \tag{2.57}$$

There are two types of drag.

2.5.1 Linear drag

Linear drag (or Stoke's drag) is experienced when a particle is moving at slow velocities, or through a viscous fluid like honey. The linear drag force is given by,

$$\mathbf{F}(\mathbf{v}) = -c\mathbf{v} = -cv\mathbf{e}_v \tag{2.58}$$

where c is a constant

Example 0

A particle with initial velocity $\mathbf{v}_0$ that starts at $x = x_0$ is moving along the x direction. It experiences a drag force $\mathbf{F}_D = -cv\mathbf{e}_x$. What is its velocity as a function of time?

Solution

By N2L we have,

$$m\ddot{x}\mathbf{e}_x = -cv\mathbf{e}_x \tag{2.59}$$

Then,

$$\frac{dv}{dt}\mathbf{e}_x = \frac{-cv}{m}\mathbf{e}_x \tag{2.60}$$

Use separation of variables and define $\kappa = \frac{c}{m}$,

$$\int \frac{1}{v}dv = -\int \kappa dt \tag{2.61}$$

$$\implies \ln|v| = -\kappa t + C \tag{2.62}$$

$$\implies v = Ae^{-\kappa t} \tag{2.63}$$

Use the initial condition $v(0) = v_0$ to get,

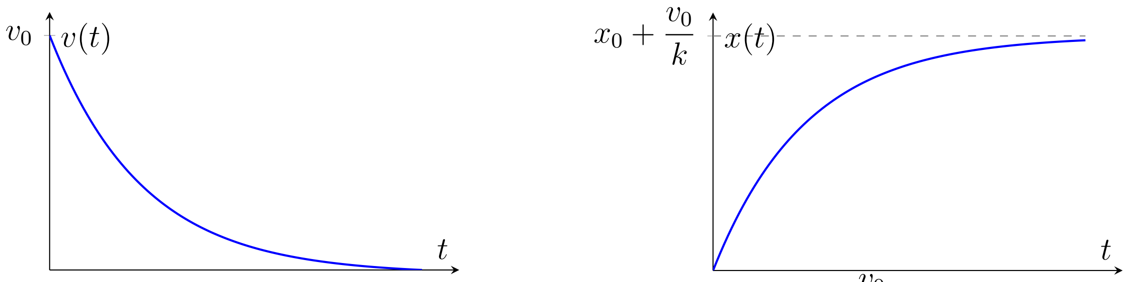
$$\mathbf{v}(t) = v(t)\mathbf{e}_x = v_0 e^{-\kappa t}\mathbf{e}_x \tag{2.64}$$

We see $\lim_{t\to\infty} v(t) = 0$ which physically makes sense since the drag force should slow the particle. Additionally, $v(t = 0) = v_0$ which is consistent with the initial conditions. κ dictates the rate of decay, a higher κ means the particle will come to rest quicker. We can also integrate to get position,

$$x(t) = -\frac{v_0}{\kappa}\left(1 - e^{-\kappa t}\right) + x_0 \tag{2.65}$$

As $t \to \infty$, the particle comes to a stop given by $x \to \frac{-v_0}{\kappa} + x_0$

In the above example, the plots for $v(t)$ and $x(t)$ are



(a) $v = v_0 e^{-kt}$

(b) $x = x_0 + \frac{v_0}{k}\left(1 - e^{-kt}\right)$

Figure 4: Velocity and position for a particle subject to linear drag.

2.5.2 Non-linear drag

Non-linear drag (or Newtonian drag) is experienced when a particle is moving at high velocities or through low viscosity fluids. The non-linear drag force is given by,

$$\mathbf{F}(\mathbf{v}) = -c\mathbf{v}^2 = -cv^2\mathbf{e}_v \tag{2.66}$$

where c is a constant.

Example 0

A particle of mass m is falling near the Earth's surface. It is subject to a non-linear drag force given by $|\mathbf{F}_D| = cv^2$. Find $v(t)$ with the initial conditions $z(0) = z_0; v(0) = 0$

Solution

The particle's weight acts in the negative $\mathbf{e}_z$ position. Then, the drag force must act in the positive $\mathbf{e}_z$ position. Thus by N2L,

$$m\ddot{z}\mathbf{e}_z = (cv^2 - mg)\mathbf{e}_z \tag{2.67}$$

Rearranging,

$$\frac{dv}{dt} = \kappa v^2 - g \tag{2.68}$$

Use separation of variables to get,

$$\int_0^v \frac{1}{\kappa v'^2 - g} dv' = \int_0^t dt' \tag{2.69}$$

This simplifies to,

$$t = -\frac{1}{\sqrt{g\kappa}} \operatorname{arctanh}(v\sqrt{g\kappa}) \tag{2.70}$$

Inverting,

$$\mathbf{v}(t) = -\sqrt{\frac{g}{\kappa}} \tanh(\sqrt{g\kappa}t)\mathbf{e}_z \tag{2.71}$$

The plot of the $v(t)$ from the example above is

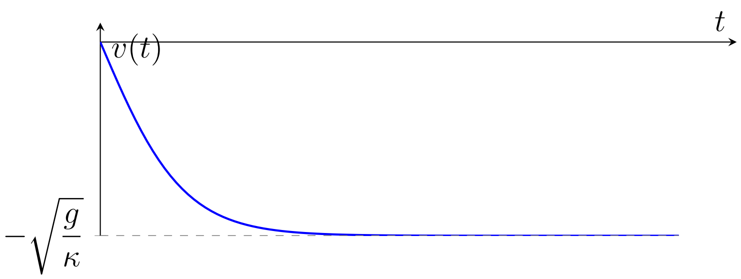


Figure 5: $v(t) = -\sqrt{\frac{g}{\kappa}} \tanh\left(\sqrt{g\kappa}\, t\right)$

This makes physical sense. Initially, the particle is in free-fall. Thus, the particle accelerates downwards causing the magnitude of its velocity to increase. This, in turn causes the drag force to increase until it cancels out with the particle's weight. At this point, there is no net external force so by N1L, the particle falls at constant velocity. This velocity is known as the terminal velocity and is given by,

$$\mathbf{v}_T = -\sqrt{\frac{g}{\kappa}}\mathbf{e}_z \tag{2.72}$$

We can see the particle is in free-fall using a Taylor expansion. We have,

$$\tanh(x) \approx x - \frac{1}{3}x^3 + ... \tag{2.73}$$

Then,

$$\tanh(\sqrt{g\kappa}t) \approx \sqrt{g\kappa t} - \frac{1}{3}(g\kappa)^{\frac{3}{2}}t^3 + ... \tag{2.74}$$

Then,

$$v(t) \approx -\sqrt{\frac{g}{\kappa}}\left(\sqrt{g\kappa}t - \frac{1}{3}(g\kappa)^{\frac{3}{2}}t^3 + ...\right) \tag{2.75}$$

$$\approx -gt + \frac{1}{3}g^2\kappa^2t^3 + ... \tag{2.76}$$

The first term shows that for small t, we can approximate the particle's motion using free-fall.

2.6 General forces

A general force can depend on position, velocity, and time $\mathbf{F} = \mathbf{F}(\mathbf{r}, \mathbf{v}, t)$. Then, by N2L we have

$$m\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}, \mathbf{v}, t) \tag{2.77}$$

In many cases, we need to resort to numerical simulations to solve such an ODE. However, there are special cases depending on the form of the force. These only work in the 1D case.

2.6.1 $F(v,t) = f(v)h(t)$

For a force $F(v,t) = f(v)h(t)$, we can use separation of variables to get,

$$\int \frac{m}{f(v)} dv = \int h(t)\, dt \tag{2.78}$$

2.6.2 $F(r,v) = f(v)h(r)$

For a force $F(r,v) = f(v)h(r)$, we can use the chain rule to get

$$mv\frac{dv}{dr} = f(v)h(r) \tag{2.79}$$

Then use separation of variables to get,

$$\int \frac{mv}{f(v)} dv = \int h(r)\, dr \tag{2.80}$$

2.7 Time-varying mass and rocket motion

2.7.1 No external forces

Consider a rocket of mass M moving in one dimension in free space (i.e. no external forces) with a speed v (note that we are not in the center of mass frame of the rocket as then we would observe $v = 0$)

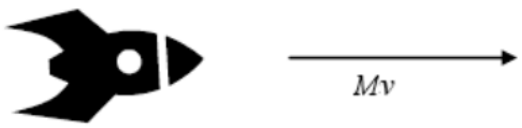


Figure 6: Rocket moving through free space

An infinitesimal amount of time dt later, the rocket has ejected a piece of fuel of mass dm at a speed u relative to the rocket

Since the rocket burns fuel, its mass is no longer constant and is a function fo time. Thus,

Figure 7: The rocket at $t + dt$

we must use the more general case of N2L. The initial momentum of the rocket is,

$$p_i = M(t)v \tag{2.81}$$

Since the burning of the fuel causes the rocket to accelerate, it will now have velocity $v + dv$. Additionally, the velocity of the piece of fuel in our reference frame is $v - u$. Then,

$$p_{fuel} = dm(v - u) \tag{2.82}$$

If the mass of the fuel is dm, then the rocket should have lost mass dm at $t + dt$. Thus, $M(t + dt) = M(t) - dm$. Then,

$$p_f = (M - dm)(v + dv) + dm(v - u) \tag{2.83}$$

Expanding this,

$$p_f = Mv + Mdv - dmv - dmdv + dmv - dmu \tag{2.84}$$

$$= Mv + Mdv - dmdv - udm \tag{2.85}$$

We can neglect $dmdv$ since it is extremely small. Thus,

$$p_f = Mv + Mdv - udm \tag{2.86}$$

According to N2L,

$$\mathbf{F} = \dot{\mathbf{P}} \tag{2.87}$$

Since there is no net external force, $\mathbf{F} = 0 \implies \mathbf{P} = \text{constant}$. Then, by conservation of momentum,

$$p_i = p_f \tag{2.88}$$

$$\implies Mv = Mv + Mdv - udm \tag{2.89}$$

$$Mdv = udm \tag{2.90}$$

$$\implies dv = \frac{u}{M}dm \tag{2.91}$$

We need an expression for dm in terms of dM. We have,

$$dM = M(t + dt) - M(t) \tag{2.92}$$

$$= (M - dm) - M \tag{2.93}$$

$$= -dm \tag{2.94}$$

Thus, the rocket loses exactly the mass of the fuel burned. Then,

$$dv = -\frac{u}{M}dM \tag{2.95}$$

We integrate to get,

$$v(t) = -u\ln(M(t)) + c \tag{2.96}$$

where c is a constant of integration. We can omit the absolute value sign since mass is always positive. Let us assume that at $v(t = 0) = v_0$ and $M(t = 0) = M_0$. Then,

$$v(0) = -u\ln(M_0) + c = v_0 \tag{2.97}$$

Then,

$$c = v_0 + u\ln(M_0) \tag{2.98}$$

Then,

$$v(t) = -u\ln(M(t)) + u\ln(M_0) + v_0 \tag{2.99}$$

Combining the logarithms,

$$v(t) = u \ln\left(\frac{M_0}{M(t)}\right) + v_0 \tag{2.100}$$

Since $M(t) < M_0$ for all $t > 0$, the natural log is positive which shows that the velocity increases.

Example 0

A rocket starting from rest is comprised of 90% fuel. What is its velocity as a function of u after all fuel is burned.

Solution

We have $v_0 = 0$ and $M_f = 0.1M_0$. Thus,

$$v = u \ln\left(\frac{M_0}{0.1M_0}\right) \approx 2.3u \tag{2.101}$$

The fact that our rocket being made of 90% fuel results in only a 2.3 times increase in speed, shows the need for all non-fuel mass (i.e. the payload of the rocket) is as small as possible.

2.7.2 Rocket in a gravitational field

Consider a rocket near the surface of the Earth. Since the rocket is in the Earth's gravitational field, it will now have a net external force acting on it, given by its weight. This is given by,

$$\mathbf{F} = -Mg\mathbf{e}_z \tag{2.102}$$

Working strictly in the z-axis, we have

$$\dot{p} = -Mg \tag{2.103}$$

The initial momentum is given by,

$$p_i = Mv \tag{2.104}$$

The final momentum is given by,

$$p_f = (M - dm)(v + dv) + dm(v - u) \tag{2.105}$$

Expanding and neglecting the $dmdv$ term gives,

$$p_f = Mv + Mdv - udm \tag{2.106}$$

By N2L we have,

$$\begin{aligned} dp &= p_f - p_i &(2.107)\\ &= Mdv - udm &(2.108)\\ &= Mdv + udM &(2.109)\end{aligned}$$

However,

$$dp = -Mgdt \tag{2.110}$$

Then,

$$Mdv + udM = -Mgdt \tag{2.111}$$

Divide by dt to get,

$$M\frac{dv}{dt} + u\frac{dM}{dt} = -Mg \tag{2.112}$$

Dividing by M,

$$\frac{dv}{dt} = -g - \frac{u}{M}\frac{dM}{dt} \tag{2.113}$$

To solve this ODE we need an expression for $\frac{dM}{dt}$. Assume the fuel is burned at a constant rate α. Then,

$$\frac{dM}{dt} = -\alpha \tag{2.114}$$

This ODE can be solved to get,

$$M = M_0 - \alpha t \tag{2.115}$$

Substitute into eq. (2.113),

$$\frac{dv}{dt} = -g + \frac{\alpha u}{M_0 - \alpha t} \tag{2.116}$$

Integrate to get,

$$\int_{v_0}^{v} dv' = \int_0^t -g + \frac{\alpha u}{M_0 - \alpha t'} dt' \tag{2.117}$$

$$\implies v - v_0 = -gt - u \ln(M_0 - \alpha t) \tag{2.118}$$

$$\tag{2.119}$$

Then,

$$v(t) = v_0 - gt - u \ln\left(\frac{M_0 - \alpha t}{M_0}\right) \tag{2.120}$$

3 Newton's Laws of Motion in 2D and 3D

As we move to higher dimensions, the type of coordinate system we use can greatly simplify our problem. The most common coordinate system is the Cartesian Coordinate System

3.1 N2L in Cartesian Coordinates

Recall that N2L is a vector equation,

$$\mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t) = m\ddot{\mathbf{r}} \tag{3.1}$$

where $\mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t)$ is the most general form of a force. In cartesian coordinates the force is,

$$\mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t) = F_x\mathbf{e}_x + F_y\mathbf{e}_y + F_z\mathbf{e}_z \tag{3.2}$$

Similarly, the position is,

$$\mathbf{r}(t) = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z \tag{3.3}$$

Thus, N2L is really the following set of ODEs,

$$F_x = m\ddot{x} \tag{3.4}$$

$$F_y = m\ddot{y} \tag{3.5}$$

$$F_z = m\ddot{z} \tag{3.6}$$

We need to solve this ODE across every component. An extended example follows. Consider a particle of mass m with a positive charge q and an initial velocity $\mathbf{v}(t = 0) = \mathbf{v}_0$ in an external magnetic field $\mathbf{B} = B\mathbf{e}_z$.

We will show that the charged particle follows a helical trajectory as shwon in the diagram.

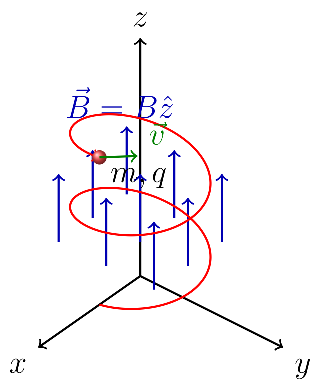


Figure 8: A charged particle of mass m and charge q undergoing helical motion in a uniform magnetic field $\vec{B} = B\hat{z}$.

The force on the particle due to the magnetic field is given by,

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \tag{3.7}$$

Thus,

$$\mathbf{F} = q(v_{\mathbf{e}x} + v\mathbf{e}_y + v\mathbf{e}_z) \times B\mathbf{e}_z \tag{3.8}$$

$$= \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ v_x & v_y & v_z \\ 0 & 0 & B \end{vmatrix} \tag{3.9}$$

$$= qB(v_y\mathbf{e}_x - v_x\mathbf{e}_y) \tag{3.10}$$

By N2L we have,

$$\mathbf{F} = m\ddot{\mathbf{r}} = q\mathbf{v} \times \mathbf{B} \tag{3.11}$$

$$\implies m(\ddot{x}\mathbf{e}_x + \ddot{y}\mathbf{e}_y + \ddot{z}\mathbf{e}_z) = qB(v_y\mathbf{e}_x - v_x\mathbf{e}_y) \tag{3.12}$$

Then breaking up the force into its components we have,

$$m\ddot{x} = qBv_y \tag{3.13}$$

$$m\ddot{y} = -qBv_x \tag{3.14}$$

$$m\ddot{z} = 0 \tag{3.15}$$

Recognising that $\ddot{x} = \dot{v}_x$, $\ddot{y} = \dot{v}_y$ and $\ddot{z} = \dot{v}_z$

$$m\dot{v}_x = qBv_y \tag{3.16}$$

$$m\dot{v}_y = -qBv_x \tag{3.17}$$

$$m\dot{v}_z = 0 \tag{3.18}$$

The velocity in the z component can easily be integrated to give,

$$v_z = v_{z0} \tag{3.19}$$

This shows that the particle moves with constant velocity in the z direction. The remaining ODEs are coupled as $\dot{v}_x \propto v_y$ and $\dot{v}_y \propto v_x$. We define the cyclotron frequency ω as

$$\omega = \frac{qB}{m} \tag{3.20}$$

Then, the remaining ODEs become

$$\dot{v}_x = \omega v_y \tag{3.21}$$

$$\dot{v}_y = -\omega v_x \tag{3.22}$$

We take the time derivative of eq. (3.21) and substitute into eq. (3.22) to get,

$$\ddot{v}_x = \omega \dot{v}_y \tag{3.23}$$

$$\implies \ddot{v}_x = -\omega^2 v_x \tag{3.24}$$

Then take the time derivative of eq. (3.22) and substitute into eq. (3.21) to get,

$$\ddot{v}_y = -\omega \dot{v}_x \tag{3.25}$$

$$\implies \ddot{v}_y = -\omega^2 v_y \tag{3.26}$$

We have now decoupled our ODEs to get two independent ODEs,

$$\ddot{v}_x = -\omega^2 v_x \tag{3.27}$$

$$\ddot{v}_y = -\omega^2 v_y \tag{3.28}$$

Rearranging,

$$\ddot{v}_x + \omega^2 v_x = 0 \tag{3.29}$$

$$\ddot{v}_y + \omega^2 v_y = 0 \tag{3.30}$$

This is the form of the simple harmonic oscillator ODE. From eq. (2.50) we have,

$$v_x = A\cos(\omega t) + B\sin(\omega t) + v_{0x} \tag{3.31}$$

$$v_y = C\cos(\omega t) + D\sin(\omega t) + v_{0y} \tag{3.32}$$

Alternatively, we could define the complex function η

$$\eta = v_x + i v_y \tag{3.33}$$

Taking the derivtaive,

$$\dot{\eta} = \dot{v}_x + i\dot{v}_y \tag{3.34}$$

$$= -\omega v_y - i\omega v_x \tag{3.35}$$

Thus,

$$\dot{\eta} = -i\omega\eta \tag{3.36}$$

By separation of variables we have

$$\ln(|\eta|) = -i\omega t + C \tag{3.37}$$

Then,

$$\eta = ce^{-i\omega t} \tag{3.38}$$

Since η was a complex function, c must also be a complex number we write

$$c = Ae^{i\phi} \tag{3.39}$$

Then,

$$\eta = Ae^{i(\phi-\omega t)} \tag{3.40}$$

Using Euler's formula

$$Ae^{i\omega t} = A(\cos(\omega t) + i\sin(\omega t)) \tag{3.41}$$

We have,

$$\eta = v_x + iv_y = A\cos(\omega t - \phi) - A\sin(\omega t - \phi) \tag{3.42}$$

Then,

$$v_x = \Re(\eta) = A\cos(\omega t - \phi) \tag{3.43}$$

$$v_y = \Im(\eta) = -A\sin(\omega t - \phi) \tag{3.44}$$

Thus, the components of our velocities are

$$v_x = A\cos(\omega t - \phi) \tag{3.45}$$

$$v_y = A\sin(\omega t - \phi) \tag{3.46}$$

$$v_z = v_{0z} \tag{3.47}$$

Our positions are then given by integration,

$$x = \int A\cos(\omega t - \phi)dt = \frac{A}{\omega}\sin(\omega t - \phi) + c_x \tag{3.48}$$

$$y = \int -A\sin(\omega t - \phi)dt = \frac{A}{\omega}\cos(\omega t - \phi) + c_y \tag{3.49}$$

$$z = \int v_{0z}dt = v_{0z}t + c_z \tag{3.50}$$

This is precisely the parameterisation of a helix.

3.2 Curvilinear Coordinates

Other coordinate systems are often better-adapted to the symmetries of a problem. In the previous example, the particle's trajectory wrapped around a cylinder. The most common are coordinate systems well-adapted to the circle, cylinder, and sphere

3.2.1 Polar Coordinates

In polar coordinates, we represent a point P using coordinates (r, φ).

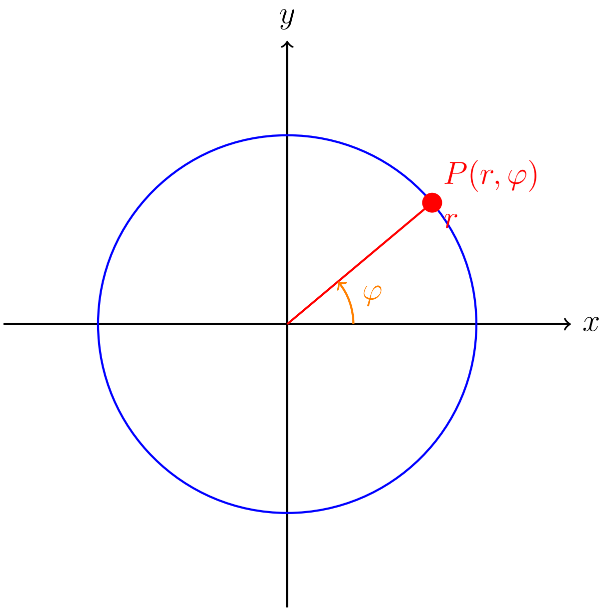


Figure 9: Polar Coordinates

r is the distance from the origin to the point and φ is the angle from the horizontal to the line joining the origin and the point. r and φ are given by,

$$r = \sqrt{x^2 + y^2} \tag{3.51}$$

$$\varphi = \arctan\left(\frac{y}{x}\right) \tag{3.52}$$

The values that r and φ can take on are,

$$r \in [0, \infty) \tag{3.53}$$

$$\varphi \in [0, 2\pi] \tag{3.54}$$

We can convert from polar to cartesian with,

$$x = r\cos(\varphi) \tag{3.55}$$

$$y = r\sin(\varphi) \tag{3.56}$$

In polar coordinates, functions of the form $r =$ constant create circles of radius r. Functions of the form $\varphi =$ constant create infinitely long lines that are angled φ radians away from the horizontal.

3.2.2 Cylindrical Coordinates

In cylindrical coordinates, we represent a point P using coordinates (r, φ, z)

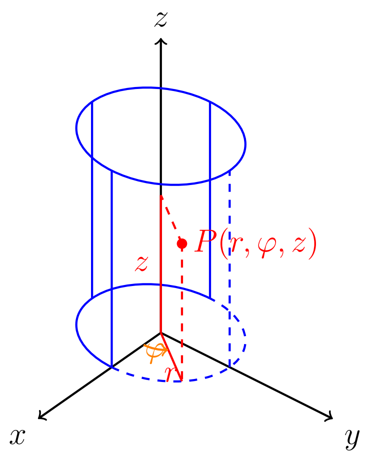


Figure 10: Caption

r is the distance from the origin to the projection of the point on the x-y plane. φ is the angle from the x axis to the line connecting the origin and the projection of the point. z is the height of the point from x-y plane. Cylindrical coordinates can be thought of as the 3D equivalent of polar coordinates as

$$r = \sqrt{x^2 + y^2} \tag{3.57}$$

$$\varphi = \arctan\left(\frac{y}{x}\right) \tag{3.58}$$

$$z = z \tag{3.59}$$

The values that ,φ and z can take on are given by

$$r \in [0, \infty) \tag{3.60}$$

$$\varphi \in [0, 2\pi] \tag{3.61}$$

$$z \in (-\infty, \infty) \tag{3.62}$$

We can convert from cylindrical to cartesian with,

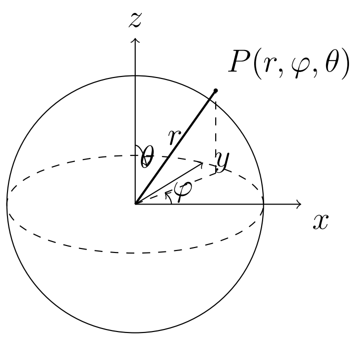
$$x = r\cos(\varphi) \tag{3.63}$$

$$y = r\sin(\varphi) \tag{3.64}$$

$$z = z \tag{3.65}$$

3.2.3 Spherical Coordinates

In spherical coordinates, a point P is represented as (r, φ, θ) r is the distance from the



origin to the point P. φ is the angle from the x axis to the projection of the point on the x-y plane. θ, known as the azimuthal angle, is the angle from the z-axis to the line from the origin to the point. r, φ, and θ are given by

$$r = \sqrt{x^2 + y^2 + z^2} \tag{3.66}$$

$$\varphi = \arctan\left(\frac{y}{x}\right) \tag{3.67}$$

$$\theta = \arccos\left(\frac{r}{z}\right) \tag{3.68}$$

The values that r, φ, and θ can take on are given by,

$$r \in [0, \infty) \tag{3.69}$$

$$\varphi \in [0, 2\pi] \tag{3.70}$$

$$\theta \in [0, \pi] \tag{3.71}$$

We can convert from spherical coordinates to cartesian with,

$$x = r\cos(\varphi)\sin(\theta) \tag{3.72}$$

$$y = r\sin(\varphi)\sin(\theta) \tag{3.73}$$

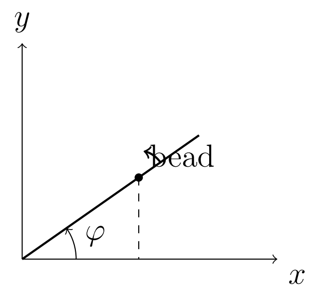
$$z = r\cos(\theta) \tag{3.74}$$

3.3 N2L in curvilinear coordinates

3.3.1 N2L in Polar Coordinates

Consider a rod that rotates only in the x-y plane. A bead, on the rod, can slide up and down the rod without friction. The rod rotates at a constant rate ω. We wanted to find the trajectory of the bead as the rod whirls around.

Since our motion is confined to the x-y plane and the rod rotates in a circle, it is natural to use polar coordinates. Let the position of the bead on the rod be defined by $r(t)$. Unlike the cartesian coordinate system which are constant, the basis vectors in polar



coordinates change with time as seen below,

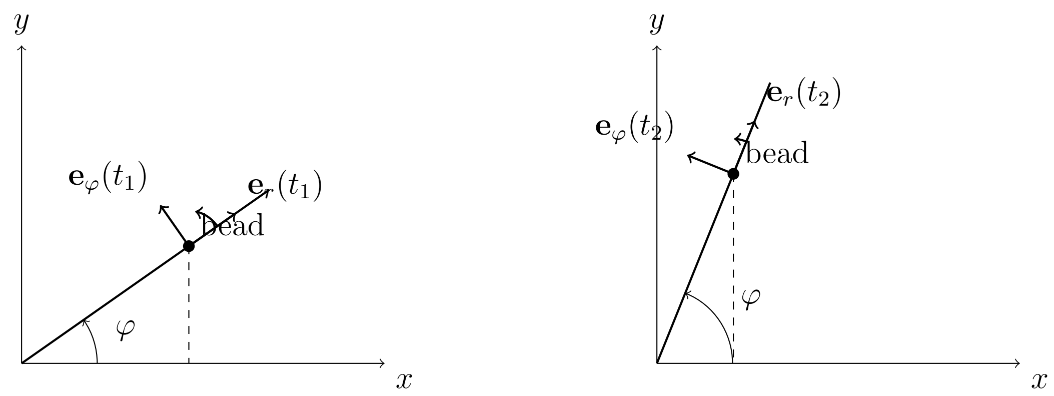


Figure 11: Basis vectors in polar coordinates

Then, the position vector of the bead is given by,

$$\mathbf{r}(t) = r(t)\mathbf{e}_r(t) \tag{3.75}$$

We can differentiate to get velocity. From the product rule,

$$\dot{\mathbf{r}}(t) = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r \tag{3.76}$$

We have,

$$\mathbf{e}_r = \cos(\varphi)\mathbf{e}_x + \sin(\varphi)\mathbf{e}_y \tag{3.77}$$

Then by the chain rule and product rule,

$$\dot{\mathbf{e}}_r = -\dot{\varphi}\sin(\varphi)\mathbf{e}_x + \cos(\varphi)\dot{\mathbf{e}}_x + \dot{\varphi}\cos(\varphi)\mathbf{e}_y + \sin(\varphi)\dot{\mathbf{e}}_y \tag{3.78}$$

However, $\dot{\mathbf{e}_x} = \dot{\mathbf{e}_y} = 0$. Then,

$$\dot{\mathbf{e}_r} = \dot{\varphi}\cos(\varphi)\mathbf{e}_y - \dot{\varphi}\sin(\varphi)\mathbf{e}_x \tag{3.79}$$

It holds that this is exactly $\dot{\mathbf{e}_\varphi}$. Then,

$$\dot{\mathbf{e}_r} = \dot{\varphi}\mathbf{e}_\varphi \tag{3.80}$$

Sinc $\mathbf{e}_\varphi$ must be orthogoal to $\mathbf{e}_r$ we have,

$$\mathbf{e}_\varphi = -\sin(\varphi)\mathbf{e}_x + \cos(\varphi)\mathbf{e}_y \tag{3.81}$$

Then,

$$\dot{\mathbf{e}_\varphi} = -\dot{\varphi}\cos(\varphi)\mathbf{e}_x - \dot{\varphi}\sin(\varphi)\mathbf{e}_y \tag{3.82}$$

Then,

$$\dot{\mathbf{e}_\varphi} = -\dot{\varphi}\mathbf{e}_r \tag{3.83}$$

Since the rod rotates at a constant rate ω, we have $\dot{\varphi} = \omega$. Then,

$$\dot{\mathbf{r}(t)} = \dot{r}\mathbf{e}_r + r\omega\mathbf{e}_\varphi \tag{3.84}$$

Differentiating again to get acceleration,

$$\begin{aligned}
\ddot{\mathbf{r}(t)} &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}_r} + \dot{r}\omega\mathbf{e}_\varphi + r\omega\dot{\mathbf{e}_\varphi} &(3.85)\\
&= \ddot{r}\mathbf{e}_r + \omega\dot{r}\mathbf{e}_\varphi + \omega\dot{r}\mathbf{e}_\varphi - r\omega^2\mathbf{e}_r &(3.86)\\
&= (\ddot{r} - \omega^2 r)\mathbf{e}_r + 2\omega\dot{r}\mathbf{e}_\varphi &(3.87)
\end{aligned}$$

Now if we neglect the gravitational force as our motion is planar along the ground, the net force acting on the bead is the normal force from the rod. This is always perpendicular to the rod. Since the basis vector $\mathbf{e}_\varphi$ is always perpendicular to the rod we have,

$$\mathbf{F}_N = F\mathbf{e}_\varphi \tag{3.88}$$

Then by N2L,

$$F_N\mathbf{e}_\varphi = m(\ddot{r} - \omega^2 r)\mathbf{e}_r + 2\omega\dot{r}\mathbf{e}_\varphi \tag{3.89}$$

Equating components we have,

$$m(\ddot{r} - \omega^2 r) = 0 \tag{3.90}$$

$$2m\omega\dot{r} = F \tag{3.91}$$

For the first ODE, the characteristic polynomial is

$$\lambda^2 - \omega^2 = 0 \tag{3.92}$$

$$\implies (\lambda + \omega)(\lambda - \omega) = 0 \tag{3.93}$$

$$\implies \lambda = \pm\omega \tag{3.94}$$

Thus,

$$r(t) = Ae^{\omega t} + Be^{-\omega t} \tag{3.95}$$

where A and B are determined by initial conditions. Let the initial conditions be $r(0) = r_0$ and $\dot{r}(0) = 0$. Then,

$$r(0) = A + B = r_0 \tag{3.96}$$

$$\dot{r}(0) = \omega(A - B) = 0 \tag{3.97}$$

Solving this systems gives,

$$A = B = \frac{r_0}{2} \tag{3.98}$$

Then,

$$r(t) = r_0\frac{e^{\omega t} - e^{-\omega t}}{2} \tag{3.99}$$

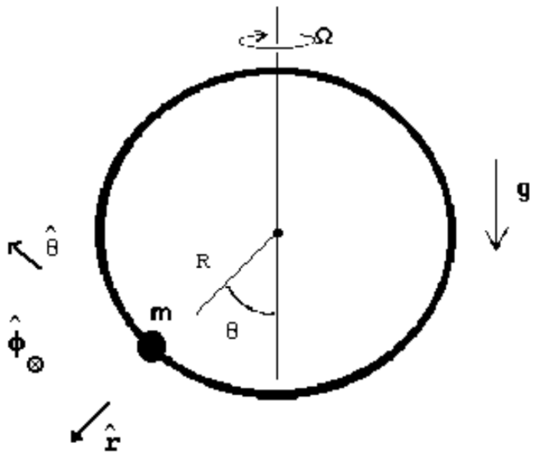


Figure 12: Spinning hoop

This is the definition of cosh. Also recognising that $\dot{\varphi} = \omega \implies \varphi = \omega t$. Thus,

$$r(t) = r_0 \cosh(\varphi(t)) \tag{3.100}$$

Then, the position vector of the bead is

$$\mathbf{r}(t) = r_0 \cosh(\varphi(t))\mathbf{e}_r \tag{3.101}$$

3.3.2 N2L in Spherical Coordinates

Consider a hoop of fixed radius R that spins around its vertical axis at a fixed rate Ω. A bead of mass m is free to move along the hoop with no friction. The setup is in a gravitational field. Since the hoop spins into the plane and the bead can move along the plane of the hoop, it is natural to use spherical coordinates. The position vector is,

$$\mathbf{r}(t) = r(t)\mathbf{e}_r \tag{3.102}$$

Then,

$$\dot{\mathbf{r}}(t) = \dot{r}\mathbf{e}_r + r\sin(\theta)\dot{\varphi}\mathbf{e}_\varphi + r\dot{\theta}\mathbf{e}_\theta \tag{3.103}$$

Once again,

$$\ddot{\mathbf{r}}(t) = (\ddot{r}-r\dot{\theta}^2-r\dot{\varphi}^2\sin\theta)\mathbf{e}_r+(r\ddot{\varphi}\sin\theta+2r\dot{\theta}\dot{\varphi}\cos\theta+2\dot{r}\dot{\varphi}\sin\theta)\mathbf{e}_\varphi+(r\ddot{\theta}+2\dot{r}\dot{\theta}-r\dot{\varphi}^2\sin\theta\cos\theta)\mathbf{e}_\theta \tag{3.104}$$

The only forces acting on the bead are the normal force and its weight. Since the bead can move along the hoop and the hoop is spinning, the normal force has a radial component and component along $\mathbf{e}_\varphi$. Thus,

$$\mathbf{F}_N = N_r\mathbf{e}_r + N_\varphi\mathbf{e}_\varphi \tag{3.105}$$

The weight always acts in the $-\mathbf{e}_z$ direction. Then,

$$\mathbf{F}_g = -mg\mathbf{e}_z = mg(\cos\theta\mathbf{e}_r - \sin\theta\mathbf{e}_\theta) \tag{3.106}$$

The net force is then,

$$\mathbf{F} = (N_r + mg\cos\theta)\mathbf{e}_r + N_\varphi\mathbf{e}_\varphi - mg\sin\theta\mathbf{e}_\theta \tag{3.107}$$

Since the bead can only move along the hoop, it is always a constant distance $r(t) = R =$ constant away from the origin. Thus $\dot{r} = \ddot{r} = 0$. Also, $\dot{\varphi} = \Omega$. Then, by comparing components,

$$N_r + mg\cos\theta = -m(R\dot{\theta}^2 - R\Omega^2\sin\theta) \tag{3.108}$$

$$N_\varphi = 2mR\Omega\dot{\theta}\cos\theta \tag{3.109}$$

$$-mg\sin\theta = m(R\ddot{\theta} - R\Omega^2\sin\theta\cos\theta) \tag{3.110}$$

By the third equation we have,

$$\ddot{\theta} = -\frac{g}{R}\sin\theta + \Omega^2\sin\theta\cos\theta \tag{3.111}$$

This ODE tells us how the bead moves along the hoop. It cannot be solved analytically.

3.4 Conservative Forces

As seen earlier, many problems can become involved. In fact, certain problems cannot be solved analytically and a numerical solution must be sought. However, certain symmetries can simplify our approach. These manifest in the form of conservation laws: conservation of energy, linear momentum, and angular momentum. These can be used to reduce the order of the differential equation we have to solve.

Definition 3.1 (Conservative Force)**.** A force that depends on position $\mathbf{F} = \mathbf{F}(\mathbf{r})$ is conservative if it can be written as

$$\mathbf{F} = -\nabla U(\mathbf{r}) \tag{3.112}$$

where $U(\mathbf{r})$ is potential energy and ∇ is the gradient.

In cartesian coordinates,

$$\nabla = \frac{\partial}{\partial x}\mathbf{e}_x + \frac{\partial}{\partial y}\mathbf{e}_y + \frac{\partial}{\partial z}\mathbf{e}_z \tag{3.113}$$

In cylindrical coordinates,

$$\nabla = \frac{\partial}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial}{\partial \varphi}\mathbf{e}_\varphi + \frac{\partial}{\partial z}\mathbf{e}_z \tag{3.114}$$

In spherical coordinates.

$$\nabla = \frac{\partial}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial}{\partial \varphi}\mathbf{e}_\varphi + \frac{1}{r\sin\varphi}\frac{\partial}{\partial \theta}\mathbf{e}_\theta \tag{3.115}$$

Lemma 3.1 (Curl of a conservative force)**.** If a force $\mathbf{F}$ is conservative, then

$$\nabla\times\mathbf{F} = \vec{0} \tag{3.116}$$

Proof. If a force is conservative,

$$\mathbf{F} = -\nabla U \tag{3.117}$$

Then,

$$\nabla\times\mathbf{F} = -\nabla\times(\nabla U) = \vec{0} \tag{3.118}$$

■

In fact, it is sufficient to check if the curl of a force is zero to determine if it is conservative. Not every force has vanishing curl. A general force can be written as,

$$\mathbf{F} = \mathbf{F}_{\text{conservative}} + \mathbf{F}_{\text{dissipative}} \tag{3.119}$$

where $\mathbf{F}_{\text{dissipative}}$ is a dissipative component of the force like friction or drag. Since this is not conservative, it cannot be written as the gradient of a potential energy. Then, by the linearity of the curl operator,

$$\begin{aligned} \nabla\times F &= \nabla\times F_{\text{conservative}} + \nabla\times F_{\text{dissipative}} && (3.120)\\ &= \vec{0} + \nabla\times F_{\text{dissipative}} && (3.121)\\ &\neq \vec{0} && (3.122) \end{aligned}$$

A consequence of conservative forces is that the work done by $\mathbf{F}$ is path-independent. In other words, the work done by $\mathbf{F}$ only depends on the endpoints.

Lemma 3.2 (Path Independence of Work)**.** If $\mathbf{F}$ is a conservative force that does work along a path between two points A and B, then the change in potential energy only depends on the endpoints,

$$W = -(U(P_2) - U(P_1)) \tag{3.123}$$

Proof. Consider the following diagram,

The work done along some path C is,

$$W = \int_C \mathbf{F}\cdot d\mathbf{r} \tag{3.124}$$

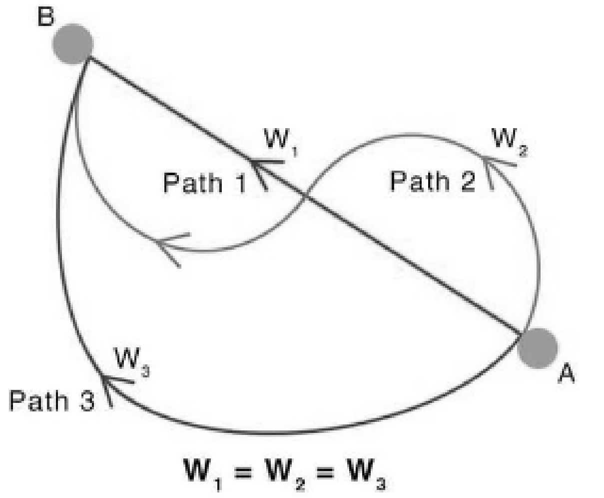


Figure 13: Possible paths from A to B

Since $\mathbf{F}$ is conservative, the integral becomes

$$W = -\int_C \nabla U \cdot d\mathbf{r} \tag{3.125}$$

Then by the fundamental theorem of line integrals,

$$W = -(U(B) - U(A)) \tag{3.126}$$

Thus,

$$W = U(A) - U(B) \tag{3.127}$$

which shows that the work done only depends on the endpoints, and not the path itself.

■

One consequence of the above is that the work done on a closed path is 0. This is because the endpoints are the same. Thus, if $\mathbf{F}$ is conservative,

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \tag{3.128}$$

Conservative forces imply the conservation of mechanical energy.

Theorem 3.3 (Conservation of Mechanical Energy)**.** If a force $\mathbf{F}$ is conservative. Then,

$$E = T + U(\mathbf{r}) = \text{constant} \tag{3.129}$$

Proof. If $\mathbf{F}$ is conservative then $\mathbf{F} = -\nabla U$. Now,

$$E = \frac{1}{2}m\mathbf{v}^2 + U(\mathbf{r}) \tag{3.130}$$

where $\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v}$. Taking the derivative of E,

$$\frac{dE}{dt} = \frac{dT}{dt} + \frac{dU}{dt} \tag{3.131}$$

Consider $\frac{dT}{dt}$,

$$\frac{dT}{dt} = m\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} \tag{3.132}$$

Since $\mathbf{a} = \frac{d\mathbf{v}}{dt}$

$$\frac{dT}{dt} = m\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = \mathbf{v} \cdot \mathbf{F} \tag{3.133}$$

Now,

$$\frac{dU(\mathbf{r})}{dt} = \frac{d\mathbf{r}}{dt} \cdot \nabla U \tag{3.134}$$

$$= \mathbf{v} \cdot \nabla U \tag{3.135}$$

Then,

$$\frac{dE}{dt} = \mathbf{v} \cdot \mathbf{F} + \mathbf{v} \cdot \nabla U \tag{3.136}$$

Since $\mathbf{F}$ is conservative, $\mathbf{F} = -\nabla U$. Then,

$$\frac{dE}{dt} = -\mathbf{v} \cdot \nabla U + \mathbf{v} \cdot \nabla U = 0 \tag{3.137}$$

Thus, there is no change in total energy which implies it is constant, and therefore conserved. ■

The term $\mathbf{v} \cdot \mathbf{F}$ can be used to get the work-energy theorem.

Theorem 3.4 (Work-Energy Theorem)**.** The change in kinetic energy is equal to the work-done.

$$\Delta T = W \tag{3.138}$$

Proof.

$$\frac{dT}{dt} = \mathbf{v} \cdot \mathbf{F} \tag{3.139}$$

$$\implies \Delta T = \int \mathbf{v} \cdot \mathbf{F} dt \tag{3.140}$$

$$= \int F \cdot d\mathbf{r} \tag{3.141}$$

$$\Delta T = W \tag{3.142}$$

■

Example 0

Given $\mathbf{F}(x, y, z) = (2xy + 1)\mathbf{e}_x + (x^2 + 2)\mathbf{e}_y$, find $U(x, y, z)$.

Solution

We first check if $\mathbf{F}$ is conservative.

$$\begin{aligned}\nabla\times\mathbf{F} &= \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy+1 & x^2+2 & 0 \end{vmatrix} \\ &= \vec{0}\end{aligned}$$

Thus $\mathbf{F}$ is conservative and we can use path independence. Since $\mathbf{F} = -\nabla U$, we have

$$U = -\int F \cdot d\mathbf{r} - \int_{(0,0,0}^{(x,y,z)} \mathbf{F} \cdot d\mathbf{r}$$

We can use path-independence and move across three paths. Define,

$$\begin{aligned}\Gamma_1 &= (0, 0, 0) \to (x, 0, 0) \\ \Gamma_2 &= (x, 0, 0) \to (x, y, 0) \\ \Gamma_3 &= (x, y, 0) \to (x, y, z)\end{aligned}$$

Then our integral breaks up into,

$$\begin{aligned}U &= -\int_{\Gamma_1} \mathbf{F} \cdot d\mathbf{r} - \int_{\Gamma_2} \mathbf{F} \cdot d\mathbf{r} - \int_{\Gamma_3} \mathbf{F} \cdot d\mathbf{r} \\ &= -\int_{\Gamma_1} \mathbf{F} \cdot \mathbf{e}_x dx - \int_{\Gamma_2} \mathbf{F} \cdot \mathbf{e}_y dy - \int_{\Gamma_3} \mathbf{F} \cdot \mathbf{e}_z dz \\ &= -\int_0^x F_x(x', 0, 0)dx' - \int_0^y F_y(x, y'0)dy' - \int_0^z F_z(x, y, z')dz' \\ &= -\int_0^x 1dx' - \int_0^y x^2 + 2dy' - \int_0^z 0dz' \\ &= -(x + x^2y + 2y) \\ \implies U(x, y, z) &= -(x + y(x^2 + 2))\end{aligned}$$

3.5 Central Forces

Definition 3.2 (Central Force)**.** A central force is a force that is always directed towasrds or away from a fixed point known as the center. The force only depends on the distance from that center.

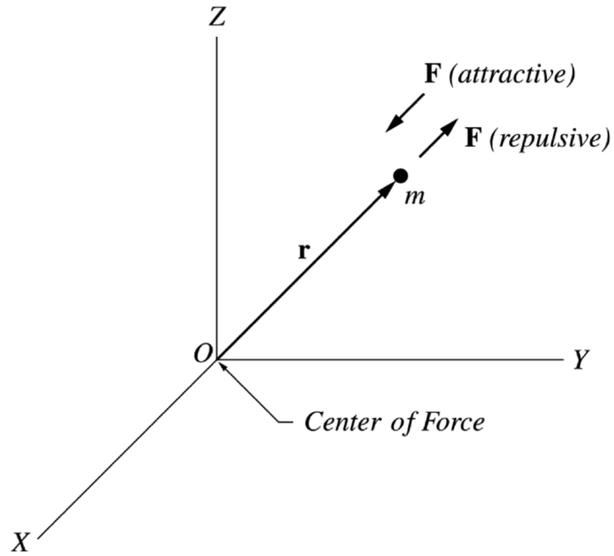
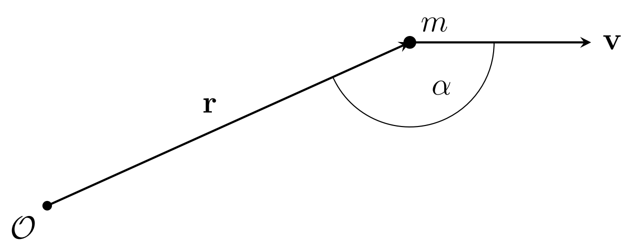


Figure 14: Central Forces

It is often convenient to work in spherical coordinates. Since the force only points towards or away from the center, it only has a radial component in the direction $\mathbf{e}_r$. Thus, a central force in spherical coordinates is expressed as

$$\mathbf{F} = F(r, \varphi, \theta)\mathbf{e}_r \tag{3.143}$$

Consider a particle of mass m a distance r away from a fixed point O moving at a velocity $\mathbf{v}$.



The angular momentum $\mathbf{L}$ of the particle with respect to $\mathcal{O}$ is defined as

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} \tag{3.144}$$

The direction of $\mathbf{L}$ is orthogonal to the $\mathbf{r}$-$\mathbf{v}$ plane and its magnitude is given by

$$|\mathbf{L}| = m|\mathbf{r}||\mathbf{v}| \sin \alpha \tag{3.145}$$

Thus, angular momentum is not only a property of rotating rigid bodies, it is also an inherent property of a particle's trajectory. Central forces imply conservation of angular momentum,

Theorem 3.5 (Conservation of Angular Momentum)**.** If $\mathcal{O}$ is the center of a central force. Then, angular momentum is conserved.

$$\mathbf{L} = \text{constant} \tag{3.146}$$

Proof. To show conservation of angular momentum, we need to show $\frac{d\mathbf{L}}{dt} = 0$.

$$\begin{aligned}
\frac{d\mathbf{L}}{dt} &= \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) && (3.147)\\
&= \frac{d\mathbf{r}}{dt} \times m\mathbf{v} + \mathbf{r} \times m\frac{d\mathbf{v}}{dt} && (3.148)\\
&= \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times \mathbf{F} && (3.149)\\
&= \vec{0} + \mathbf{r} \times \mathbf{F} && (3.150)\\
&= r\mathbf{e}_r \times F(r, \varphi, \theta)\mathbf{e}_r && (3.151)\\
&= \vec{0} && (3.152)
\end{aligned}$$

where the last equality comes from $\mathbf{e}_r \times \mathbf{e}_r = \vec{0}$. ∎

Since $\mathbf{L} =$ constant, both the magnitude and direction of angular momentum is constant. One consequence of the conservation of angular momentum is that we can reduce a 3D problem to a 2D problem.

Since the direction of angular momentum is constant, it is always orthogonal to the **r**-**v** plane. Thus, the particle is always confined to the **r**-**v** plane. We can now rotate the **r**-**v** plane such that it aligns with $\theta = \frac{\pi}{2}$. In other words, we now rotate the **r**-**v** plane to be the x-y plane. Since θ is fixed, $\dot{\theta} = 0$. In spherical coordinates, the **r**-**v** plane becomes the r-φ plane. Since **L** is now perpendicular to the r-φ plane we have,

$$\mathbf{L} = L\mathbf{e}_z \tag{3.153}$$

We can show this by computing **L** in spherical coordinates.

$$\mathbf{r} = r\mathbf{e}_r \tag{3.154}$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\sin\theta\mathbf{e}_\varphi + r\dot{\theta}\mathbf{e}_\theta \tag{3.155}$$

$$\tag{3.156}$$

Since $\theta = \frac{\pi}{2}$, $\sin\theta = 1$ and $\dot{\theta} = 0$. Then,

$$\mathbf{r} = r\mathbf{e}_r \tag{3.157}$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi \tag{3.158}$$

Then,

$$m\mathbf{v} = m\dot{r}\mathbf{e}_r + mr\dot{\varphi}\mathbf{e}_\varphi \tag{3.159}$$

Thus,

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} \tag{3.160}$$

$$= \begin{vmatrix} \mathbf{e}_r & \mathbf{e}_\varphi & \mathbf{e}_\theta \\ r & 0 & 0 \\ m\dot{r} & mr\dot{\varphi} & 0 \end{vmatrix} \tag{3.161}$$

$$= mr^2\dot{\varphi}\mathbf{e}_\theta \tag{3.162}$$

But, $\mathbf{e}_r \times \mathbf{e}_\varphi = -\mathbf{e}_\theta = \mathbf{e}_z$. Thus,

$$\mathbf{L} = mr^2\dot{\varphi}\mathbf{e}_z \tag{3.163}$$

Thus the angular momentum is fixed in the $\mathbf{e}_z$ direction. Also,

$$|L| = L = mr^2\dot{\varphi} \tag{3.164}$$

Then,

$$\dot{\varphi} = \frac{d\varphi}{dt} = \frac{L}{mr^2} \tag{3.165}$$

This shows how we reduced our problem from 3D problem (we had r, φ and θ) to a 2D problem (we only had r and φ)

3.6 Conservative Central Forces

Definition 3.3 (Conservative Central Forces)**.** A conservative central force F is a force that is both conservative and central. Namely, it is a force $\mathbf{F}$ that always points towards or away from a given point (known as the center) that can also be written as the negative gradient of a potential energy function. In spherical coordinates,

$$\mathbf{F} = F(r, \varphi, \theta)\mathbf{e}_r = -\nabla U \tag{3.166}$$

Since we are in spherical coordinates, we have

$$F(r, \varphi, \theta)\mathbf{e}_r = -\left(\frac{\partial U}{\partial r}\mathbf{e}_r + \frac{1}{r\sin\theta}\frac{\partial U}{\partial \varphi}\mathbf{e}_\varphi + \frac{1}{r}\frac{\partial U}{\partial \theta}\mathbf{e}_\theta\right) \tag{3.167}$$

By comparing components we have,

$$\frac{1}{r\sin\theta}\frac{\partial U}{\partial \varphi} = 0 \implies \frac{\partial U}{\partial \varphi} = 0 \tag{3.168}$$

$$\frac{1}{r}\frac{\partial U}{\partial \theta} = 0 \implies \frac{\partial U}{\partial \theta} = 0 \tag{3.169}$$

Thus, U is only a function of r ie $U = U(r)$ and so the force also only depends on r.

Thus, a conservative central force can be expressed as

$$\mathbf{F} = F(r)\mathbf{e}_r \tag{3.170}$$

Then,

$$U(r) = -\int F(r)dr \tag{3.171}$$

Famous conservative central forces are inverse square laws, these are forces of the form

$$\mathbf{F} = -\frac{k}{r^2}\mathbf{e}_r \tag{3.172}$$

where k is a positive constant. The potential energy of an inverse square law is then,

$$U(r) = \int \frac{k}{r^2}dr = -\frac{k}{r} + C \tag{3.173}$$

We typically specify the potential is zero at infinity, leading to $C = 0$. Then,

$$U(r) = -\frac{k}{r} \tag{3.174}$$

Conservative central forces imply both the conservation of energy and angular momentum. We can use these laws to greatly simplify our problem from 3D to 1D.

Proof. In 3D spherical coordinates (r, φ, θ), we have

$$\mathbf{r} = r\mathbf{e}_r \tag{3.175}$$

Then, the particles velocity is

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\sin\theta\mathbf{e}_\varphi + r\dot{\theta}\mathbf{e}_\theta$$

Since we have conservation of angular momentum, the particle is always confined to the **r**-**v** plane. Rotating this such that it aligns with the x-y plane by setting $\theta = \frac{\pi}{2} \implies \dot{\theta} = 0$

we have,

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi \tag{3.177}$$

Thus, we have reduced our 3D problem to a 2D problem with conservation of angular momentum. Now, we apply conservation of energy. Recall that,

$$T = \frac{1}{2}m\mathbf{v}^2 = \frac{1}{2}m(\mathbf{v} \cdot \mathbf{v}) \tag{3.178}$$

Then,

$$\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v} = \dot{r}^2 + r^2\dot{\varphi}^2 \tag{3.179}$$

$$\implies T = \frac{1}{2}m\mathbf{v}^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) \tag{3.180}$$

From eq. (3.165) we have,

$$T = \frac{1}{2}m(\dot{r}^2 + \frac{L^2}{m^2r^2}) \tag{3.181}$$

Expanding,

$$T = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} \tag{3.182}$$

We then apply conservation of energy to get,

$$E = T + U = \text{constant} \tag{3.183}$$

$$\implies E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + U(r) = \text{constant} \tag{3.184}$$

The first term is the kinetic energy in the radial direction. We define the other two terms to be the effective potential U_{eff} of our system.

Definition 3.4 (Effective Potential)**.** For a system of mass m acted on by a conservative central force $\mathbf{F} = F(r)\mathbf{e}_r$, the effective potential is defined as,

$$U_{\text{eff}} = \frac{L^2}{2mr^2} + U(r) \tag{3.185}$$

Then, by conservation of energy we have

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}} = E \tag{3.186}$$

Rearranging gives us

$$\dot{r} = \pm\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\right)} \tag{3.187}$$

This is clearly in 1D. We simply integrate $\dot{r}$ to get r. Then, the trajectory of the particle is

$$\mathbf{r} = r\mathbf{e}_r \tag{3.188}$$

■

3.6.1 Effective Potential for Inverse Square Laws

For an inverse square law we have

$$U(r) = -\frac{k}{r} \tag{3.189}$$

The effective potential is then

$$U_{\text{eff}} = \frac{L^2}{2mr^2} - \frac{k}{r} \tag{3.190}$$

Plotting the effective potential gives We see that it is possible for U_{eff} to be negative.

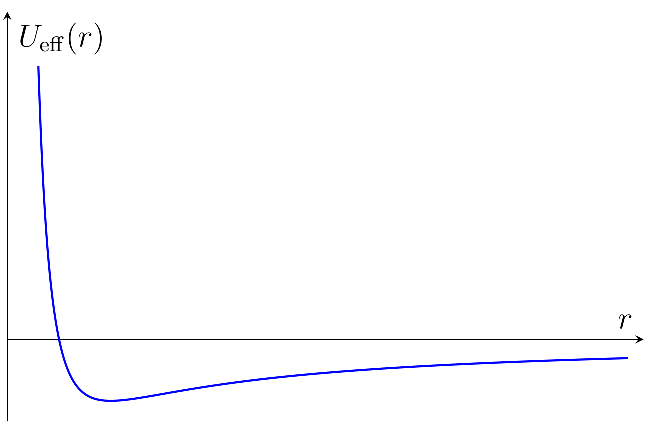


Figure 15: Plot of U_{eff} against r

Thus, if the radial kinetic energy is zero, it is possible for a particle to have negative energy. This comes from the fact that we chose $U(\infty) = 0$. If we chose some other constant of integration, then the total energy would be $E \to E + C$ The sign of the energy determines the type of orbit the system will have

1. $E > 0$. In this case, Ueff > 0. The particle comes in from ∞, experiences some sort of potential barrier, causing the particle to turn around and escape back to ∞. If our conservative central force was gravitation, this motion would correspond to a hyperbolic orbit This is also known as a scattering orbit.

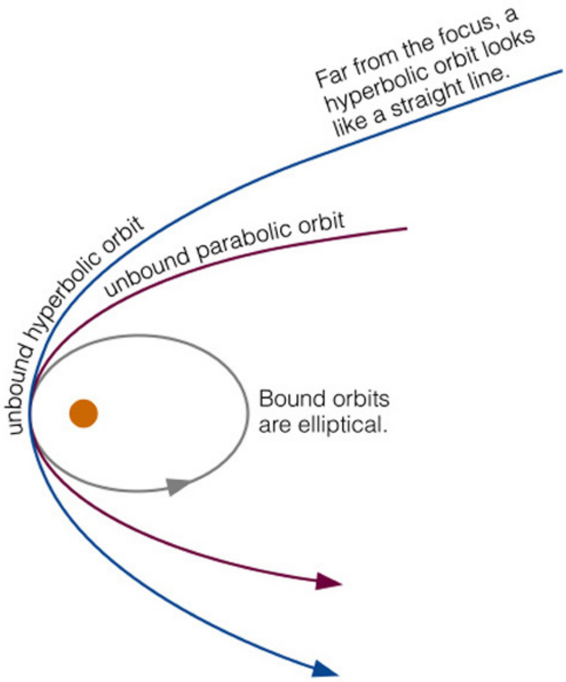


Figure 16: Types of orbits

2. $E = 0$. This corresponds to an unbound parabolic trajectory. This is the limiting case for the object to leave the gravitational field

3. $E < 0$. Here, the particle is bound between two turning points in the bottom half of the graph. It corresponds to a bound elliptical orbit

One special case occurs when $E < 0$ and U_{eff} is at its minimum. This corresponds to a bound stable circular orbit. Let the minimum occur at $r = r_c$. Then,

$$\frac{dU_{\text{eff}}}{dr} = -\frac{L^2}{mr^3} + \frac{k}{r^2} \tag{3.191}$$

At $r = r_c$,

$$\frac{k}{r_c^2} = \frac{L^2}{mr_c^3} \tag{3.192}$$

$$\implies r_c = \frac{L^2}{mk} \tag{3.193}$$

We can confirm this orbit is stable by checking we have a minimum

$$\frac{d^2U_{\text{eff}}}{dr^2} = \frac{3L^2}{mr^4} - \frac{2k}{r^3} \tag{3.194}$$

At $r = r_c$,

$$\frac{d^2U_{\text{eff}}}{dr^2}|_{r=r_c} = \frac{3L^2}{m\frac{L^8}{m^4k^4}} - \frac{2k}{\frac{L^6}{m^3r^3}} \tag{3.195}$$

$$= \frac{3m^3k^4}{L^6} - \frac{2m^3k^4}{L^6} \tag{3.196}$$

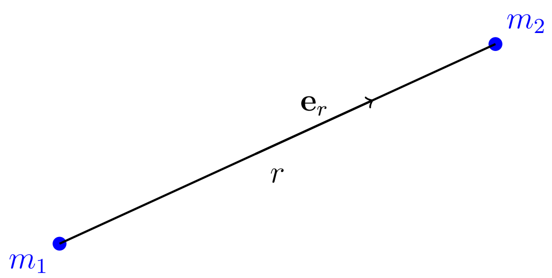
$$= \frac{m^3k^4}{L^6} \tag{3.197}$$

Since $m > 0, k > 0, L > 0$, we have $\frac{d^2U_{\text{eff}}}{dr^2}|_{r=r_c} > 0$. Thus, we have a stable orbit

4 Gravitation and Orbits

4.1 Gravitation

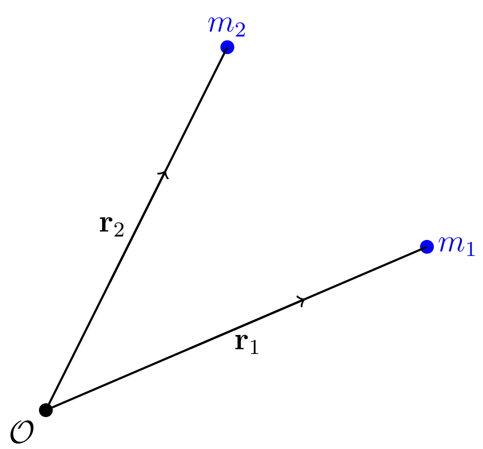
Consider two objects with mass m_1 and m_2 at a separation r from each other. The



gravitational force between these two objects is given by Newton's law of gravitation

$$\mathbf{F} = -G\frac{m_1 m_2}{r^2}\mathbf{e}_r \tag{4.1}$$

where $G = 6.67 \times 10^{-11}$ Nm^{2}kg^{-2}. By N3L, the gravitational force that m_1 exerts on m_2 is equal in magnitude and opposite in direction to the force m_2 exerts on m_1. If we have two masses m_1 and m_2 a separation $\mathbf{r}_1$ and $\mathbf{r}_2$ away from a fixed reference point $\mathcal{O}$. The



separation between the two masses is given by

$$\mathbf{r}_2 - \mathbf{r}_1 \tag{4.2}$$

Then, the unit vector in this direction of this separation is given by,

$$\frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|} \tag{4.3}$$

Thus, the gravitational force between the two masses is

$$\mathbf{F} = -G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^2}\frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|} = -G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1) \tag{4.4}$$

The law of gravitation is also valid for extended masses (masses that are not point masses) At the distance $\mathbf{r}_1$, we have an infintesimal mass dm_1. The total mass is the sum of all

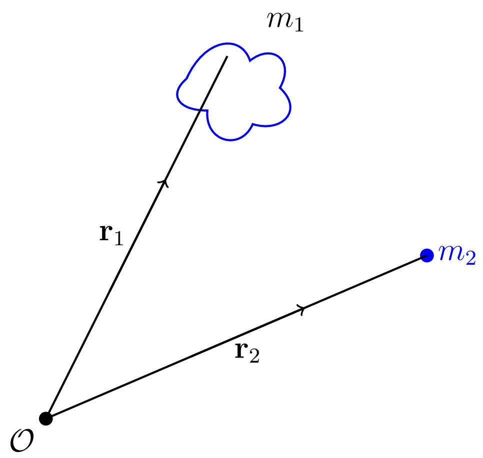


Figure 17: Extended mass system

these infinitesimal masses

$$m_1 = \int dm_1 \tag{4.5}$$

Assume we know the density $\rho(\mathbf{r})$ of m_1. Then,

$$m_1 = \int \rho(\mathbf{r}_1) d^3 r_1 \tag{4.6}$$

where $d^3 r_1$ is the volume element. The gravitational force is then,

$$\mathbf{F} = -Gm_2 \int \frac{\rho(\mathbf{r}_1)}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1) d^3 r_1 \tag{4.7}$$

The gravitational force gives rise to a gravitational field.

Definition 4.1 (Gravitational Field)**.** The gravitational field $\mathbf{g}$ created by m_1 at $\mathbf{r}_2$ is defined as

$$\mathbf{g} = \frac{\mathbf{F}_g}{m_2} = -G \int \frac{\rho(\mathbf{r}_1)}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1) d^3 r_1 \tag{4.8}$$

If m_1 is a point mass separated r away from m_2, the above simplifies to

$$\mathbf{g} = \frac{\mathbf{F}_g}{m_2} = -\frac{Gm_1}{r^2}\mathbf{e}_r \tag{4.9}$$

The gravitational field propagates the gravitational force. It is through the field that m_2 fields a gravitational force from m_1. Similarly, it is through the gravitational field created by m_2 that m_1 fields a gravitational force from m_2. Since $\mathbf{F}_g$ is a conservative central force, we have

$$\nabla \times \mathbf{F} = 0 \tag{4.10}$$

Then, the gravitational force can be written as

$$\mathbf{F} = -\nabla U \tag{4.11}$$

Since $\nabla \times \mathbf{F} = 0$, it holds

$$\nabla \times \mathbf{g} = 0 \tag{4.12}$$

Then we can write $\mathbf{g}$ as

$$\mathbf{g} = -\nabla \Phi \tag{4.13}$$

Φ is known as the gravitational potential.

Definition 4.2 (Gravitational Potential)**.** The gravitational potential Φ is the potential energy per unit mass due to the gravitational field created by m_1

$$\Phi = \frac{U}{m_2} \tag{4.14}$$

Inverting eq. (4.13) and recognising $\mathbf{g}$ only depends on r to get

$$\Phi = -\int \mathbf{g}(\mathbf{r}) \cdot d\mathbf{r} = -\int g(r) dr \tag{4.15}$$

Then

$$\Phi = -\frac{Gm_1}{r} \tag{4.16}$$

Plotting the gravitational potential, For an extended object,

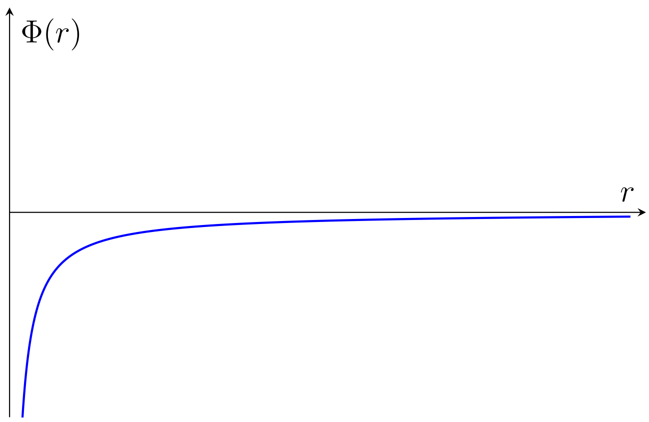


Figure 18: Plot of gravitational potential

$$\Phi = -G \int \frac{\rho(\mathbf{r}_1)}{|\mathbf{r}_2 - \mathbf{r}_1|} d^3 r_1 \tag{4.17}$$

An extended example is as follows:

4.1.1 Example: Gravitational Potential of a Spherical Shell

Consider a uniform hollow spherical shell of mass M, radius R and thickness $h = dr$. A point mass m_2 is a distance $x = |\mathbf{r}|$ away from the origin of the shell. We wish to find an expression for the gravitational potential Φ at the position of m_2 due to M.

We know that

$$dM = \rho dV \tag{4.18}$$

where dV is an infinitesimal volume element. Also,

$$d\Phi = -\frac{G}{|\mathbf{r} - \mathbf{r}_1|} dM \tag{4.19}$$

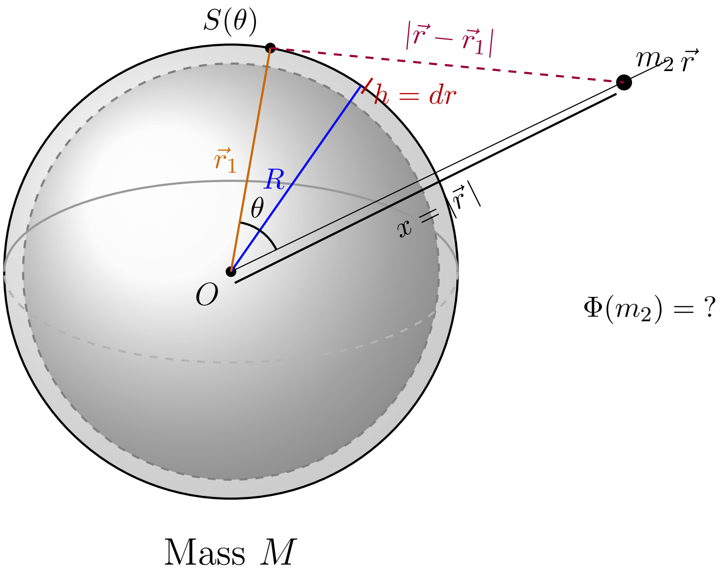


Figure 19: A hollow spherical shell of mass M, radius R, and thickness $h = dr$. A point mass m_2 is at distance $x = |\vec{r}|$ from the center O. The point $S(\theta)$ on the shell is parameterised by the azimuthal angle θ, and $|\vec{r}-\vec{r}_1|$ is the distance from $S(\theta)$ to m_2.

This is the infinitesimal gravitational potential element created by the infinitesimal mass element dM on the surface of the sphere. $|\mathbf{r}-r_1|$ depends where we are on the sphere. We parameterise this with the azimuthal angle θ and define $S(\theta) = |\mathbf{r}-\mathbf{r}_1|$ We can combine the two to give,

$$\Phi = \int d\Phi \tag{4.20}$$

The volume element in spherical coordinates is

$$dV = r^2 \sin\theta dr d\varphi d\theta \tag{4.21}$$

Since the sphere has constant radius and thickness we have

$$r = R \tag{4.22}$$

$$dr = h \tag{4.23}$$

Substituting,

$$dV = R^2 h \sin\theta d\varphi d\theta \tag{4.24}$$

Putting everything together we have

$$dM = \rho R^2 h \sin\theta d\varphi d\theta \tag{4.25}$$

Then,

$$\phi = \int -\frac{G}{S(\theta)} dM \tag{4.26}$$

Substituting and pulling out the constants,

$$-GR^2\rho h \int_0^{\pi} \int_0^{2\pi} \frac{\sin\theta'}{S(\theta')} d\varphi d\theta' \tag{4.27}$$

Evaluating the inner integral gives,

$$-2\pi GR^2\rho h \int_0^{\pi} \frac{\sin\theta'}{S(\theta')} d\theta' \tag{4.28}$$

We have that

$$S(\theta) = |\mathbf{r} - \mathbf{r}_1| \tag{4.29}$$

We can square this to give,

$$S^2(\theta) = |\mathbf{r} - \mathbf{r}_1|^2 = (\mathbf{r} - \mathbf{r}_1) \cdot (\mathbf{r} - \mathbf{r}_1) \tag{4.30}$$

Evaluating the dot product,

$$S^2(\theta) = \mathbf{r}^2 - 2\mathbf{r} \cdot \mathbf{r}_1 + \mathbf{r}_1^2 \tag{4.31}$$

From the diagram we see that $\mathbf{r}_1^2 = R^2$ and $\mathbf{r}^2 = x^2$. Then,

$$S^2(\theta) = x^2 - 2\mathbf{r} \cdot \mathbf{r}_1 + R^2 \tag{4.32}$$

Using the general expression for a dot product,

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta \tag{4.33}$$

We have

$$S^2(\theta) = x^2 - 2Rx\cos\theta + R^2 \tag{4.34}$$

Now, our integral is

$$-2\pi GR^2\rho h \int_0^{\pi} \frac{\sin\theta'}{S(\theta')} d\theta' = -2\pi GR^2\rho h \int_0^{\pi} \frac{\sin\theta'}{\sqrt{x^2 - 2Rx\cos\theta' + R^2}} d\theta' \tag{4.35}$$

Let $u = x^2 - 2Rx\cos\theta' + R^2 \implies \frac{du}{d\theta'} = 2Rx\sin\theta' \implies d\theta' = \frac{du}{2Rx\sin\theta'}$ Our integral then becomes

$$-2\pi GR^2\rho h \int_{x^2-2Rx+R^2}^{x^2+2Rx+R^2} \frac{\sin\theta'}{\sqrt{u}} \frac{du}{2Rx\sin\theta'} = -\frac{G\pi R\rho h}{x} \int_{(x-R)^2}^{(x+R)^2} \frac{1}{\sqrt{u}} du \tag{4.36}$$

Evaluating this gives,

$$-\frac{G\pi R\rho h}{x} \int_{(x-R)^2}^{(x+R)^2} \frac{1}{\sqrt{u}} du = -\frac{2G\pi R\rho h}{x} \left(\sqrt{(x+R)^2} - \sqrt{(x-R)^2}\right) \tag{4.37}$$

Since $R + x$ is positive, the negative root is unphysical. So

$$\sqrt{(x+R)^2} = x + R \tag{4.38}$$

However, $(x - R)^2$ can have a negative root depending on whether $x < R$ (inside the sphere) or $x > R$ (outside the sphere)

- Case I: $x = R$

  In this case

$$\sqrt{(x+R)^2} - \sqrt{(x-R)^2} = 2R \tag{4.39}$$

  Then the integral becomes,

$$-\frac{2G\pi R\rho h}{x}(2R) = -\frac{4G\pi R^2\rho h}{x} \tag{4.40}$$

  Since $x = R$

$$\Phi = -\frac{4G\pi R^2\rho h}{R} \tag{4.41}$$

However, consider the volume of the shell

$$V = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi(R-h)^3 \tag{4.42}$$
$$= \frac{4}{3}\pi(R^3 - (R^3 - 3R^2h + 3Rh^2 - h^3)) \tag{4.43}$$
$$\approx 4\pi R^2 h \ \ (\text{since } h << R) \tag{4.44}$$

Then,

$$M = 4\pi R^2 h\rho \tag{4.45}$$

Thus,

$$\Phi = -\frac{GM}{R} \tag{4.46}$$

- Case II: $x < R$

  When $x < R$,

$$\sqrt{(x+R)^2} - \sqrt{(x-R)^2} = x + R + x - R \tag{4.47}$$
$$= 2x \tag{4.48}$$

Then, the integral becomes

$$-\frac{2G\pi R\rho h}{x}(2x) = -4G\pi R\rho h \tag{4.49}$$

Also, $M = 4\pi R^2 h\rho \implies 4\pi R\rho h = \frac{M}{R}$. Then,

$$\Phi = -\frac{GM}{R} = -4\pi G\rho h \tag{4.50}$$

Thus we see that $\Phi = \text{constant}$ for $0 \leq x \leq R$.

Case III: $x > R$

In this case, $x - R$ is also positive so we neglect the negative root. Thus,

$$\sqrt{(x+R)^2} - \sqrt{(x-R)^2} = (x+R) - (x-R) \tag{4.51}$$

$$= 2R \tag{4.52}$$

Then, the integral becomes

$$-\frac{2G\pi R\rho h}{x}(2R) = -\frac{4G\pi R^2\rho h}{x} \tag{4.53}$$

Substituting $M = 4\pi R^2 h\rho$,

$$\Phi = -\frac{GM}{x} \tag{4.54}$$

Thus, at points outside the shell, the gravitational potential due to shell behaves as if the shell is a point mass.

We can plot the gravitational potential for all three cases:

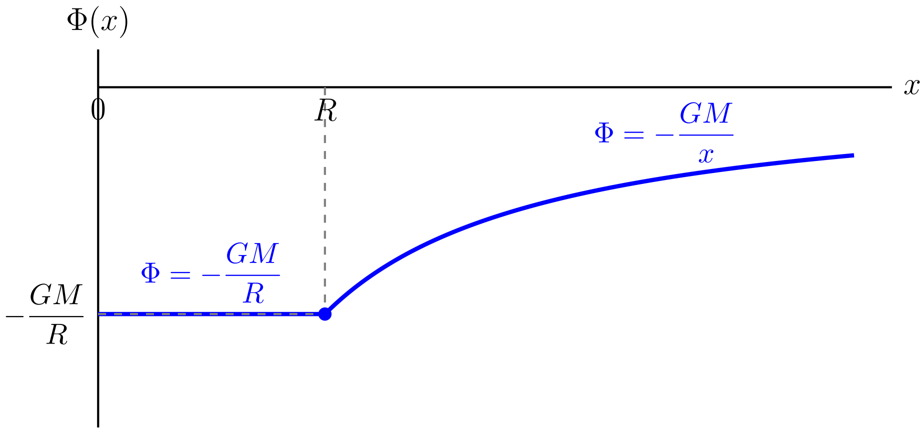


Figure 20: Gravitational potential $\Phi(x)$ due to a uniform spherical shell of mass M and radius R. For $x \leq R$, the potential is constant; for $x > R$, it falls off as $-GM/x$.

4.1.2 Example: Gravitational Potential of a Uniform Whole Sphere

We wish to find the gravitational potential Φ produced by a uniform sphere. Consider a solid sphere of radius R and mass M and a constant density ρ given by

$$\rho(r) = \begin{cases} \rho & \text{if } \ r < R \\ 0 & \text{if } \ r > R \end{cases} \tag{4.55}$$

We can think of our uniform sphere as being composed of infinitely many spherical shells of infinitesimal thickness dr.

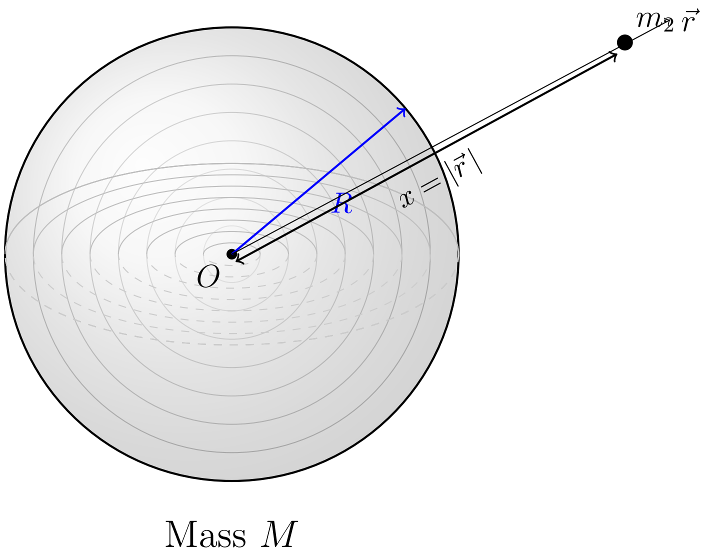


Figure 21: A uniform solid sphere of mass M and radius R, shown as a collection of concentric spherical shells. A point mass m_2 is located at distance $x = |\vec{r}|$ from the center O.

For each shell we have $h \to dr$ and $R \to r$. Then, the infinitestimal gravitational potential due to each infinitesimally small shell is

$$d\Phi = \begin{cases} -\frac{4\pi G r^2 \rho}{x} dr & \text{if } \ x > R \\ -4\pi G \rho r dr & \text{if } \ x < R \end{cases} \tag{4.56}$$

We once again break it up into cases

- Case I: $x > R$

We have

$$\begin{aligned}
\Phi &= \int d\Phi && (4.57)\\
&= -\frac{4\pi G\rho}{x}\int_0^R r^2 dr + \int_R^x 0dr \quad \text{(since } \rho = 0 \text{ for } x > R && (4.58)\\
&= -\frac{4\pi\rho R^3 G}{3x} && (4.59)
\end{aligned}$$

Since $M = \frac{4}{3}\pi R^3\rho$,

$$\Phi = -\frac{GM}{x} \tag{4.60}$$

Again, this looks like the gravitational due to a point mass.

- Case II: $x \leq R$ For $x < R$, we will have some spherical shells that contain x and some spherical shells that are smaller than x. Thus, some shells will behave like a point mass at the point x, while other spherical shells will have x inside it. Thus,

$$\begin{aligned}
\Phi &= \int d\Phi && (4.61)\\
&= -\frac{4\pi G\rho}{x}\int_0^x r^2 dr - 4\pi G\rho\int_x^R r dr && (4.62)\\
&= -\frac{4\pi G\rho}{3}x^2 - 2\pi G\rho(R^2 - x^2) && (4.63)
\end{aligned}$$

$M = \frac{4}{3}\pi R^3\rho \implies \rho = \frac{3M}{4\pi R^3}$ Substituting in for ρ and simplying gives,

$$\Phi = -\frac{GM}{2R}\left(3 - \frac{x^2}{R^2}\right) \tag{4.64}$$

Thus,

$$\Phi = \begin{cases} -\frac{GM}{x} & \text{if } x > R \\ -\frac{GM}{2R}\left(3 - \frac{x^2}{R^2}\right) & \text{if } x < R \end{cases} \tag{4.65}$$

Plotting this gives,

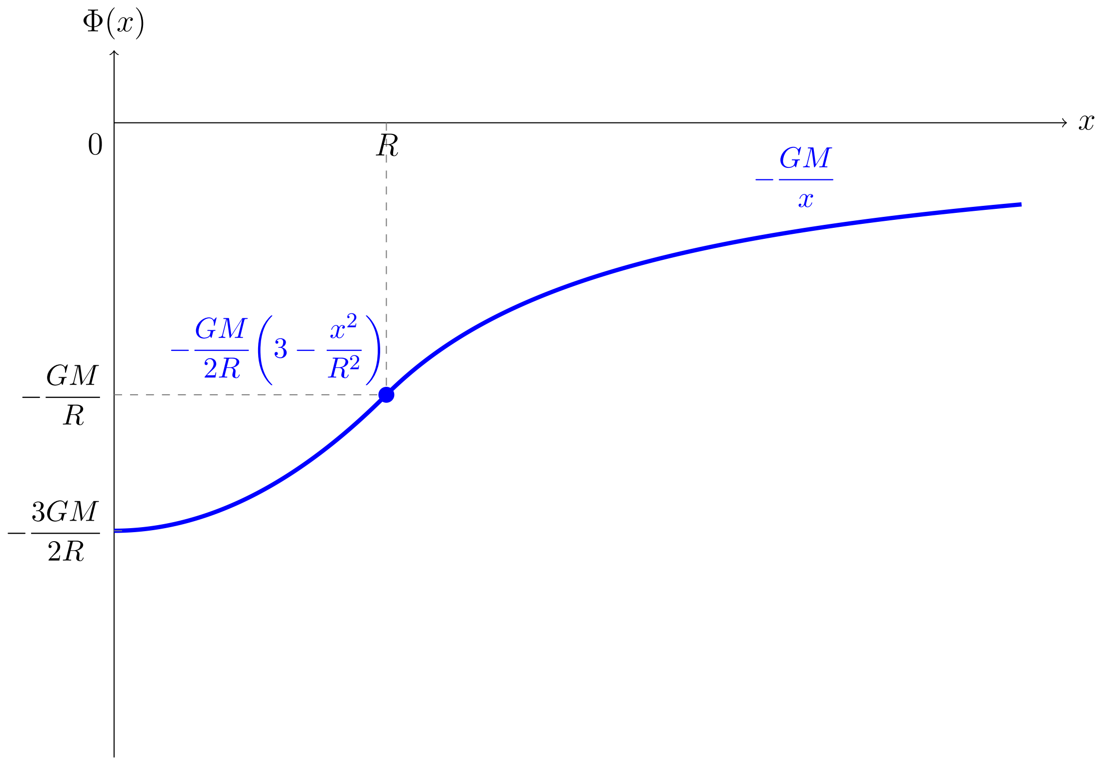


Figure 22: Gravitational potential $\Phi(x)$ of a uniform solid sphere of mass M and radius R. Inside the sphere ($x < R$), the potential is parabolic; outside ($x > R$), it falls off as $-GM/x$.

From both examples, we see that the gravitational potential outside and far away from a mass distribution is the same as that of a point mass

4.2 Poisson Equation for Gravitational Potential

We wish to obtain the gravitational potential for a general mass density $\rho(\mathbf{r})$

4.3 Orbits in Gravitational Fields

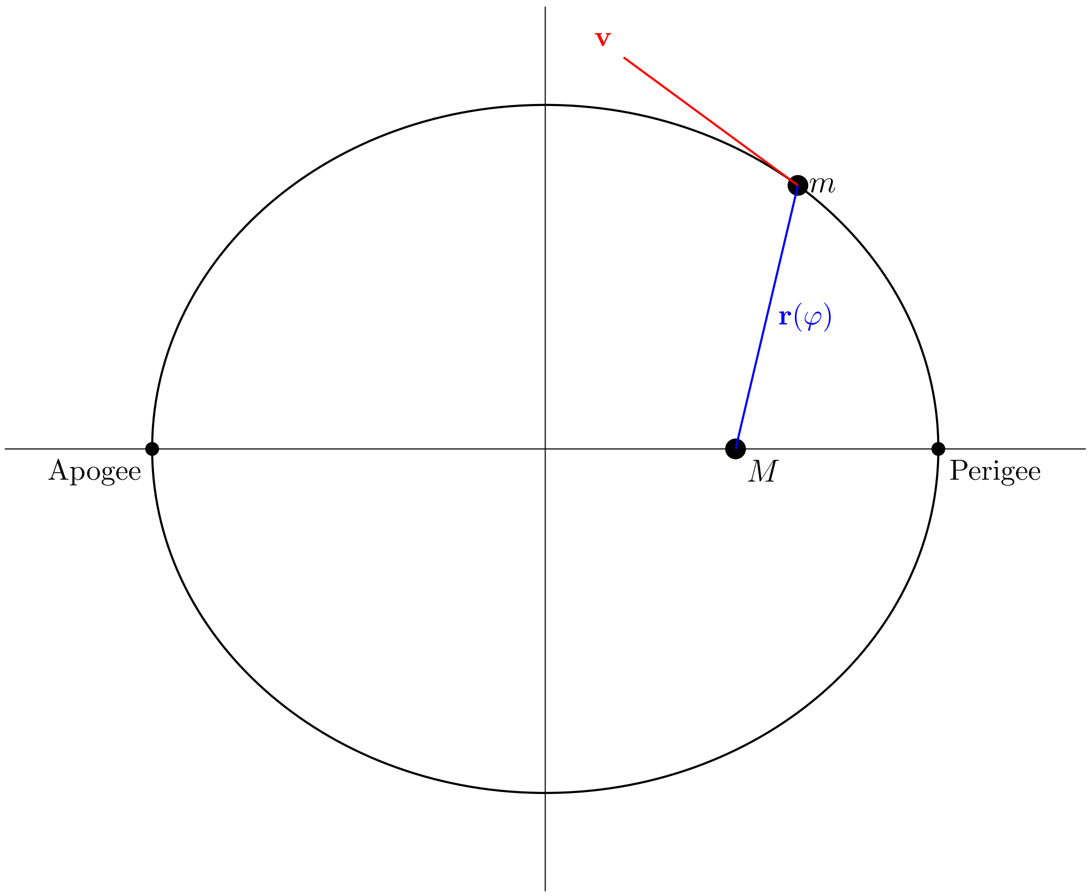
Orbits in Gravitational Fields follow Kepler's Laws

Definition 4.3 (Kepler's Laws)**.** Kepler's Laws describe the orbits of planets around the sun

- Planetary orbits around the sun are planar ellipses with the sun at one of the foci

- The line between the sun and a planet sweeps out equal areas in equal times
- The square of the period of a planet's orbit P is proportional to the cube semi-major axis a

$$P^2 \propto a^3 \tag{4.66}$$



The angle between $\mathbf{r}$ and $\mathbf{v}$ is α. The two vectors are not always orthogonal as we will see. Since the gravitational force is central and conservative, we can use the conservation of angular momentum and energy. However, we find it more convenient to work with the angular momentum per unit mass, and the energy per unit mass of the orbiting particle m. The angular momentum per unit mass is defined as

$$\mathbf{l} = \frac{\mathbf{L}}{m} = \mathbf{r} \times \mathbf{v} \tag{4.67}$$

Since angular momentum is conserved, angular momentum per unit mass must also be conserved since they only have a factor of a scalar. Thus, the magnitude and direction of $\mathbf{l}$ is constant. We can rotate our frame such that the azimuthal angle $\theta = \frac{\pi}{2} \implies \dot{\theta} = 0$.

This means that our motion is confined purely to the x-y plane or r-φ plane. Thus,

$$\mathbf{l} = l\mathbf{e}_z = r^2\dot{\varphi}\mathbf{e}_z = \text{constant} \tag{4.68}$$

We can rearrange this to get

$$\dot{\varphi} = \frac{l}{r^2} \tag{4.69}$$

We can now use conservation of energy to define the energy per unit mass (which is also conserved),

$$\varepsilon = \frac{E}{m} = \frac{T}{m} + \frac{U}{m} \tag{4.70}$$

Since $\frac{U}{m}$ is defined to be the gravitational potential due to M, we have

$$\varepsilon = \frac{T}{m} + \Phi = \frac{1}{2}|\mathbf{v}|^2 + \Phi \tag{4.71}$$

Also, recall that

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi \tag{4.72}$$

Then,

$$|\mathbf{v}|^2 = \mathbf{v}\cdot\mathbf{v} = \dot{r}^2 + r^2\dot{\varphi}^2 \tag{4.73}$$

$$= \dot{r}^2 + \frac{l^2}{r^2} \tag{4.74}$$

Substituting $\Phi = -\frac{GM}{r}$ gives

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.75}$$

The first term is the kinetic energy per unit mass in the radial direction and the last two terms form the effective potential per unit mass. We can rearrange this to get

$$\dot{r} = \sqrt{2\left(\varepsilon - \frac{l^2}{2r^2} + \frac{GM}{r}\right)} \tag{4.76}$$

We define two special points in the orbit

Definition 4.4 (Perigee)**.** The point in the particle's orbit that is closest to M. Labelled appropriately in section 4.3. The distance to the perigee is r_p

Definition 4.5 (Apogee)**.** The point in the particle's orbit that is furthest from M. Labelled appropriately in section 4.3. The distance to the apogee is r_a

It holds that at the apogee and perigee, the position and velocity vectors are orthogonal. This is because the velocity has no radial component at these points. Thus,

$$\mathbf{v}_{a,p} = r\dot{\varphi}\mathbf{e}_\varphi \quad (\mathbf{r}_{a,p} \text{ and } \mathbf{v}_{a,p} \text{ refer to the vectors at the apogee and perigee}) \tag{4.77}$$

Then,

$$\mathbf{r}_{a,p} \cdot \mathbf{v}_{a,p} = 0 \tag{4.78}$$

Also, the angular momentum per unit mass at the apogee and perigee are

$$\mathbf{l} = \mathbf{r}_{a,p} \times \mathbf{v}_{a,p} \tag{4.79}$$

$$\implies l = |\mathbf{r}_{a,p}||\mathbf{v}_{a,p}| \sin\left(\frac{\pi}{2}\right) \tag{4.80}$$

$$= r_{a,p} v_{a,p} \tag{4.81}$$

Since $\mathbf{l}$ is conserved we have

$$r_a v_a = r_p v_p \tag{4.82}$$

Since $r_a > r_p$, we must have $v_a < v_p$. Thus, the speed of the particle at the apogee is slower than at the perigee. Further, the energy per unit mass is

$$\varepsilon = \frac{1}{2}|\mathbf{v}_{a,p}|^2 - \frac{GM}{r_{a,p}} \tag{4.83}$$

$$= \frac{1}{2}v_{a,p}^2 - \frac{GM}{r_{a,p}} \tag{4.84}$$

$$= \frac{1}{2}v_{a,p}^2 - \frac{GM}{l}v_{a,p} \tag{4.85}$$

Thus, the speed at the apogee or perigee can be found via the quadratic formula

$$v_{a,p} = \frac{GM}{l} \pm \sqrt{\left(\frac{GM}{l^2} + 2\varepsilon\right)} \tag{4.86}$$

Since the speed at the apogee is slower at the perigee, the speed at the apogee corresponds to the negative root while the speed at the perigee corresponds to the positive root. The distance to the apogee and perigee is given by

$$r_{a,p} = \frac{l}{v_{a,p}} = \frac{l}{\frac{GM}{l} \pm \sqrt{\left(\frac{GM}{l^2} + 2\varepsilon\right)}} \tag{4.87}$$

4.4 Effective Potential Per Unit Mass and Orbits

The energy per unit mass of the particle is given by

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \Phi_{\text{eff}}(r) \tag{4.88}$$

The effective potential per unit mass is given by

$$\Phi_{\text{eff}}(r) = \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.89}$$

If the second term is sufficiently large, Φ_{eff} can be negative. This can also cause energy per unit mass to be negative. The sign of the energy per unit mass determines the type of orbit of the particle of mass m. Plotting the effective potential we have

- Case I: Circular Orbits

  Circular orbits occur at the minimum of the effective potential. Let this occur at $r = r_c$. Then,

$$\frac{d\Phi_{\text{eff}}}{dr}|_{r=r_c} = 0 \tag{4.90}$$

  It is stable if

$$\frac{d^2\Phi_{\text{eff}}}{dr^2}|_{r=r_c} > 0 \tag{4.91}$$

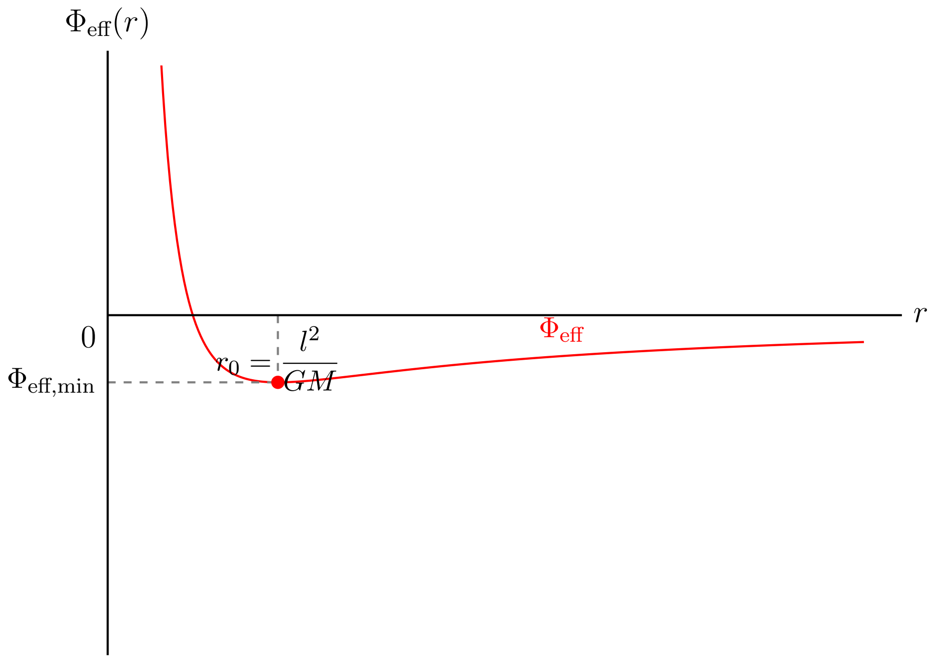


Figure 23: The effective potential $\Phi_{\text{eff}}(r) = \frac{l^2}{2r^2} - \frac{GM}{r}$. The minimum occurs at $r_0 = l^2/GM$.

Computing the first derivative,

$$\frac{d\Phi_{\text{eff}}}{dr} = -\frac{l^2}{r^3} + \frac{GM}{r^2} \tag{4.92}$$

At $r = r_c$ we have

$$\frac{l^2}{r_c^3} = \frac{GM}{r_c^2} \tag{4.93}$$

$$\implies r_c = \frac{l^2}{GM} \tag{4.94}$$

Computing the second derivative

$$\frac{d^2\Phi_{\text{eff}}}{dr^2} = \frac{3l^2}{r^4} - \frac{2GM}{r^3} \tag{4.95}$$

Then,

$$\frac{d^2\Phi_{\text{eff}}}{dr^2}|_{r=r_c} = \frac{3l^2}{r_c^4} - \frac{2GM}{r_c^3} \tag{4.96}$$

$$= \frac{3l^2}{\left(\frac{l^2}{GM}\right)^4} - \frac{2GM}{\left(\frac{l^2}{GM}\right)^3} \tag{4.97}$$

$$= \frac{3G^4M^4}{l^6} - \frac{2G^4M^4}{l^6} \tag{4.98}$$

$$= \frac{G^4M^4}{l^6} > 0 \tag{4.99}$$

We want to find the velocity of the particle in a circular orbit. We have $r_c = \frac{l^2}{GM} =$ constant. Then,

$$\dot{r}|_{r=r_c} = 0 \tag{4.100}$$

Thus at r_c,

$$\mathbf{v} = \dot{r}_c\mathbf{e}_r + r_c\dot{\varphi}\mathbf{e}_\varphi = r_c\dot{\varphi}\mathbf{e}_\varphi \tag{4.101}$$

We have

$$\dot{\varphi} = \frac{l}{r^2} \tag{4.102}$$

We can refer to this as ω. Then, the velocity vector at r_c becomes

$$\mathbf{v} = r_c\omega\mathbf{e}_\varphi \tag{4.103}$$

Then, taking the derivative gives

$$\mathbf{a} = -r_c\dot{\varphi}^2\mathbf{e}_r \tag{4.104}$$

$$= -\frac{v^2}{r_c}\mathbf{e}_r \tag{4.105}$$

We want to express our velocity in terms of the constants of the problem (G, M, r_c). We can use N2L to get

$$-\frac{GMm}{r_c^2}\mathbf{e}_r = -\frac{mv^2}{r_c}\mathbf{e}_r \tag{4.106}$$

This gives,

$$\mathbf{v} = \sqrt{\frac{GM}{r_c}}\mathbf{e}_\varphi \tag{4.107}$$

This is the velocity a particle of mass m has when it is orbiting a particle of mass M at a constant distance r_c away from M. The period of this orbit is given by

$$P = \frac{2\pi}{\omega} = \frac{2\pi r_c}{v} \tag{4.108}$$

Then,

$$P = 2\pi\sqrt{\frac{r_c^3}{GM}} \tag{4.109}$$

At circular orbits, the velocity vector is always orthogonal to the position vector. Then,

$$l = r_c v \tag{4.110}$$

Thus, the velocity and position is

$$v = \frac{GM}{l} \tag{4.111}$$

$$r_c = \frac{l}{GM} \tag{4.112}$$

The energy per unit mass is

$$\varepsilon = -\frac{1}{2}\left(\frac{GM}{l^2}\right)^2 \tag{4.113}$$

If we want to leave the circular orbit, we need out energy to come entirely from Φ_{eff}. Then,

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.114}$$

Since there should be no kinetic energy

$$\varepsilon = \frac{l^2}{2r_c^2} - \frac{GM}{r_c} \tag{4.115}$$

Recall $l = r_c v$. Then,

$$\frac{1}{2}v_{esc} - \frac{GM}{r_c} = 0 \tag{4.116}$$

Thus, the escape velocity is

$$v_{esc} = \sqrt{\frac{2GM}{r_c}} \tag{4.117}$$

- Case II: Elliptical Orbits

  For elliptical orbits, we need $\varepsilon < 0$. This corresponds to any point with $\Phi_{\text{eff}} < 0$ but not at the minimum. Thus,

$$-\frac{1}{2}\left(\frac{GM}{l^2}\right)^2 < \varepsilon < 0 \tag{4.118}$$

  We will have two turning points of motion which occur exactly at the apogee and perigee of the ellipse

- Case III: Parabolic Orbits Paraobolic orbits occur at $\varepsilon = 0$. These orbits are considered marginally unbound. The turning point occurs at the distance of closest approach (see fig. 16). Since $\varepsilon = 0$ and the velocity is completely tangential at the perigee ($l = r_p v_p$) we have

$$\frac{1}{2}v_p^2 - \frac{GM}{r_p} = 0 \tag{4.119}$$

$$\implies v_p = \sqrt{\frac{2GM}{r_p}} = \frac{2GM}{l} \tag{4.120}$$

  Also,

$$r_p = \frac{l^2}{2GM} \tag{4.121}$$

  At the apogee $v_a \to 0$ so $r_a = \frac{l}{v_a} \to \infty$ which matches fig. 16

- Case IV: Hyperbolic Orbits

  Hyperbolic orbits occur at $\varepsilon > 0$ which are unbound orbits. Recall that

$$v_{a,p} = \frac{GM}{l} \pm \sqrt{\left(\frac{GM}{l^2}\right)^2 + 2\varepsilon} \tag{4.122}$$

  Since $\varepsilon > 0$, the square root term is always greater than the other term. Thus, the negative root is unphysical and so $v > 0$ and $r > 0$

4.5 Two Body Problem in Newtonian Gravity

We now consider the situation where instead of having one mass and one point mass, we have two similar sized masses orbiting a common center (such as a binary star system). This leads us to formalise the two-body problem. To determine the equations of motion

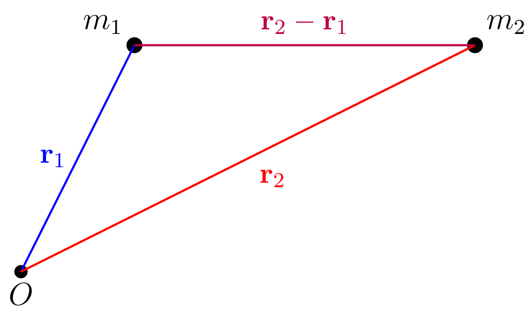


Figure 24: Two masses m_1 and m_2 with position vectors $\vec{r}_1$ and $\vec{r}_2$ from a fixed origin O. The relative position vector is $\vec{r}_2 - \vec{r}_1$.

for m_1 and m_2, we write N2L for each mass

$$m_1\ddot{\mathbf{r}}_1 = -G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^2}\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_2 - \mathbf{r}_1|} \tag{4.123}$$

$$m_2\ddot{\mathbf{r}}_2 = -G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^2}\frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|} \tag{4.124}$$

These are 6 coupled differential equations which are difficult to solve. However, if we consider the center of mass frame, we can reduce our system to just one differential equation. In the center of mass frame, our system behaves as if we have one giant particle of mass M being orbitted by a much smaller particle of mass μ.

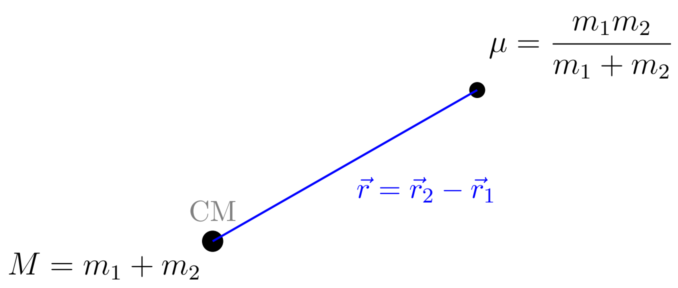


Figure 25: The two-body problem reduced to an equivalent one-body problem in the center of mass frame: a reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ orbiting the total mass $M = m_1 + m_2$ at the center of mass, separated by the relative position vector $\vec{r} = \vec{r}_2 - \vec{r}_1$.

M is known as the total mass and μ is known as the reduced mass, which are both given by

$$M = m_1 + m_2 \tag{4.125}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \tag{4.126}$$

The position of the center of mass is given by

$$\mathbf{C} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{M} \tag{4.127}$$

We now wan tto reduce the description of our problem from $(\mathbf{r}_1, \mathbf{r}_2, m_1, m_2)$ to $(\mathbf{C}, \mathbf{r}, M, \mu)$ We want to verify that the center of mass is stationary. Computing the second derivative we have

$$\begin{aligned}
\ddot{\mathbf{C}} &= \frac{1}{M}(m_1 \ddot{\mathbf{r}_1} + m_2 \ddot{\mathbf{r}_2}) && (4.128)\\
&= \frac{1}{M}\left(-G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_1 - \mathbf{r}_2) - G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1)\right) && (4.129)\\
&= \frac{1}{M}\left(-G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_1 - \mathbf{r}_2)_G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_1 - \mathbf{r}_2)\right) && (4.130)\\
&= 0 && (4.131)
\end{aligned}$$

Thus, the center of mass does not accelerate, implying there is no external force acting on the center of mass. We now want to find an equation of motion for $\mathbf{r}$ which is the trajectory of the reduced mass μ in the gravitational field of M. We compute $m_2\ddot{\mathbf{r}_2} - m_1\ddot{\mathbf{r}_1}$ to get,

$$\ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^2}\frac{\mathbf{r}}{|\mathbf{r}|} \tag{4.132}$$

This looks like the equation of motion of a test particle of mass μ in the gravitational field of a mass M. In the two body system, the energy is given by

$$E = T + U = \frac{1}{2}m_1|\dot{\mathbf{r}_1}|^2 + \frac{1}{2}m_2|\dot{\mathbf{r}_2}|^2 - G\frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|} \tag{4.133}$$

In the center of mass frame, this becomes

$$E = \frac{1}{2}\mu^2 - G\frac{M\mu}{|\mathbf{r}|} \tag{4.134}$$

4.5.1 Equations of Motion in Center of Mass frame

We can use angular momentum conservation and enegry conservation (since the gravitational force is central and conservative) to get,

$$\mathbf{l} = \frac{\mathbf{L}}{\mu} = \mathbf{r} \times \mathbf{v} = \text{constant} \tag{4.135}$$

Since this is constant, its direction and magnitude remains unchanged. Thus, we can rotate our frame such that motion of the test mass μ is completely within the r-φ plane. Thus, $\theta = \frac{\pi}{2} \implies \dot{\theta} = 0$. Then,

$$\mathbf{l} = l\mathbf{e}_z = r^2\dot{\varphi}\mathbf{e}_z \tag{4.136}$$

Thus,

$$\dot{\varphi} = \frac{l}{r^2} \tag{4.137}$$

Now we apply energy conservation with energy per unit mass,

$$\varepsilon = \frac{E}{\mu} \tag{4.138}$$

$$= \frac{T}{\mu} + \frac{U}{\mu} \tag{4.139}$$

$$= \frac{T}{\mu} + \Phi \tag{4.140}$$

Our velocity vector in the r-φ plane is

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi \tag{4.141}$$

Then,

$$
\begin{aligned}
|\mathbf{v}|^2 &= \mathbf{v} \cdot \mathbf{v} && (4.142)\\
&= \dot{r}^2 + r^2\dot{\varphi}^2 && (4.143)\\
&= \dot{r}^2 + \frac{l^2}{2r^2} && (4.144)
\end{aligned}
$$

Thus, our energy per unit mass is

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r} = \text{constant} \tag{4.145}$$

We can then rearrange to get,

$$\dot{r} = \frac{dr}{dt} = \sqrt{2\left(\varepsilon - \frac{l^2}{2r^2} + \frac{GM}{r}\right)} \tag{4.146}$$

4.5.2 Keplerian Orbits

We now want to parameterise $\mathbf{r}$ the orbit of μ using the polar angle φ. We start by defining recalling that the energy per unit mass is conserved. Thus,

$$\varepsilon = \text{constant} \implies \dot{\varepsilon} = 0 \tag{4.147}$$

We can use this constraint to have

$$
\begin{aligned}
\dot{\varepsilon} &= \frac{d}{dt}\left(\frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r}\right) && (4.148)\\
&= \dot{r}\ddot{r} - \frac{l^2}{r^3}\dot{r} + \frac{GM}{r^2}\dot{r} && (4.149)\\
&= \dot{r}\left(\ddot{r} - \frac{l^2}{r^3} + \frac{GM}{r^2}\right) = 0 && (4.150)
\end{aligned}
$$

Since $\dot{r} \neq 0$ ($\dot{r} = 0$ only at the perigee or apogee. However, the equation must hold true everywhere by the continuity of the orbit),

$$\ddot{r} + \frac{l^2}{r^3} + \frac{GM}{r^2} \tag{4.151}$$

This is a non-linear ODE. However, we can simplify it by defining $u(\varphi)$ as

$$u(\varphi) = \frac{1}{r(\varphi)} \tag{4.152}$$

We want to relate $\frac{du}{d\varphi}$ and $\frac{d^2u}{d\varphi^2}$ to $\dot{r}$ and $\ddot{r}$. Compute,

$$\begin{aligned} \dot{r} &= \frac{d}{dt}\left(\frac{1}{u(\varphi)}\right) \\ &= -\frac{1}{u^2}\frac{du}{d\varphi}\frac{d\varphi}{dt} \end{aligned}$$

Since $\frac{d\varphi}{dt} = \dot{\varphi} = \frac{l}{r^2}$, we have

$$\dot{\varphi} = lu^2 \tag{4.153}$$

Then,

$$\dot{r} = -\frac{1}{u^2}\frac{du}{d\varphi}lu^2 \tag{4.154}$$

$$= -l\frac{du}{d\varphi} \tag{4.155}$$

Then,

$$\ddot{r} = \frac{d}{dt}\left(-l\frac{du}{d\varphi}\right) \tag{4.156}$$

$$= -l\frac{d^2u}{d\varphi^2}\dot{\varphi} \tag{4.157}$$

$$= -l^2u^2\frac{d^2u}{d\varphi^2} \tag{4.158}$$

Inserting into eq. (4.151),

$$-l^2u^2\frac{d^2u}{d\varphi^2} - l^2u^3 + GMu^2 = 0 \tag{4.159}$$

$$\implies u^2\left(-l^2\frac{d^2u}{d\varphi^2} - l^2u + GM\right) = 0 \tag{4.160}$$

$$\implies \frac{d^2u}{d\varphi^2} + u - \frac{GM}{l^2} = 0 \tag{4.161}$$

The resulting second order ODE is called Binet's equation. We can solve this ODE to get,

$$u(\varphi) = D\cos(\varphi - \varphi_0) + \frac{GM}{l^2} \tag{4.162}$$

where D and φ_0 are integration constants fixed by initial conditions. We can choose the initial position of μ to be at the perigee to get $\varphi_0 = 0$. D is obtained from energy conservation. We have,

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.163}$$

At the perigee, $r = r_p$ and $\dot{r} = 0$. Then, we eventually get

$$D = \frac{GM}{l^2}\sqrt{1 + \frac{2\varepsilon}{G^2M^2}} \tag{4.164}$$

We then have,

$$u(\varphi) = \frac{GM}{l^2}\left(\sqrt{1 + \frac{2\varepsilon}{G^2M^2}}\cos(\varphi) + 1\right) \tag{4.165}$$

We can invert this to get $r(\varphi)$. However, this quickly turns messy. Thus we define two new terms: the latus rectum α and the eccentricity ϵ as

$$\alpha = \frac{l^2}{GM} \tag{4.166}$$

$$\epsilon = \sqrt{1 + \frac{2\varepsilon l^2}{G^2M^2}} \tag{4.167}$$

Then we get,

$$r(\varphi) = \frac{\alpha}{1 + \epsilon\cos(\varphi)} \tag{4.168}$$

The position vector of μ is then,

$$\mathbf{r}(\varphi) = \frac{\alpha}{1 + \epsilon\cos(\varphi)}\mathbf{e}_r \tag{4.169}$$

The equation for r is the equation for conic sections with origin in a focal point that includes ellipses, parabolas, and hyperbolas The perigee occurs at $\varphi = 0$ (from our initial

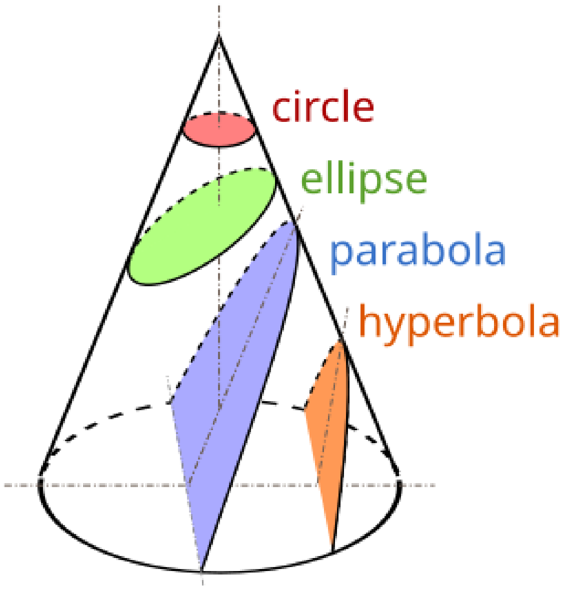


Figure 26: Conic Sections

condition). Then,

$$r_p = r(\varphi = 0) = \frac{\alpha}{1 + \epsilon} \tag{4.170}$$

The apogee occurs at $\varphi = \pi$ (given that our perigee is at $\varphi = 0$). Then,

$$r_a = r(\varphi = \pi) = \frac{\alpha}{1 - \epsilon} \tag{4.171}$$

The eccentricity characterises the type of orbit we have

- Case I: Hyperbola ($\epsilon > 1$)

  Since the hyperbola only occurs. Then, $\varepsilon > 0$ and

$$\epsilon = \sqrt{1 + \frac{2\varepsilon l^2}{G^2 M^2}} > 1 \tag{4.172}$$

γ is known as the scattering angle. φ_{crit} is the angle for r to go to infinity. This

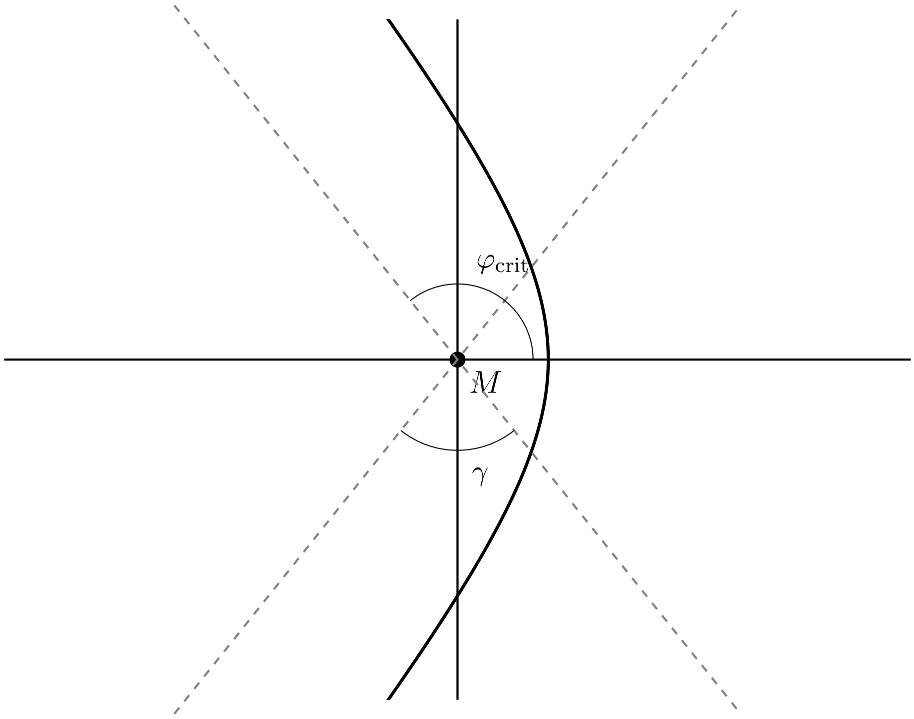


Figure 27: A hyperbolic orbit (right branch) with mass M at the focus. The angle φ_{crit} is measured from the positive x-axis to the asymptote in the second quadrant. The scattering angle γ is between the asymptotes in the third and fourth quadrants.

occurs when

$$1 + \epsilon\cos(\varphi_{\text{crit}}) = 0 \tag{4.173}$$

$$\implies \cos(\varphi_{\text{crit}}) = -\frac{1}{\epsilon} \tag{4.174}$$

- Case II: Parabolic Orbits ($\epsilon = 1$) A parabolic orbit occurs at $\varepsilon = 0$. Then,

$$\epsilon = \sqrt{1 + \frac{2\varepsilon l^2}{G^2M^2}} = 1 \tag{4.175}$$

The trajectory is then given by

$$r(\varphi) = \frac{1}{1+\cos(\varphi)} \tag{4.176}$$

The distance of closest approach is the perigee of the orbit. This occurs at $\varphi = 0$.

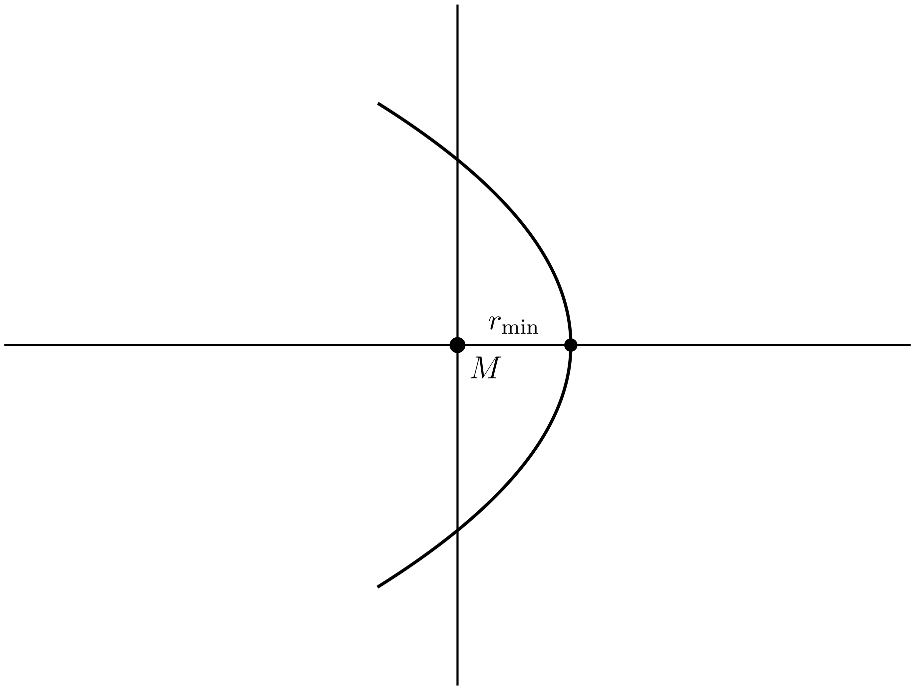


Figure 28: A shifted inverted (left-opening) parabolic orbit with M placed to the left of the closest-approach point; the distance of closest approach is r_{min}.

Then,

$$r(\varphi = 0) = \frac{\alpha}{2} \tag{4.177}$$

The furthest distance occurs at $\varphi = \frac{\pi}{2} \implies r \to \infty$ which matches that of a parabola

- Case III: Circular Orbits ($\epsilon = 0$)

  Circular orbits occur at the minimum of effective potential. We have

$$\varepsilon = \frac{1}{2}\dot{r}^2 + \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.178}$$

  Then,

$$\Phi_{\text{eff}} = \frac{l^2}{2r^2} - \frac{GM}{r} \tag{4.179}$$

  Take the derivative to get,

$$\frac{d\Phi_{\text{eff}}}{dr} = -\frac{l^2}{r^3} + \frac{GM}{r^2} \tag{4.180}$$

Let the minimum occur at $r = r_c$. Then,

$$r_c = \frac{l^2}{GM} \tag{4.181}$$

This is precisely the latus rectum. Thus,

$$r_c = \alpha \tag{4.182}$$

Further, the energy in a circular orbit is given by,

$$\varepsilon_c = \frac{1}{2}\dot{r_c}^2 + \frac{l^2}{2r_c^2} - \frac{GM}{r_c} \tag{4.183}$$

$$\implies \varepsilon_c = -\frac{1}{2}\left(\frac{GM}{l}\right)^2 \tag{4.184}$$

Substituting this into

$$\epsilon = \sqrt{1 + \frac{2\varepsilon_c l^2}{G^2 M^2}} = 0 \tag{4.185}$$

At a circular orbit, the trajectory is always a constant value away from the origin (as the radius of a circle is constant)

$$r = \alpha \tag{4.186}$$

- Case IV: Elliptic Orbits ($0 < \epsilon < 1$)

  With elliptic orbits we have energy greater than that of a circular orbit but still negative. Thus,

$$\varepsilon_c < \varepsilon < 0 \tag{4.187}$$

$$\implies -\frac{1}{2}\left(\frac{GM}{l}\right)^2 < \varepsilon < 0 \tag{4.188}$$

  Substituting this into eccentricity gives,

$$0 < \epsilon < 1 \tag{4.189}$$

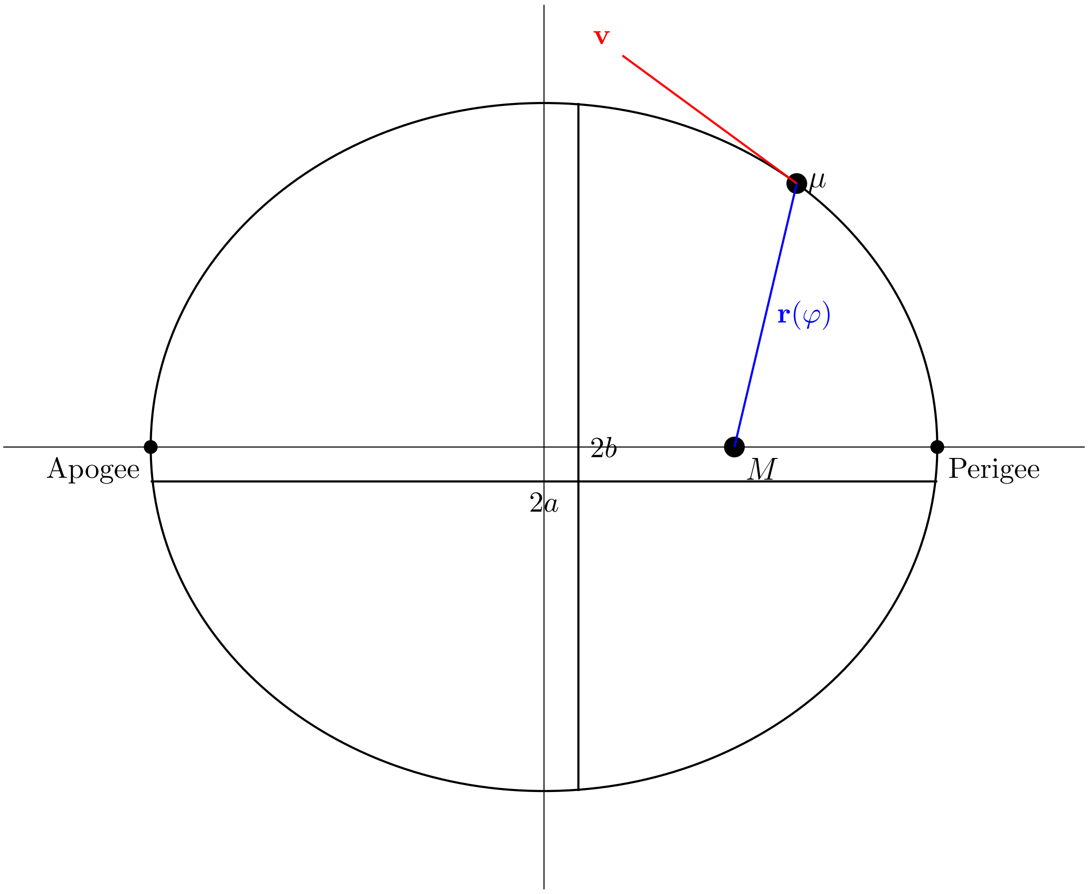


Figure 29: Elliptical Orbit

The trajectory of μ is then

$$r(\varphi) = \frac{\alpha}{1 + \epsilon \cos(\varphi)} \tag{4.190}$$

The elliptical orbit will have M at one of the foci of the ellipse We have an elliptical orbit with the perigee and apogee at

$$r_p = \frac{\alpha}{1 + \epsilon} \tag{4.191}$$

$$r_a = \frac{\alpha}{1 - \epsilon} \tag{4.192}$$

If M was the sun, the perigee and the apogee would be called the perihelion and aphelion respectively. The semimajor axis is the distance from the apogee to the origin. The semiminor axis is the distance from the origin to the vertical end of the ellipse. The semimajor axis is given by,

$$
\begin{aligned}
a &= \frac{r_p + r_a}{2} && (4.193)\\
&= \frac{\alpha}{1+\epsilon} + \frac{\alpha}{1-\epsilon} && (4.194)\\
&= \frac{GM}{2|\varepsilon|} && (4.195)
\end{aligned}
$$

The semiminor axis is given by

$$
\begin{aligned}
b &= a\sqrt{1-\epsilon^2} && (4.196)\\
&= \sqrt{\alpha a} && (4.197)\\
&= \frac{l}{\sqrt{2|\varepsilon|}} && (4.198)
\end{aligned}
$$

We can now connect everything we have derived with Kepler's Laws

- Law I: Orbits of planets around the sun are planar ellipses witht he sun at a focal point of the ellipse

  We have precisely derived this with the parameterisation of r as

  $$r = \frac{\alpha}{1+\epsilon\cos\varphi} \tag{4.199}$$

  Further, this ellipse is planar (confined to the same plane), as we saw that the motion of the μ is confined to the r-φ plane due to conservation of angular momentum

- Law II: The line between the sun and a planet sweeps out equal areas in equal times

  The infinitesimal area swept out is given by the area of a sector with angle $d\varphi$

  $$dA = \frac{1}{2}r^2 d\varphi \tag{4.200}$$

Then,

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\varphi} \tag{4.201}$$
$$= \frac{1}{2}r^2\frac{l}{r^2} \tag{4.202}$$
$$= \frac{l}{2} \tag{4.203}$$

This is a constant. Thus, for equal times, the area swept out does not depend on r

- Law III: The square of the period of a planet's orbit is proportional to the cube of the semimajor axis of the ellipse

  We have
  $$\frac{dA}{dt} = \frac{l}{2} \tag{4.204}$$

  Then, the area swept out in one period is

  $$A = \int_0^P \frac{l}{2}dt = \frac{lP}{2} \tag{4.205}$$

  However, from geometry, the area of an ellipse with semimajor axis a and semiminor axis b is
  $$A = \pi ab \tag{4.206}$$

  Substituting eq. (4.195) and eq. (4.198) gives,

  $$A = \pi\sqrt{\alpha}a^{\frac{3}{2}} \tag{4.207}$$

  Then,

  $$\frac{lP}{2} = \pi\sqrt{\alpha}a^{\frac{3}{2}} \tag{4.208}$$
  $$\implies P^2 = \frac{4\pi^2}{GM}a^3 \tag{4.209}$$

  Then,
  $$\frac{P^2}{a^3} = \frac{4\pi^2}{GM} = \text{constant} \tag{4.210}$$

5 Oscillations and Oscillators

Oscillations are present everywhere from pendulums to vibrations and sound. Additionally, we have seen that we can describe the minimum of a potential with a simple harmonic oscillator. Thus, it is worth studying oscillators and oscillations in-depth.

5.1 Simple Harmonic Oscillator

Definition 5.1 (Simple Harmonic Oscillator)**.** A simple harmonic oscillator is a system that obeys simple harmonic motion.

5.1.1 Equations of Motion

We start by studying the most famous simple harmonic oscillator: the block on a spring. Consider a block of mass m attached to a spring of spring constant k and equilibrium length L. The position of the block is described by the function $x(t)$, where $x = 0$

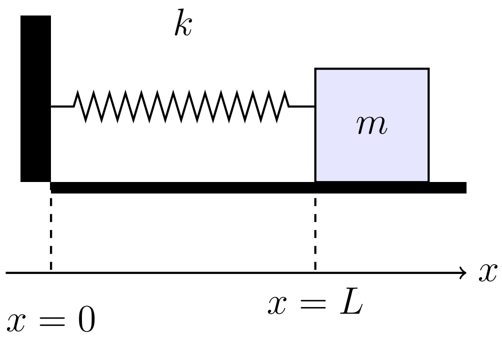


Figure 30: Mass on Spring

represents the wall and $x = L$ represents the equilibrium length of the spring. We assume the mass slides along the floor with no friction or drag. We can derive the equations of motion through N2L or energy conservation. First, we consider N2L. The spring force exerts a on the block given by

$$\mathbf{F} = -k(x(t) - L)\mathbf{e}_x \tag{5.1}$$

where $[k]$ =Nm^{-1} = kgs^{-2}. The minus sign indicates it is a restoring force. Namely, pulling the block (extending the spring to the right) will cause the force to point to the left towards equilibrium while pushing the block (compressing the spring to the right) will cause the force to point to the right towards equilibrium. A restoring force always points towards some "equilibrium" point. In this case, the equilibrium point is determined by the point where the length of the spring is exactly L. Applying N2L,

$$-k(x(t) - L)\mathbf{e}_x = m\ddot{x}\mathbf{e}_x \tag{5.2}$$

$$\implies m\ddot{x} + k(x(t) - L) = 0 \tag{5.3}$$

This is the ODE for the simple harmonic oscillator. Since the spring force is conservative, We can derive the equation of motion using energy conservation. The total energy is given by

$$E = T + U \tag{5.4}$$

Via conservation of energy,

$$\dot{E} = 0 \tag{5.5}$$

The kinetic energy is,

$$T = \frac{1}{2}m\mathbf{v}^2 = \frac{1}{2}m\dot{x}^2 \tag{5.6}$$

The potential energy is given by

$$\begin{aligned}
U &= \frac{1}{2}k(\Delta x)^2 && (5.7)\\
&= \frac{1}{2}k(x(t) - L)^2 && (5.8)\\
&= \frac{1}{2}k(x^2 - 2xL + L^2) && (5.9)\\
&= \frac{1}{2}kx^2 - xL + \frac{1}{2}L^2 && (5.10)
\end{aligned}$$

Then,

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 - xL + \frac{1}{2}L^2 \tag{5.11}$$

$$\implies \dot{E} = m\dot{x}\ddot{x} + kx\dot{x} - L\dot{x} = 0 \tag{5.12}$$

Although $\dot{x} = 0$ at the turning points of motion, by the continuity of the motion of the block, we can factor $\dot{x}$ and divide to get

$$m\ddot{x} + k(x - L) = 0 \tag{5.13}$$

To solve this ODE, we can translate our reference frame to the right by L such that the origin of our new reference frame is at the equilibrium length of the spring. Recall that the translation of an inertial reference frame is still a reference frame. The new reference frame is defined by,

$$y = x - L \quad (y = 0 \text{ at } x = L) \tag{5.14}$$

$$\implies \dot{y} = \dot{x} \tag{5.15}$$

$$\implies \ddot{y} = \ddot{x} \tag{5.16}$$

In our new reference frame, the equation of motion is the more familiar,

$$m\ddot{y} + ky = 0 \tag{5.17}$$

We can divide by m and define the term $\omega^2 = \frac{k}{m}$ to get,

$$\ddot{y} + \omega^2 y = 0 \tag{5.18}$$

The characteristic polynomial of this ODE is

$$r^2 + \omega^2 = 0 \tag{5.19}$$

$$\implies r = \pm i\omega \tag{5.20}$$

Then, the solution is given by

$$y(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t) \tag{5.21}$$

where $c_1, c_2 \in \mathbb{C}$ are fixed by initial conditions. We can also express this as a cosine function with a phase

$$y(t) = C \cos(\omega t - \phi) \tag{5.22}$$

where $, \phi \in \mathbb{R}$. C is the amplitude of the oscillation and ϕ is the phase. Regardless of how we express the solution, the period of a single oscillation is given by

$$P = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \tag{5.23}$$

5.1.2 Energy of the Simple Harmonic Oscillator

For simplicity, we will use the form $y(t) = C\cos(\omega t - \phi)$. The energy is given by

$$E = T + U \tag{5.24}$$

$$= \frac{1}{2}m\dot{y}^2 + \frac{1}{2}ky^2 \tag{5.25}$$

$$= \frac{1}{2}mC^2\omega^2 \sin^2(\omega t - \phi) + \frac{1}{2}kC^2\cos^2(\omega t - \phi) \tag{5.26}$$

Thus,

$$T = \frac{1}{2}mC^2\omega^2 \sin^2(\omega t - \phi) \tag{5.27}$$

$$U = \frac{1}{2}kC^2\cos^2(\omega t - \phi) \tag{5.28}$$

We can see that $T \geq 0$ and $U \geq 0$ The potential energy is maximised when,

$$\cos^2(\omega t - \phi) = 1 \tag{5.29}$$

$$\implies \omega t - \phi = n\pi, \;\; n \in \mathbb{Z} \tag{5.30}$$

The turning points occur when $T = 0$ (at the points where potential energy is maximised. This also occurs when

$$\omega t - \phi = n\pi, \;\; n \in \mathbb{Z} \tag{5.31}$$

The kinetic energy is maximised when

$$\sin^2(\omega t - \phi) = 1 \tag{5.32}$$

$$\implies \omega t - \phi = \frac{n\pi}{2}, \;\; n \in \mathbb{Z} \tag{5.33}$$

The maximum kinetic and potential energy is given by

$$T_{max} = \frac{1}{2}mC^2\omega^2 \tag{5.34}$$

$$U_{max} = \frac{1}{2}kC^2 \tag{5.35}$$

Substituting $\omega^2 = \frac{k}{m}$,

$$T_{max} = \frac{1}{2}kC^2 \tag{5.36}$$

$$U_{max} = \frac{1}{2}kC^2 \tag{5.37}$$

Thus, a plot of the energy is as follows We now consider examples of simple harmonic oscillators

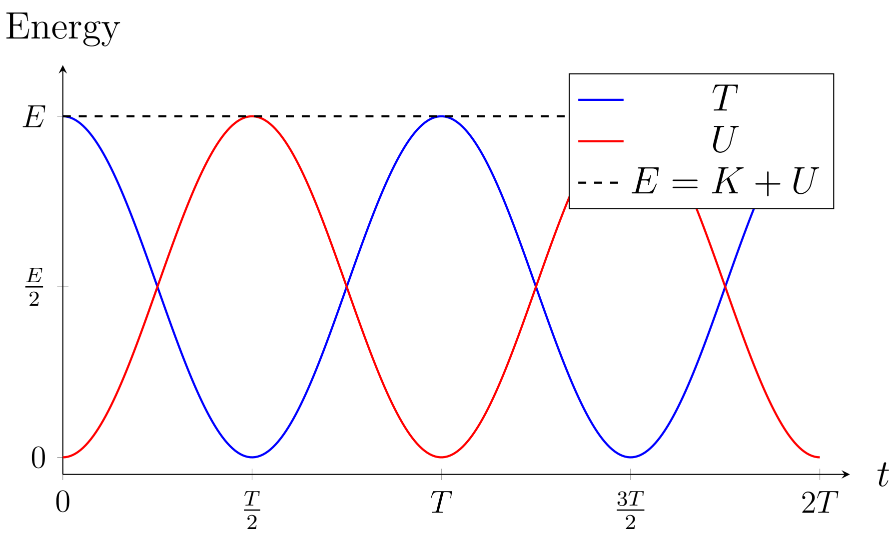


Figure 31: Kinetic energy, potential energy, and total energy of a simple harmonic oscillator over two full periods

5.1.3 Example 1: Simple Pendulum

Consider a simple pendulum system created by a mass m placed at the end of a massless rigid rod of length L in a constant gravitational field $\mathbf{g}$ on Earth. We want to find the equation of motion and a form for ω. Since the rod is massless, the only force

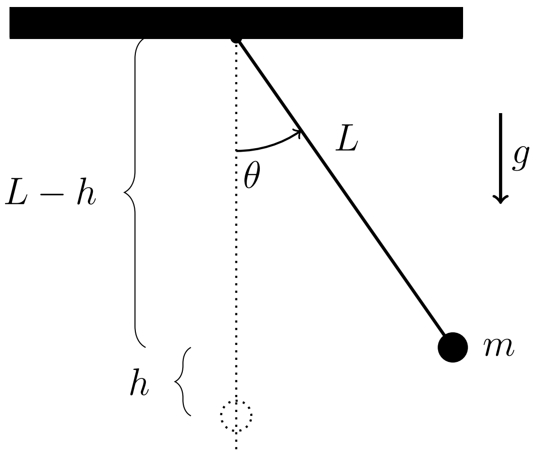


Figure 32: Simple pendulum displaced by θ: the mass rises by $h = L - L\cos\theta$ relative to the lowest point.

acting on the mass is the gravitational force, which is conservative. Thus, we can use energy conservation. We choose to work in polar coordinates where r is always constant

$r = L \implies \dot{r} = 0$ and $\theta = \theta(t)$. The total energy is,

$$E = T + U \tag{5.38}$$

Near the surface of the Earth, the potential energy can be approximated as

$$U = mgh \tag{5.39}$$

From the diagram, $h = L - L\cos(\theta) = L(1 - \cos(\theta))$. Thus,

$$U = mgL(1 - \cos(\theta))) \tag{5.40}$$

Further,

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) \tag{5.41}$$

$$= \frac{1}{2}mL^2\theta^2 \tag{5.42}$$

Then,

$$E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta) \tag{5.43}$$

By conservation of energy, we have

$$\dot{E} = mL^2\dot{\theta}\ddot{\theta} + mgL\dot{\theta}\sin\theta = 0 \tag{5.44}$$

$$\implies \dot{\theta}(mL^2\ddot{\theta} + mgL\sin\theta) = 0 \tag{5.45}$$

$$\implies mL^2\ddot{\theta} + mgL\sin\theta = 0 \tag{5.46}$$

$$\implies \ddot{\theta} + \frac{g}{L}\sin\theta = 0 \tag{5.47}$$

This ODE does not have a closed-form solution. Thus, we can instead consider how the system behaves when we displace the pendulum by a small angle instead. We can use the Maclaurin series for $\sin\theta$,

$$\sin\theta \approx \theta + O(\theta^3) \tag{5.48}$$

to write our equation of motion as

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \tag{5.49}$$

Thus, $\omega^2 = \frac{g}{L}$. The period of oscillation is then,

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{5.50}$$

Interestingly, the period of the pendulum does not depend on the mass of the pendulum bob!

5.1.4 Example 2: Cart on Spring in Bowl

Consider a mass m attached to a spring of spring constant k. The mass can move in a bowl, without friction, whose height from the ground is described by the quadratic function $h = \alpha(S - x)^2$. The system is near the Earth. We start by assuming that the

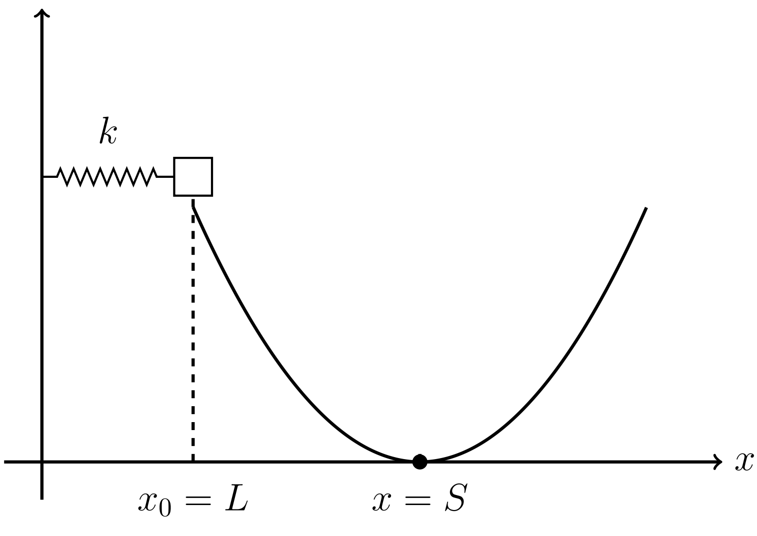


Figure 33: The spring–block system on a bowl

height of the bowl is much smaller than the width of the bowl. This allows us to neglect the vertical extension of the spring, but ensures that we do not treat the problem as the simple mass on a spring. Since the forces acting on the mass are the spring and the gravitational force, both of which are conservative, we can use conservation of energy. The potential energy on the mass comes from the spring potential and gravitational potential.

Thus,

$$
\begin{aligned}
U &= U_{spring} + U_{grav} && (5.51)\\
&= \frac{1}{2}k(\Delta x)^2 + mgh && (5.52)\\
&= \frac{1}{2}k(x-L)^2 + mg\alpha(S-x)^2 && (5.53)\\
&= \frac{1}{2}k(x^2 - 2xL + L^2) + mg\alpha(S^2 - 2Sx + x^2) && (5.54)\\
&= \frac{1}{2}kx^2 - kxL + \frac{1}{2}kL^2 + mg^2 - 2mg\alpha Sx + mg\alpha x^2 && (5.55)
\end{aligned}
$$

Then, the total energy is

$$
\begin{aligned}
E &= T + U && (5.56)\\
&= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 - kxL + \frac{1}{2}kL^2 + mg^2 - 2mg\alpha Sx + mg\alpha x^2 && (5.57)\\
\implies \dot{E} &= m\dot{x}\ddot{x} + kx\dot{x} - kL\dot{x} - 2mg\alpha S\dot{x} + 2mg\alpha x\dot{x} = 0 && (5.58)\\
&= \dot{x}(m\ddot{x} + kx - kL - 2mg\alpha S + 2mg\alpha x) = 0 && (5.59)\\
&\implies m\ddot{x} + (2mg\alpha + k)x - kL - 2mg\alpha S = 0 && (5.60)
\end{aligned}
$$

Dividing by m,

$$\ddot{x} + \left(2g\alpha + \frac{k}{m}\right)x - \frac{kL}{m} - 2g\alpha S = 0 \tag{5.61}$$

We can now define a new inertial reference frame whose origin is at the equilibrium position x_0. However, we must find the equilibrium position. Since we now have gravitational potential, this is not necessarily at $x = L$. Let the equilibrium position be $x_{eq} \implies \ddot{x}_{eq} = 0$ since x_{eq} is a constant. Then,

$$
\begin{aligned}
\left(2g\alpha + \frac{k}{m}\right)x_{eq} - \frac{kL}{m} - 2g\alpha S = 0 && (5.62)\\
\implies x_{eq} = \frac{2g\alpha S + \frac{kL}{m}}{2g\alpha + \frac{k}{m}} && (5.63)
\end{aligned}
$$

Now define a new inertial reference frame whose origin is at x_{eq}. Thus, $y = x - x_{eq}$. Our ODE then becomes

$$\ddot{y} + \left(2g\alpha + \frac{k}{m}\right)\left(y + \frac{2g\alpha S + \frac{kL}{m}}{2g\alpha + \frac{k}{m}}\right) - \frac{kL}{m} - 2g\alpha S = 0 \tag{5.64}$$

$$\implies \ddot{y} + \left(2g\alpha + \frac{k}{m}\right) y = 0 \tag{5.65}$$

Thus,

$$\omega^2 = 2g\alpha + \frac{k}{m} \tag{5.66}$$

Since $\omega^2 = \frac{k}{m}$, the term $2mg\alpha + k$ is known as the effective spring constant of the system.

5.1.5 Example 3: Physical Pendulums

Now consider an extended mass The body behaves as if all of its mass is concentrated at

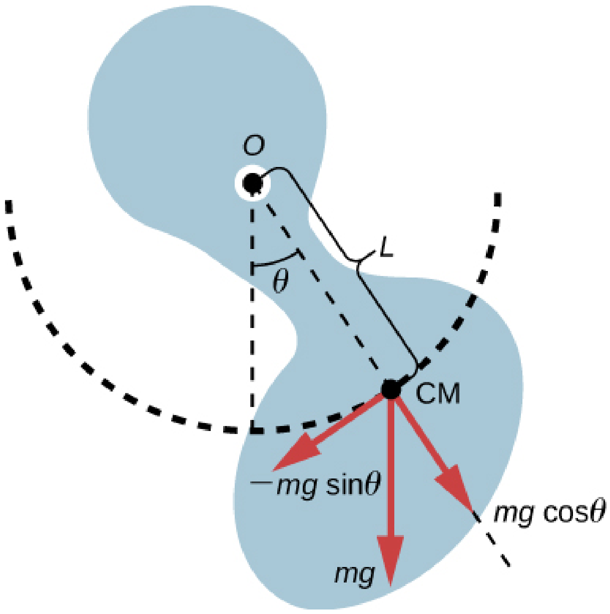


Figure 34: Physical Pendulum

its center of mass. Let L be the distance from the pivot to the center of mass. In polar coordinates, $r = L \implies \dot{r} = 0$. Also, $\theta = \theta(t)$. Then,

$$\mathbf{v} = L\dot{\theta}\mathbf{e}_\theta \tag{5.67}$$

The total energy is,

$$E = T + U \tag{5.68}$$
$$= \frac{1}{2}I\dot{\theta}^2 + mgL(1 - \cos(\theta)) \tag{5.69}$$

Applying conservation of energy, we have

$$I\ddot{\theta} + mgL\sin\theta = 0 \tag{5.70}$$

For small angles, we have $\sin\theta \approx \theta$. Then,

$$I\ddot{\theta} + mgL\theta = 0 \tag{5.71}$$

Thus,

$$\omega = \sqrt{\frac{mgL}{I}} \tag{5.72}$$

For the special case the mass is a sphere, we have $I = mL^2$. Then ω simplifies to the expression for that of a simple pendulum

5.2 Damped Harmonic Oscillator

In real life, we typically have to account for friction or drag. Thus, energy is no longer conserved. Consider the mass on a spring from example 1. We now introduce a linear drag $\mathbf{F} = -c\mathbf{v}$. The total energy is given by

$$E = T + U \tag{5.73}$$
$$= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k(x - L)^2 \tag{5.74}$$
$$\tag{5.75}$$

The change in total energy is then

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + k(x - L)\dot{x} \tag{5.76}$$

$$= \dot{x}(m\ddot{x} + k(x - L)) \tag{5.77}$$

From N2L, we have

$$m\ddot{x} = -k(x - L) - c\dot{x} \tag{5.78}$$

Substituting this into our expression for the change in energy gives,

$$\frac{dE}{dt} = \dot{x}(-k(x - L) - c\dot{x} + k(x - L)) \tag{5.79}$$

$$= -c\dot{x}^2 \tag{5.80}$$

Equating eq. (5.77) and eq. (5.80) gives,

$$\dot{x}(m\ddot{x} + k(x - L)) = -c\dot{x}^2 \tag{5.81}$$

We can rearrange this to get

$$m\ddot{x} + c\dot{x} + k(x - L) = 0 \tag{5.82}$$

We can shift our reference frame to the right such that the origin is at $x = L$. Thus,

$$y = x - L \implies \ddot{y} = \ddot{x} \tag{5.83}$$

Then,

$$m\ddot{y} + c\dot{y} + ky = 0 \tag{5.84}$$

This is the standard form of the ODE for the damped harmonic oscillator. We can divide by m to get

$$\ddot{y} + \frac{c}{m}\dot{y} + \frac{k}{m}y = 0 \tag{5.85}$$

We define the natural frequency (sometimes referred to as the eigenfrequency) of the system ω_n as the frequency of the system if no damping was present. This is given by

$$\omega_n = \sqrt{\frac{k}{m}} \tag{5.86}$$

We define the dimensionless damping parameter ζ as

$$\zeta = \frac{c}{2m\omega_n} = \frac{c}{2m}\sqrt{\frac{m}{k}} \tag{5.87}$$

We can then write our ODE for the damped harmonic oscillator as

$$\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = 0 \tag{5.88}$$

The characteristic polynomial of the ODE is

$$r^2 + 2\zeta\omega_n r + \omega_n^2 = 0 \tag{5.89}$$

The roots are given by

$$\begin{aligned} r &= \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} && (5.90) \\ &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} && (5.91) \end{aligned}$$

Our solution will depend on the sign of the discriminant $\zeta^2 - 1$

- Case I: $\zeta = 0$ (Undamped oscillator)

  If $\zeta = 0$, then our discriminant simplifies to -1 and the $-\zeta\omega_N$ term is 0. Thus, the roots to our characteristic polynomial are

$$r = \pm i\omega_N \tag{5.92}$$

The solution is then given by

$$y(t) = A\cos(\omega_N t) + B\sin(\omega_N t) \tag{5.93}$$

where A and B are constants determined by initial conditions. This is the solution of the undamped oscillator that we have seen before

- $\zeta > 1$ (Overdamped Oscillator)

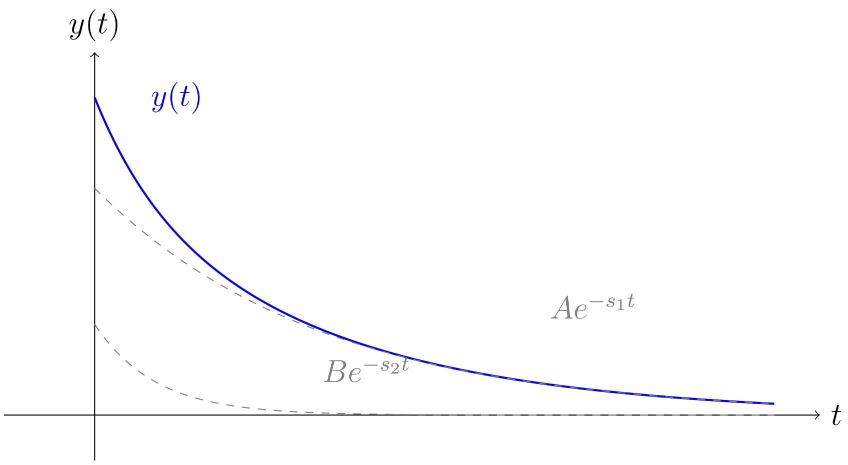
  If $\zeta > 1$, our discriminant is positive and we get two real roots given by

$$r_{1,2} = -\zeta\omega_N \pm \omega_N\sqrt{\zeta^2 - 1} \tag{5.94}$$

  The solution is then given by

$$y(t) = Ae^{-\zeta\omega_N + \omega_N\sqrt{\zeta_2 - 1}} + Be^{-\zeta\omega_N - \omega_N\sqrt{\zeta^2 - 1}} \tag{5.95}$$

  This is known as the overdamped solution. Our solution will simply decay with time, indicating that the damping outweights the oscillations.



- : $\zeta = 1$ (Critical Damping)

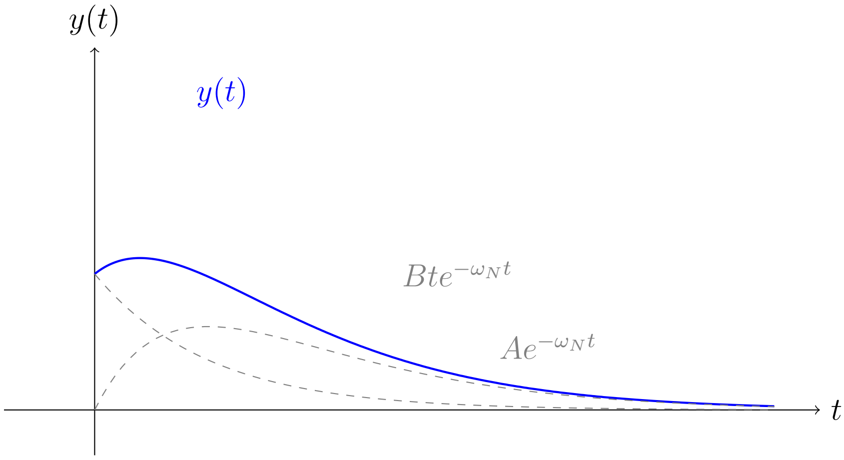
  In the case $\zeta = 1$, our discriminant simplifies to 0. The roots of our characteristic polynomial are

$$r = -\omega_N \tag{5.96}$$

The solution is given by

$$y(t) = Ae^{-\omega_N t} + Bte^{-\omega_N t} \tag{5.97}$$

This is known as the critically damped solution. It's the boundary between oscillatory and non-oscillatory decay. $\zeta = 1$ means the damping is exactly enough to prevent any oscillation, and the system returns to equilibrium in the shortest possible time without overshooting



- $0 < \zeta < 1$ (Underdamped Oscillator)

  If $0 < \zeta < 1$, our discriminant will be negative as $0 < \zeta^2 < 1$. Thus, the roots of our characteristic polynomial are

$$r = -zeta\omega_N \pm i\omega_N\sqrt{1-\zeta^2} \tag{5.98}$$

  We define the damped frequency of our oscillator ω_d as

$$\omega_d = \omega_N\sqrt{1-\zeta^2},\ 0 < \omega_d < \omega_N \tag{5.99}$$

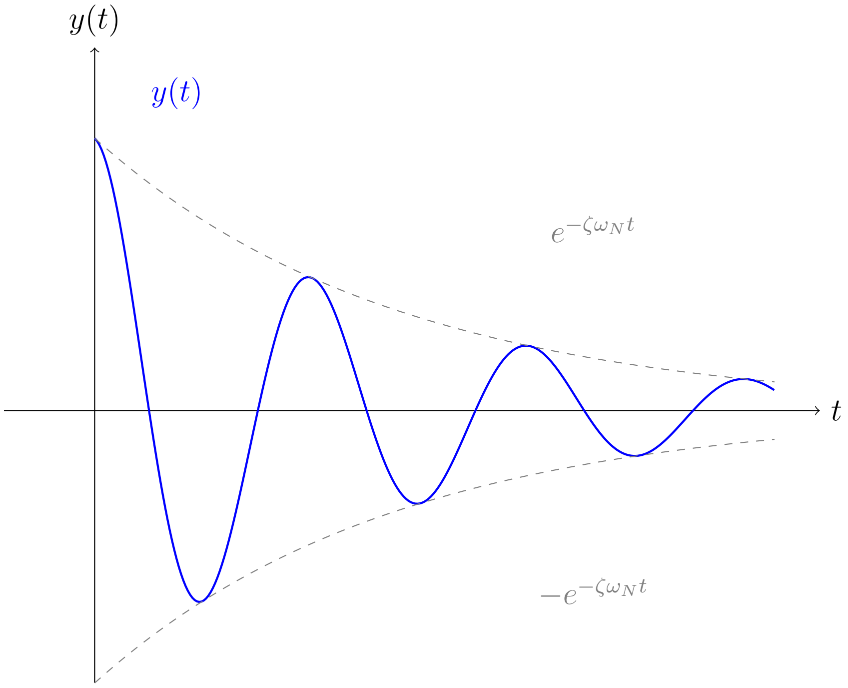
  Then, the roots of our characteristic polynomial are

$$-\zeta\omega_N \pm i\omega_d \tag{5.100}$$

The solution is given by

$$y(t) = Ae^{-\zeta\omega_N t}\left(A\cos(\omega_d t) + B\sin(\omega_d t)\right) \tag{5.101}$$

This is known as the underdamped solution. Here, the oscillator oscillates around the equilibrium while decaying. The exponential decay in time occurs with frequency ω_N



while the oscillator oscillates with frequency ω_d. We can rewrite the underdamped solution using an amplitude C and phase ϕ,

$$y(t) = Ce^{-\zeta\omega_N t}\cos(\omega_d t - \phi) \tag{5.102}$$

We will typically use the underdamped solution for the rest of this section.

5.3 Forced Oscillators

Now consider our oscillator being subject to a driving force $\mathbf{F}(t)$. The equation of motion is then

$$m_{\text{eff}}\ddot{\mathbf{x}} + c_{\text{eff}}\dot{\mathbf{x}} + k_{\text{eff}}\mathbf{x} = \mathbf{F}(t) \tag{5.103}$$

Example I: Forced Simple Pendulum

Consider an undamped simple pendulum with mass m at the end of a massless rod of length L with an external driving force $\mathbf{F}$. We want to find the equation of motion of such a system We work in polar coordinates as it greatly simplies our problem. We know

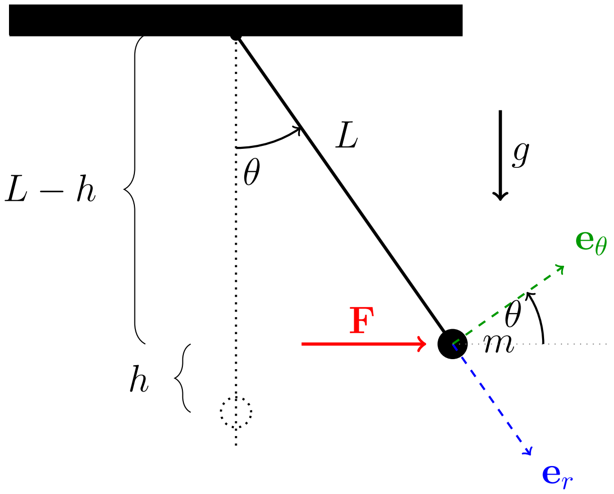


Figure 35: Simple pendulum displaced by θ with basis vectors $\hat{e}_r$ (radial, outward) and $\hat{e}_\theta$ (tangential, direction of increasing θ). A horizontal force $\mathbf{F}$ acts on the mass from the left.

that $r = L \implies \dot{r} = 0$. Further, $\theta = \theta(t) \implies \dot{\theta} = \dot{\theta}(t)$. Thus, $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta = L\dot{\theta}\mathbf{e}_\theta$. Also, $\cos\theta = \frac{L-h}{h} \implies h = L(1-\cos\theta)$. We decide to use the modified energy method to get,

$$\dot{E} - P_{\text{diss}} = \mathbf{F}\cdot\mathbf{v} \tag{5.104}$$

Since we have undamped motion, there is no power dissipated so $P_{\text{diss}} = 0$. Thus,

$$\dot{E} = \mathbf{F}\cdot\mathbf{v} \tag{5.105}$$

The total energy is given by,

$$
\begin{aligned}
E &= T + U && (5.106)\\
&= \frac{1}{2}m\mathbf{v}^2 + mgh && (5.107)\\
&= \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1-\cos\theta) && (5.108)\\
\implies \dot{E} &= mL^2\dot{\theta}\ddot{\theta} + mgL\dot{\theta}\sin\theta && (5.109)
\end{aligned}
$$

Also,

$$
\begin{aligned}
\mathbf{F}\cdot\mathbf{v} &= \mathbf{F}\cdot L\dot{\theta}\mathbf{e}_\theta && (5.110)\\
&= L\dot{\theta}F\cdot\mathbf{e}_\theta && (5.111)\\
&= FL\dot{\theta}\cos(\theta) && (5.112)
\end{aligned}
$$

since $\mathbf{a}\cdot\mathbf{b} = |a||b|\cos\theta$ and $|\mathbf{e}_\theta| = 1$. Thus,

$$mL^2\dot{\theta}\ddot{\theta} + mgL\dot{\theta}\sin\theta = FL\dot{\theta}\cos\theta \tag{5.113}$$

Our equation of motion is then

$$mL\ddot{\theta} + mg\sin\theta = F\cos\theta \tag{5.114}$$

If we assume we only have small angles, we can use the following Taylor expansions,

$$
\begin{aligned}
\sin\theta &\approx \theta && (5.115)\\
\cos\theta &\approx 1 && (5.116)
\end{aligned}
$$

Our equation of motion then reduces to

$$mL\ddot{\theta} + mg\theta = F \tag{5.117}$$

Here,

$$m_{\text{eff}} = mL \tag{5.118}$$

$$k_{\text{eff}} = mg \tag{5.119}$$

Example 2: Mass on a moving cart

Consider a moving cart that is allowed to accelerate. There is a mass m on the cart, that is attached to a spring of equilibrium length L and spring constant k. The overall displacement of the cart is $y(t)$, the displacement of the mass with respect to the cart is $x(t)$ and there is a stationary observer in the lab frame. Even though there is no external

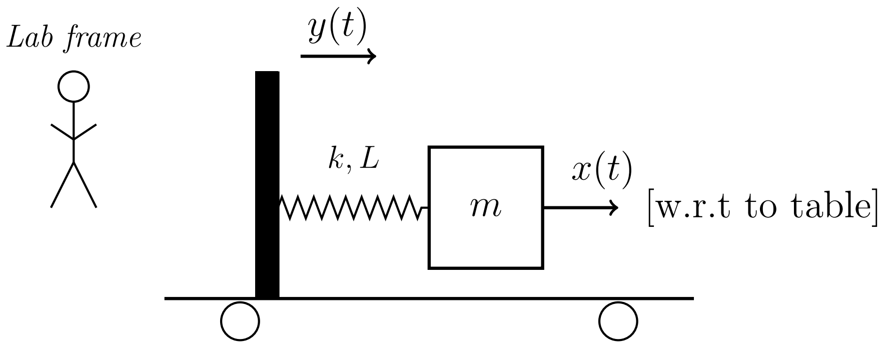


Figure 36: A mass m connected by a spring (k, natural length L) to a wall on a rolling table. $y(t)$ is the table position in the lab frame; $x(t)$ is the mass displacement w.r.t. the table.

force on the mass-cart system, the acceleration of the cart gives rise to an effective spring force on the mass. The displacement of the mass in the lab frame is given by

$$z(t) = x(t) + y(t) \tag{5.120}$$

The only force acting on the mass is the effective spring force, which depends on the position of the mass with respect to the cart

$$\mathbf{F} = -kx\mathbf{e}_x \tag{5.121}$$

If we apply N2L in the lab frame,

$$m\ddot{z} = -kx \tag{5.122}$$

$$\implies m(\ddot{x} + \ddot{y}) = -kx \tag{5.123}$$

$$\implies m\ddot{x} + kx = -m\ddot{y} \tag{5.124}$$

The right side $-m\ddot{y}$ is what we'd interpret as the driving force. However, it's not a real force applied to the mass. It's an pseudo-force arising because x in a non-inertial reference frame. The cart accelerates, and from the cart's perspective, the mass feels a fictitious push $-m\ddot{y}$ in addition to the spring.

5.3.1 General Solution of Forced Oscillators

The equations of motion of a forced oscillator are second-order linear inhomogenous differential equations. It is worthwhile to introduce some theory about linear differential equations and linear differential operators.

Interlude: Linear Differential Operators and Equations

An operator $\mathscr{L}$ is a map between functions. An operator maps one function to another function.

$$\mathscr{L} : f(t,x) \to g(t,x) \tag{5.125}$$

where we write

$$\mathscr{L}[f(t,x)] = g(t,x) \tag{5.126}$$

An operator is linear if it satisfies the rules of additivity and homogeneity. Namely, an operator is linear if

$$\mathscr{L}\left[af_1(t,x) + bf_2(t,x)\right] = a\mathscr{L}\left[f_1(t,x)\right] + b\mathscr{L}\left[f_2(t,x)\right] \tag{5.127}$$

where a and b are constants and f_1 and f_2 are functions. Examples of operators include

$$\mathscr{L} = \frac{d}{dx} \implies \mathscr{L}[f] = \frac{df}{dx} \tag{5.128}$$

$$\mathscr{L} = m\frac{d^2}{dt^2} \implies \mathscr{L}[f] = m\frac{d^2 f}{dt^2} \tag{5.129}$$

For the second example,

$$\mathscr{L}[af_1(t,x) + bf_2(t,x)] = am\frac{\partial^2 f_1}{\partial t^2} + bm\frac{\partial^2 f_2}{\partial t^2} \tag{5.130}$$

The damped forced oscillator obeys the following differential equation

$$m_{\text{eff}}\ddot{x} + c_{\text{eff}}\dot{x} + k_{\text{eff}}x = F(t) \tag{5.131}$$

The linear operator for the damped forced harmonic oscillator is then given by

$$\mathscr{L} = m_{\text{eff}}\frac{d^2}{dt^2} + c_{\text{eff}}\frac{d}{dt} + k \tag{5.132}$$

Thus, we can compactly write our ODE as

$$\mathscr{L}[x] = F \tag{5.133}$$

A general solution to the ODE $\mathscr{L}[x] = F$ is given by the theory of ordinary differential equations to be

$$x(t) = x_h(t) + x_p(t) \tag{5.134}$$

where $x_h(t)$ is the solution to the homogenous equation $\mathscr{L}[x] = 0$ and $x_p(t)$ is the particular solution whose form depends on F. If we have an undamped oscillator, we typically have

$$x_h(t) = A\cos(\omega_N t) + B\sin(\omega_N t) \tag{5.135}$$

If we have a damped oscillator, we typically have

$$x_h(t) = e^{-\zeta\omega_N t}\left(A\cos(\omega_d t) + B\sin(\omega_d t)\right) \tag{5.136}$$

where $\omega_N = \sqrt{\frac{k_{\text{eff}}}{m_{\text{eff}}}}$ is the natural frequency of the oscillator, $\zeta = \frac{c_{\text{eff}}}{2m_{\text{eff}}\omega_N}$ is the dimensionless damping parameter and $\omega_d = \omega_N\sqrt{1-\zeta^2}$ is the frequency of the damped oscillator. A and B are determined by initial conditions on $x(t)$ (not just $x_h(t)$). We need to use the method of undetermined coefficients to determine the form of $x_p(t)$.

Example: Finding $x(t)$ for a forced harmonic oscillator

Assume an external force $F(t) = bt^2$ is applied to a harmonic oscillator where $m_{\text{eff}} = k_{\text{eff}} = 1 and c_{\text{eff}} = 0 and \mathrm{b} = 1. The ODE is then, \ddot{x} + x = t^2 (5.137)$ where $x(0) = \dot{x}(0) = 0$ We use the form $x_h(t) = A\cos(t) + B\sin(t)$ since $\omega_N = 1$ and $c = 0$. Now, we guess $x_p = At^2 + Bt + C$. Then,

$$\dot{x}_p = 2At + B \tag{5.138}$$

$$\dot{x}_p = 2A \tag{5.139}$$

Then,

$$At^2 + Bt + (2A + C) = t^2 \tag{5.140}$$

Thus by comparing coefficients,

$$A = 1 \tag{5.141}$$

$$B = 0 \tag{5.142}$$

$$C = -2A = -2 \tag{5.143}$$

Then,

$$x(t) = A\cos(t) + B\sin(t) + t^2 - 2 \tag{5.144}$$

We now apply the initial conditions

$$x(0) = A - 2 = 0 \implies A = 2 \tag{5.145}$$

$$\dot{x} = -A\sin(t) + B\cos(t) + 2t \tag{5.146}$$

$$\implies \dot{x}(0) = B = 0 \tag{5.147}$$

$$\implies B = 0 \tag{5.148}$$

Thus,

$$x(t) = 2\cos(t) + t^2 - 2 \tag{5.149}$$

5.3.2 Harmonic Forced Oscillators

A harmonic force is a force of the form

$$F(t) = F_0\cos(\omega t - \theta) \tag{5.150}$$

where ω is the frequency of the force θ is the phase of the force. We can then write our ODE as

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = F_0\cos(\omega t - \phi) \tag{5.151}$$

We can either use the ansatz $x_p(t) = D\cos(\omega t - \phi)$ and use the method of undetermined functions. However, we can also use complex functions which we see will give us a more general solution. Recall that the complex exponential is given by

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t) \tag{5.152}$$

We can break this up into its real and imaginary component as

$$e^{i\omega t} = \Re\left(e^{i\omega t}\right) + \Im\left(e^{i\omega t}\right) \tag{5.153}$$

We can further write the force $F(t)$ as the real part of the complex exponential

$$F(t) = F_0 \cos(\omega t - \theta) = \Re\left(e^{i(\omega t - \theta)}\right) \tag{5.154}$$

We can now use the ansatz

$$x_p(t) = \Re\left(Xe^{i\omega t}\right), X \in \mathbb{C} \tag{5.155}$$

We don't know a-priori if x has a phase. The phase, if any, is encoded in X. We rewrite our ODE as $\mathscr{L}[x] - F = 0$ and insert our ansatz into our linear operator to get

$$\Re\left(-m\omega^2 Xe^{i\omega t} + ic\omega Xe^{i\omega t} + kXe^{i\omega t} - F_0 e^{i(\omega t - \theta)}\right) = 0 \tag{5.156}$$

Note that the $\Re$ operator and $\frac{d}{dt}$ commute (since $\Re$ is a linear operation and X is constant). If the entire expression above is 0, then

$$e^{i\omega t}\left(-mX\omega^2 + Xic\omega + kX\right) - F_0 e^{-i\theta} = 0 \tag{5.157}$$

We can then solve for X to get,

$$X = \frac{F_0 e^{-i\theta}}{-m\omega^2 + ic\omega + k} \tag{5.158}$$

We can express this in terms of the natural frequency of the oscillator $\omega_N = \sqrt{\frac{k}{m}}$ and the dimensionless damping parameter $\zeta = \frac{c}{2m\omega_N}$

$$X = \frac{F_0 e^{-i\theta}}{k}\left(1 + 2i\zeta\frac{\omega}{\omega_N} - \frac{\omega^2}{\omega_N^2}\right)^{-1} \tag{5.159}$$

We define the complex frequency response function $\tilde{G}(\omega)$ of the damped harmonic oscillator as

$$\tilde{G}(\omega) = \frac{1}{k}\left(1 + 2i\zeta\frac{\omega}{\omega_N} - \frac{\omega^2}{\omega_N^2}\right)^{-1} \tag{5.160}$$

This encodes how the oscillator responds at each driving frequency ω Then, X is

$$X = \tilde{G}(\omega)F_0 e^{-\theta} \tag{5.161}$$

Since the complex frequency response function is well complex, it can be written using Euler's form

$$\tilde{G}(\omega) = G(\omega)e^{-i\phi(\omega)} \tag{5.162}$$

where $G(\omega)$ and $\phi(\omega)$ are real functions. $G(\omega)$ is the magnitude of the complex response function and is termed the response function. $\phi(\omega)$ is the phase; we use a negative sign as convention to make the phase positive as the driven-damped oscillator always lags behind the driving force. We can see this lag by defining $u = 1 - \frac{\omega^2}{\omega_N^2}$ and $v = 2\zeta\frac{\omega}{\omega_N}$

$$\tilde{G}(\omega) = \frac{1}{k(u+iv)} = \frac{u-iv}{k(u^2+v^2)} \tag{5.163}$$

Since $v > 0$, $\Im(\tilde{G}) < 0 \implies \arg\left(\tilde{G}\right) < 0$ implying a lag. The magnitude ands the phase of the complex response frequency function is given by

$$G(\omega) = \frac{1}{k}\left[\left(1 - \frac{\omega}{\omega_N}\right)^2 + \left(2\zeta\frac{\omega}{\omega_N}\right)^2\right]^{-\frac{1}{2}} \tag{5.164}$$

$$\phi(\omega) = \arctan\left(\frac{2\zeta\omega\omega_N}{\omega_N^2 - \omega^2}\right), 0 \leq \phi < \pi \tag{5.165}$$

We can plot $G(\omega)$ and $\phi(\omega)$ to get

We can insert our expression for $\tilde{G}(\omega)$ into $X = \tilde{G}(\omega)$, and then insert X into our original ansatz to get our particular solution as

$$x_p(t) = \Re\left(Xe^{i\omega t}\right) \tag{5.166}$$

$$= \Re\left(G(\omega)e^{-i\phi(\omega)}F_0 e^{-i\theta}e^{i\omega t}\right) \tag{5.167}$$

$$= F_0 G(\omega)\cos(\omega t - \theta - \phi(\omega)) \tag{5.168}$$

Now, our solution to the forced, damped, harmonic oscillator is then the sum of our

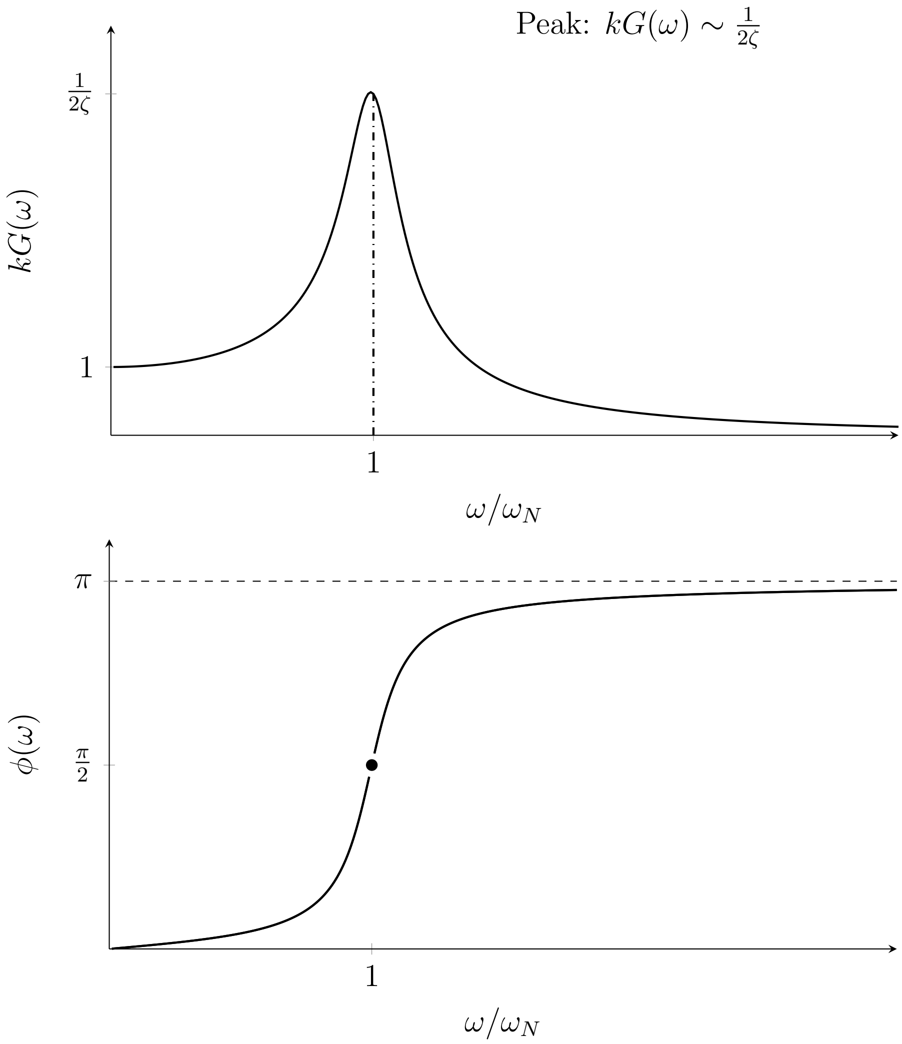


Figure 37: Magnitude $kG(\omega)$ and phase $\phi(\omega)$ of the complex frequency response function for $\zeta = 0.1$.

complementary and particular solution

$$x(t) = x_h(t) + x_p(t) \tag{5.169}$$

$$\implies x(t) = e^{-\zeta\omega_N t}\left(A\cos(\omega_d t) + B\sin(\omega_d t)\right) + F_0 G(\omega)\cos\left(\omega t - \theta - \phi(\omega)\right) \tag{5.170}$$

where $\omega_d = \omega_N\sqrt{1-\zeta^2}$ is the frequency of oscillations of the damped oscillator.

Remark. The particular solution does not decay in time but the homogenous solution does. $x_h(t)$ decays as $e^{-\zeta\omega_N t} = e^{-\frac{t}{\tau}}$ where $\tau = \frac{1}{\zeta\omega_N}$. At late times $t >> 1$, the oscillator oscillates as the particular solution. To that end, we typically call $x_h(t)$ as the transient solution and $x_p(t)$ the steady-state solution. The steady state solution $x_p(t)$ oscillates with the same frequency ω of the driving force. The amplitude and phase of $x_p(t)$ also depend on ω. If our force was a sine instead of cosine, we could replace the cosine in our particular solution with a sine.

5.3.3 Analysis of the Response Function and Resonance

Recall that our steady state solution is

$$x_p(t) = F_0 G(\omega)\cos(\omega t - \theta - \phi(\omega)) \tag{5.171}$$

where the response function and phase is

$$G(\omega) = \frac{1}{k}\left[\left(1 - \frac{\omega}{\omega_N}\right)^2 + \left(2\zeta\frac{\omega}{\omega_N}\right)^2\right]^{-\frac{1}{2}} \tag{5.172}$$

$$\phi(\omega) = \arctan\left(\frac{2\zeta\omega\omega_N}{\omega_N^2 - \omega^2}\right), 0 \leq \phi < \pi \tag{5.173}$$

If we plot our response function against ω

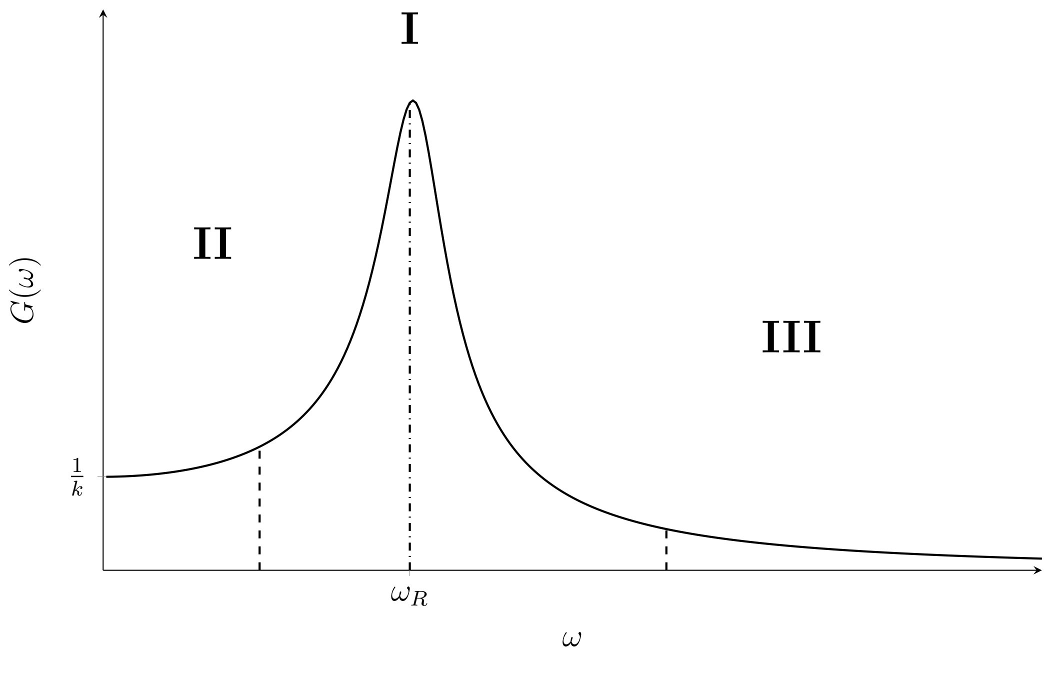


Figure 38: Magnitude $G(\omega)$ showing the resonance peak near ω_R with the bandwidth (hatched region) between half-power frequencies.

- Region I: Resonance ($\omega = \omega_R$)

Resonance is the situation in which there is a peak in our response function. The frequency at which the peak occurs is known as the resonant frequency ω_R which satisfies

$$\frac{dG}{d\omega}|\omega = \omega_R = 0 \tag{5.174}$$

The resonant frequency is given by

$$\omega_R = \omega_N\sqrt{1 - 2\zeta^2} \tag{5.175}$$

The magnitude of the response function at the resonant frequency is

$$G(\omega = \omega_R) = \frac{1}{2k\zeta\sqrt{1 - \zeta^2}} \tag{5.176}$$

- Region II: Stiffness controlled region ($\omega < \omega_R = \omega_N\sqrt{1 - 2\zeta^2}$) We start by defining the parameter ϵ as

$$\epsilon = \frac{\omega}{\omega_N} \tag{5.177}$$

We can then expand our response function for small ω i.e. $\epsilon << 1$

$$G(w) = \frac{1}{k} + \frac{1 - 2\zeta^2}{k^2}\epsilon^2 \approx \frac{1}{k} \tag{5.178}$$

This region is dominated by the spring constant k (also called the stiffness of the spring), and hence the region is called the stiffness controlled region. The phase of the response is given by

$$\phi(\omega) \approx 2\zeta\epsilon \approx 0 \tag{5.179}$$

Thus, the steady state solution in the stiffness controlled region is approximately

$$x_p(t) \approx \frac{F_0}{k}\cos(\omega t - \theta) \tag{5.180}$$

- Region III: Mass controlled region $\left(\omega > \omega_R = \omega_N\sqrt{1 - 2\zeta^2}\right)$

We expand for $\omega >> \omega_R$ so that $\frac{\omega}{\omega_N}$ is large. Thus,

$$G(w) \approx \frac{1}{m\omega^2} \tag{5.181}$$

$$\phi(\omega) \approx n\pi, n \in \mathbb{Z} \tag{5.182}$$

The steady-state solution is given by

$$x_p(t) \approx \frac{F_0}{m\omega^2}\cos(\omega t - \theta - \pi) \tag{5.183}$$

The steady state solution is behind the driving force by a phase π. Further, its position is also determined by its mass.

5.4 Periodic Forced Oscillators and Fourier Series

Our analysis of harmonic forces will help us study more general forces. Namely, we consider a periodic force. A periodic force is a force that satisfies

$$F(t+T) = F(t) \tag{5.184}$$

where T is the period. The frequency of the periodic force is

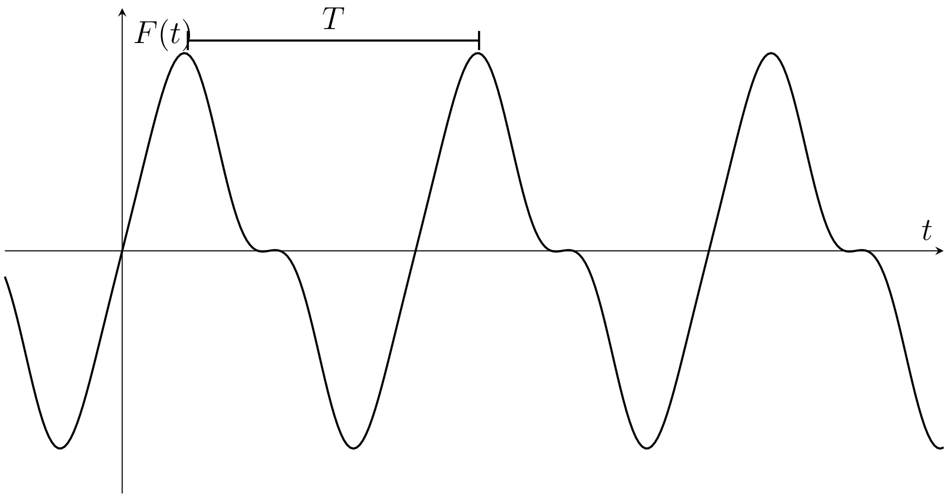


Figure 39: A general periodic forcing function $F(t)$ with period T.

$$\Omega = \frac{2\pi}{T} \tag{5.185}$$

For complicated periodic functions, there is often no closed form. Thus, we turn to the Fourier series to be able write down an approximation of our force.

5.4.1 Fourier Series

Any sufficiently well-behaved periodic function can be written as a superposition of cosine and sine functions as below

$$F(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t) + \sum_{n=1}^{\infty} b_n \sin(n\Omega t) \tag{5.186}$$

where a_0, a_n and b_n are termed the Fourier coefficients. Since cosine is an even function, $\sum_{n=1}^{\infty} a_n \cos(n\Omega t)$ is the even part of the function $F(t)$. Since sine is an odd function, $\sum_{n=1}^{\infty} b_n \sin(n\Omega t)$ is the odd part of the function $F(t)$. The Fourier coefficients are given by the following integrals

$$a_0 = \frac{2}{T}\int_T F(t)dt \tag{5.187}$$
$$a_n = \frac{2}{T}\int_T F(t)\cos(n\Omega t)dt \tag{5.188}$$
$$b_n = \frac{2}{T}\int_T F(t)\sin(n\Omega t)dt \tag{5.189}$$

$\int_t$ refers to the integral over a period. Thus, our bounds of integration could be $[0, T], [T, 2T]$ or even $\left[-\frac{T}{2}, \frac{T}{2}\right]$ as long as we integrate over one period. Here, $\{1, \cos(n\Omega t), \sin(n\Omega t)\}$ are known as basis functions and the equations for the Fourier coefficients come from projecting $F(t)$ along the basis function. The inner product we use is defined as

$$\langle f(t), g(t)\rangle = \int_T f(t)g(t)dt \tag{5.190}$$

We look to verify that our basis functions are orthogonal. Define

$$I = \int_T \cos(m\Omega t)\sin(n\Omega t)dt \tag{5.191}$$

We can rewrite the integrand using the identity $\cos(A)\cos(B) = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$. Thus,

$$I = \frac{1}{2}\int_T \cos((m+n)\Omega t)dt + \frac{1}{2}\int_T \cos((m-n)\Omega t)dt \tag{5.192}$$

$$= \frac{1}{2}I_1 + \frac{1}{2}I_2 \tag{5.193}$$

Evaluating each integral

$$I_1 = \frac{1}{(m+n)\Omega}\left[\sin((m+n)\Omega t)\right]_0^T \tag{5.194}$$

$$= \frac{1}{(m+n)\Omega}\left[\sin((m+n)\Omega \cdot \frac{2\pi}{\Omega}) - \sin(0)\right] \tag{5.195}$$

$$= 0 \tag{5.196}$$

Now, we can show a similar step for I_2 to see that

$$I_2 = \begin{cases} 0, & \text{if} \\ m-n \neq 0 \\ \int_T \cos(0) = T, & \text{if} \\ m-n = 0 \end{cases} \tag{5.197}$$

Thus,

$$I = \begin{cases} 0, & \text{if} \\ m \neq n \\ \frac{T}{2}, & \text{if} \\ m = n \end{cases} \tag{5.198}$$

We can express this compactly using the Kronecker-Delta defined as

$$\delta_{mn} = \begin{cases} 1, & \text{if} \\ m = n \\ 0, & \text{if} \\ m \neq n \end{cases} \tag{5.199}$$

Thus,

$$I = \delta_{mn} T \tag{5.200}$$

We can apply a similar process to see that the other orthogonality relationships follow

$$\int_T \cos(m\Omega t)\cos(n\Omega t)dt = \frac{T}{2}\delta_{mn} \tag{5.201}$$

$$\int_T \sin(m\Omega t)\sin(n\Omega t)dt = \frac{T}{2}\delta_{mn} \tag{5.202}$$

$$\int_T \cos(m\Omega t)\sin(n\Omega t)dt = 0 \tag{5.203}$$

Example: Square Wave

Consider a periodic function $F(t)$ described by

$$F(t) = \begin{cases} 1, & \text{if} \\ -\frac{T}{2} \leq t < 0 \\ -1, & \text{if} \\ 0 \leq \frac{T}{2} \end{cases} \tag{5.204}$$

and then periodic We wan to find the Fourier coefficients and write $F(t)$ as a Fourier Series. We can see that on $\left[0, \frac{T}{2}\right)$, $F(t) = 1$ while on $\left[-\frac{T}{2}, 0\right)$, $F(t) = -1$. Thus, $F(-t) = -F(t)$ so F is an odd function. Consequently,

$$a_0 = \int_T F(t)dt = 0 = a_n = \int_T F(t)\cos(n\Omega t)dt = 0 \tag{5.205}$$

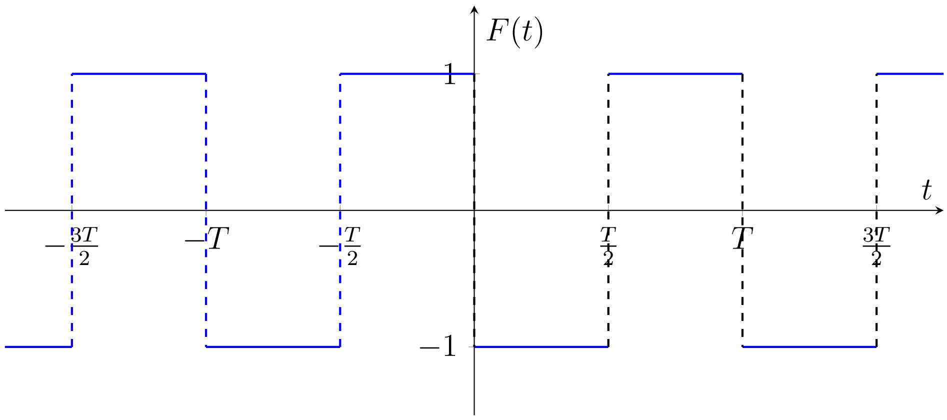


Figure 40: Square wave $F(t)$ with period T: $F(t) = 1$ on $[-T/2, 0)$ and $F(t) = -1$ on $[0, T/2)$.

since the integrand of both integrals are odd functions, and the integral of an odd function over a symmetric interval is 0. Now,

$$b_n = \frac{2}{T}\int_T F(t)\sin(n\Omega t)dt \tag{5.206}$$

$$= \frac{2}{T}\int_{-\frac{T}{2}}^{0} \sin(n\Omega t)dt + \frac{2}{T}\int_0^{\frac{T}{2}} -\sin(n\Omega t)dt \tag{5.207}$$

$$= \frac{2}{T}\left[-\frac{1}{n\Omega}\cos(n\Omega t)\right]_{-\frac{T}{2}}^{0} + \frac{2}{T}\left[\frac{1}{n\Omega}\cos(n)\right]_0^{\frac{T}{2}} \tag{5.208}$$

$$= \frac{2}{T}\left[-\frac{1}{n\Omega} + \frac{1}{n\Omega}\cos(\pi n) + \frac{1}{n\Omega}\cos(\pi n) - \frac{1}{n\Omega}\right] \tag{5.209}$$

$$= \frac{2}{T}\left[-\frac{2}{n\Omega} + \frac{2}{n\Omega}\cos(n\pi)\right] \tag{5.210}$$

$$= -\frac{2}{n\Omega} + \frac{2}{n\Omega}\cos(n\pi) \tag{5.211}$$

$$\implies b_n = \begin{cases} 0, & \text{if} \\ n \text{ even} & \\ -\frac{4}{n\pi}, & \text{if} \\ n \text{ odd} & \end{cases} \tag{5.212}$$

Thus,

$$F(t) = \sum_{n=1,3,\ldots}^{\infty}\left(-\frac{4}{n\pi}\sin(n\Omega t)\right) \tag{5.213}$$

5.4.2 Damped Harmonic Oscillator with Periodic Driving Forces

Now consider our forced damped harmonic oscillator where we have a periodic driving force. Our equation of motion is then

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = F(t) \tag{5.214}$$

where $F(t+T) = F(t)$. Since F is periodic, we can use its Fourier series representation to get

$$\mathscr{L}[x] = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t) + \sum_{n=1}^{\infty} b_n \sin(n\Omega t) = \sum_{p=1}^{\infty} F_n(t) \tag{5.215}$$

We can clearly see that $F(t)$ is a linear superposition of harmonic forces. Since $\mathscr{L}$ is a linear operator, the particular solution must be a linear superposition of the particular solution to each F_n. We can write this as

$$x(t) = \sum_{n=1}^{\infty} x_{p,n}(t) \tag{5.216}$$

where each $x_{p,n}(t)$ solves $\mathscr{L}[x_{p,n}(t)] = F_n(t)$. We already know the solution for each $F_n(t)$, it is given by our solution to the harmonic force

$$x_{p,n}(t) = f_n G(n\Omega) \cos(n\Omega t - \theta - \phi(n\Omega)) \tag{5.217}$$

where f_n is related to the Fourier coefficients. Our initial force had $\theta = 0$. Thus,

$$\begin{aligned} x_p(t) &= \sum_{n=1}^{\infty} x_{p,n}(t) && (5.218) \\ &= \frac{a_0}{2k} + \sum_{n=1}^{\infty} a_n G(n\Omega) \cos(n\Omega t - \phi(n\Omega)) + \sum_{n=1}^{\infty} b_n G(n\Omega) \sin(n\Omega t - \phi(n\Omega)) && (5.219) \end{aligned}$$

where

$$G(n\Omega) = \frac{1}{k}\left[\left(1-\left(\frac{n\Omega}{\omega_N}\right)^2\right)^2 + \left(2\zeta\frac{n\Omega}{\omega_N}\right)^2\right]^{-\frac{1}{2}} \tag{5.220}$$

$$\phi(n\Omega) = \arctan\left(\frac{2\zeta\omega_N n\Omega}{\omega_N^2 - (n\Omega)^2}\right) \tag{5.221}$$

Note that $\Omega = \frac{2\pi}{T}$ is the frequency of the driving force, $\omega_N = \sqrt{\frac{k}{m}}$ is the natural frequency of the oscillator and $\zeta = \frac{c}{2m\omega_N}$ is the dimensionless damping parameter. The full solution is then

$$x(t) = x_h(t) + x_p(t) \tag{5.222}$$

$$= e^{-\zeta\omega_N t}\left[A\cos(\omega_d t) + B\sin(\omega_d t)\right] + \frac{a_0}{2k} + \sum_{n=1}^{\infty} a_n G(n\Omega)\cos(n\Omega t - \phi(n\Omega)) + \tag{5.223}$$

$$+\sum_{n=1}^{\infty} b_n G(n\Omega)\sin(n\Omega t - \phi(n\Omega)) \tag{5.224}$$

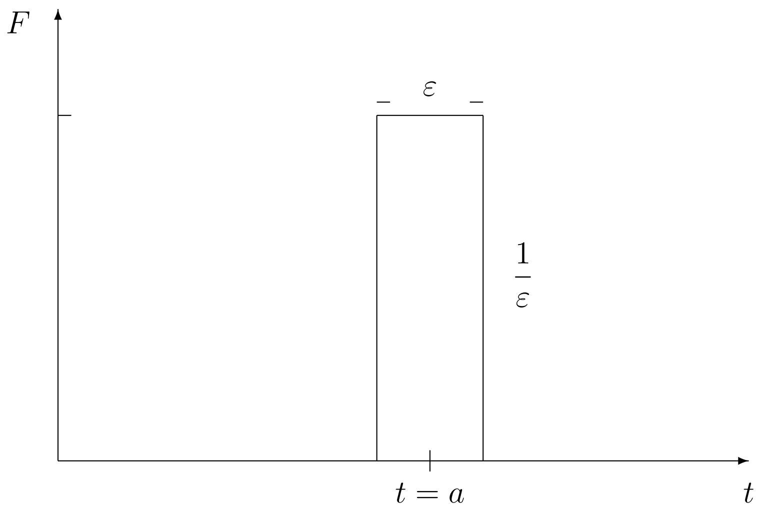
We get A, B from initial conditions applied to $x(t)$. Resonance occurs at $n\Omega = \omega_R = \omega_N\sqrt{1-2\zeta^2}$

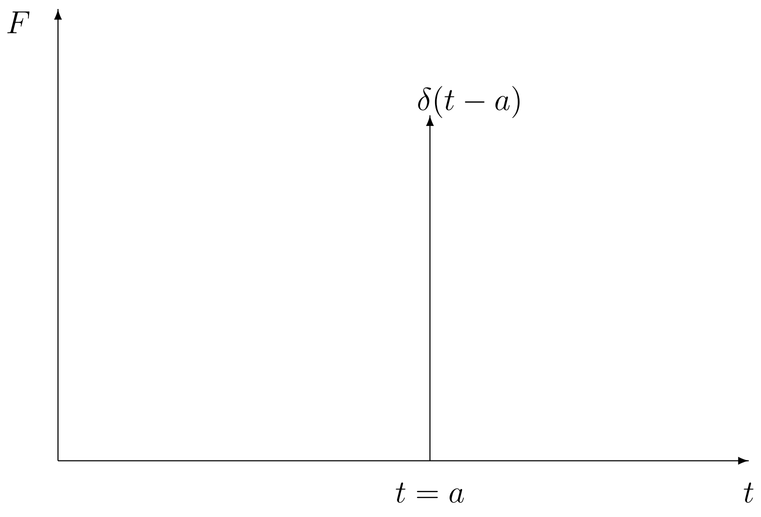
5.4.3 Impulse Responses and Green's Functions

An impulse is a force applied instantaneously. It can be described by the Dirac Delta distribution, also known as the Delta function.

5.4.4 Dirac Delta Distribution

We want to consider a force that acts for a short time. We describe it by a narrow rectangle at $t = a$ with a width of ϵ and height $\frac{1}{\epsilon}$ Now consider the limit as $\epsilon \to 0$, the result is the Dirac delta distribution The dirac delta distribution is defined more rigorously using the theory of distributions. However, we take a more operational approach. The dirac delta distribution centred at a, $\delta(t-a)$ is defined by its sifting property. Consider a well-behaved function $f(t)$ that is continuous at a and some nice interval B where $a \in B$.





Then,

$$\int_B f(t)\delta(t-a)dt = f(a) \tag{5.225}$$

Typically, we let $B = (-\infty, \infty)$, although any interval where a is in the interior is valid. Using the real line, the sifting property of the dirac delta distribution is

$$\int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a) \tag{5.226}$$

Example 0

$$\int_{-\infty}^{\infty} \delta(t-a)dt = 1 \;\; (f(t=a) = 1 \text{ for } f(t) = 1) \tag{5.227}$$

Example 0

$$\int_{-\infty}^{\infty} \sin(t)\delta\left(t - \frac{3\pi}{2}\right) dt = \sin\left(\frac{3\pi}{2}\right) \tag{5.228}$$

$$= -1 \tag{5.229}$$

Example 0

Consider

$$\int_{-\infty}^{\infty} \sin(t)\delta(t^3 - 8)dt \tag{5.230}$$

This integral is not in the standard form. Thus, we need to use substitution. Let $u = t^3 - 8 \implies du = 3t^2 dt$. Thus,

$$\int_{-\infty}^{\infty} \sin(t)\delta(t^3 - 8)dt = \int_{-\infty}^{\infty} \frac{\sin\left(u^{\frac{1}{3}}\right)}{3u^{\frac{2}{3}}}\delta(u-8)du \tag{5.231}$$

$$= \frac{\sin(2)}{12} \tag{5.232}$$

We also need to consider a way we can treat the derivative of δ. Consider

$$\int_{-\infty}^{\infty} f(t)\frac{d}{dt}\left(\delta(t-a)\right) dt \tag{5.233}$$

We use integration by parts where $u = f(t) \implies du = f'(t)$ and $dv = \frac{d}{dt}\left(\delta(t-a)\right) \implies$

$\delta(t-a)$

$$\int_{-\infty}^{\infty} f(t)\frac{d}{dt}\left(\delta(t-a)\right)dt = \overset{0}{\cancel{\left[f(t)\delta(t-a)\right]_0^\infty}} - \int_{-\infty}^{\infty} f'(t)\delta(t-a)dt \tag{5.234}$$

$$= -\int_{-\infty}^{\infty} f'(t)\delta(t-a)dt \tag{5.235}$$

$$= -f'(a) \tag{5.236}$$

Where the boundary terms cancel since $\delta(t-a) = 0 \forall t \neq a$

$$\int_{-\infty}^{\infty} f(t)\frac{d}{dt}\left(\delta(t-a)\right)dt = -f'(a) = -\dot{f}(a) \tag{5.237}$$

We now want to consider a force described by an impulse and how an oscillator would respond to such a force. We can describe the force using a delta function.

5.4.5 Impulse Force

Consider a large force applied over a short duration to our oscillator which we define as

$$\Delta t << \frac{1}{\omega_N} \tag{5.238}$$

We can write this force as $F(t) = I\delta(t)$ where I is an impulse. Thus,

$$\int_{-\infty}^{\infty} F(t)dt = \int_{-\infty}^{\infty} I\delta(t)dt = I \tag{5.239}$$

Further, assume our oscillator is initially at rest and at $x=0$. Since our time period is much shorter than the natural period of the oscillator, the system has no time to move appreciably during the impulse. Thus, the position of the oscillator does not change but its momentum changes. This is given by

$$mv_f - mv_i = \int_{-\infty}^{\infty} F(t)dt \tag{5.240}$$

Since $v_i = 0$, the initial conditions of the oscillator after applying the force is

$$x(t=0) = 0 \tag{5.241}$$

$$\dot{x}(t=0) = v_f = \frac{I}{m} \tag{5.242}$$

This is known as a quiescent condition. Since we modelled the impulse with the delta distribution at $t = 0$, the impulse is instantaneous such that the velocity at $t = 0$ is given by the final velocity of the oscillator. Thus, the ODE that describes the harmonic oscillator is

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = F(t) = I\delta(t) \tag{5.243}$$

$$x(0) = 0 \tag{5.244}$$

$$\dot{x} = \frac{I}{m} \tag{5.245}$$

The solution to this is given by

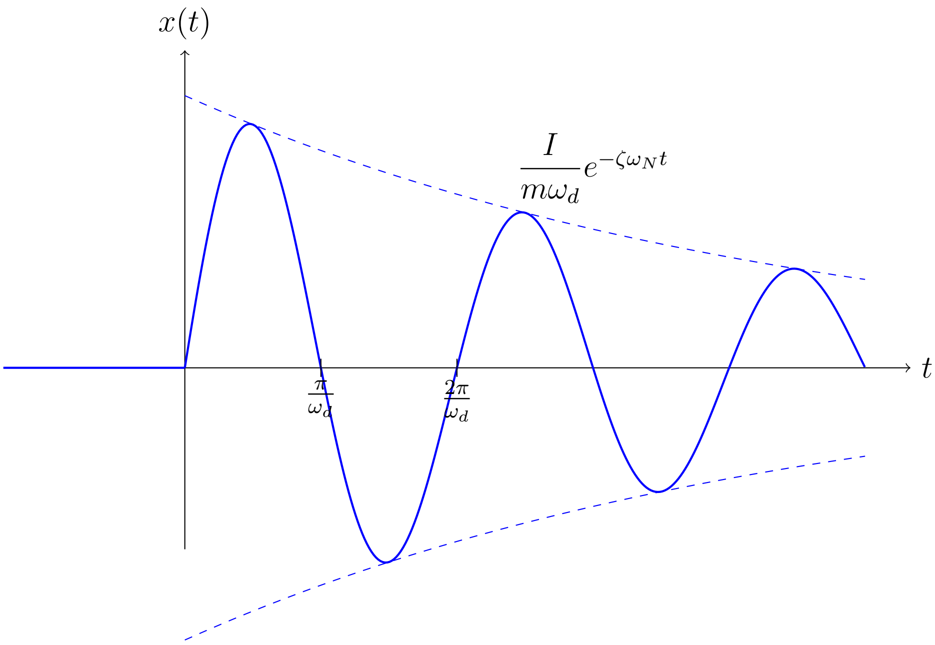
$$x(t) = \begin{cases} 0, & t < 0 \\ \frac{I}{m\omega_d} e^{-\zeta\omega_N t} \sin(\omega_d t), & t \geq 0 \end{cases} \tag{5.246}$$

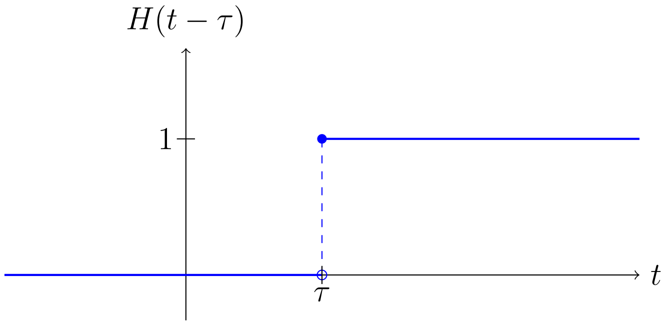
A plot of the solution is as follows, We can model this sudden jump in the oscillator solution using the Heaviside step function, which is defined as

$$H(t-\tau) = \begin{cases} 0, & t < \tau \\ 1, & t \geq \tau \end{cases} \tag{5.247}$$

Thus, since the jump happens at $t = 0$, we have $\tau = 0$ and our solution can be compactly written as

$$x(t) = \frac{I}{m\omega_d} e^{-\zeta\omega_N t} \sin(\omega_d t) H(t) \tag{5.248}$$





If instead of $t = 0$, our impulse force was applied at some $t = \tau$, we would have

$$x(t) = \frac{I}{m\omega_d} e^{-\zeta\omega_N (t-\tau)} \sin(\omega_d(t-\tau))H(t-\tau) \tag{5.249}$$

We don't have a particular solution as we do not have external forcing after the impulse is added. The impulse encodes the forcing entirely into the initial conditions of the oscillator.

5.4.6 Green's Function

We define the Green's function $G(t-\tau)$ as the solution to the damped harmonic oscillator subject to an impulse force at $t = \tau$

$$\mathscr{L}[G(t-\tau)] = m\ddot{G}(t-\tau) + c\dot{G}(t-\tau) + kG(t-\tau) = \delta(t-\tau) \tag{5.250}$$

Explicitly, the Green's Function is

$$G(t-\tau) = \frac{1}{m\omega_d} e^{-\zeta\omega_N(t-\tau)} \sin(\omega_d(t-\tau))H(t-\tau) \tag{5.251}$$

For $\tau = 0$,

$$G(t) = \frac{1}{m\omega_d} e^{-\zeta\omega_N t} \sin(\omega_d)H(t) \tag{5.252}$$

Thus, the solution to

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = I\delta(t) \tag{5.253}$$

is given by

$$x(t) = IG(t) \tag{5.254}$$

5.4.7 Arbitrary Forcing and Convolution

The power of the Green's function is the ability to write down solutions to systems subject to arbitrary forces. Since our differential equation is linear, our solution will be a superposition of Green's functions.

Consider a force described by multiple impulse forces where each impulse is applied over a very short duration

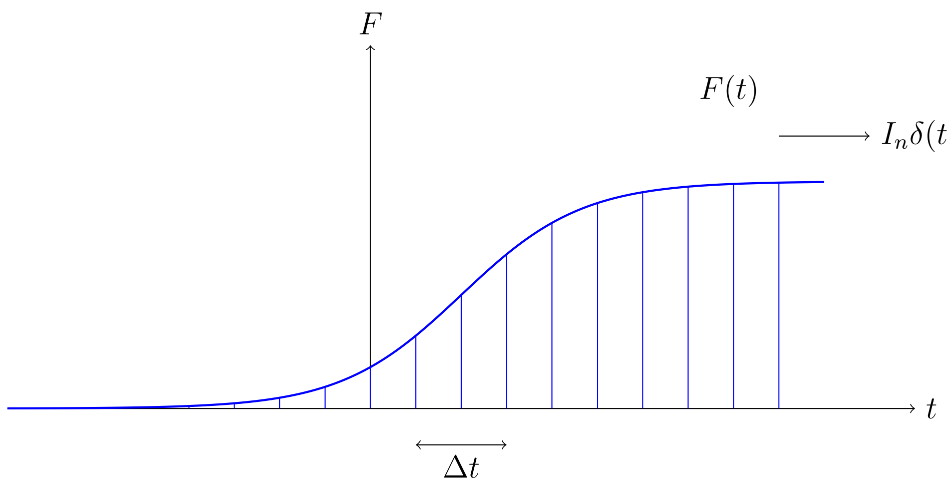
$$F(t) = \sum_n I_n \delta(t - t_n) \tag{5.255}$$

where

$$I_n = F(t_n)\delta t \tag{5.256}$$

Our force can then be described by

$$F(t) = \sum_n F(t_n)\Delta t \delta(t - t_n) \tag{5.257}$$



We would expect our solution to this sort of forcing to be a superposition of Green's functions given by

$$x(t) = \sum_n I_n G(t-t_n) = \sum_n F(t_n)\Delta t G(t-t_n) \tag{5.258}$$

To properly describe the force, we take the limit as $n \to \infty$. Then, our force is

$$F(t) = \int_{-\infty}^{\infty} F(\tau)\delta(t-\tau)d\tau \tag{5.259}$$

and our solution becomes

$$x(t) = \int_{-\infty}^{\infty} F(\tau)G(t-\tau)d\tau \tag{5.260}$$

This is known as a convolution between F and G. We represent it as

$$x(t) = F(t) * G(t) = \int_{-\infty}^{\infty} F(\tau)G(t-\tau)d\tau \tag{5.261}$$

We can insert our explicit representation of the Green's function to get

$$x(t) = \int_{-\infty}^{\infty} \frac{F(\tau)}{m\omega_d} e^{-\zeta\omega_N(t-\tau)} \sin(\omega_d(t-\tau))H(t-\tau)d\tau \tag{5.262}$$

Since the Heaviside function is 0 for $t < \tau$, the bounds of the integral become

$$x(t) = \int_{-\infty}^{t} \frac{F(\tau)}{m\omega_d} e^{-\zeta\omega_N(t-\tau)} \sin(\omega_d(t-\tau))d\tau \tag{5.263}$$

Thus, if we have a damped harmonic oscillator subject with quiescent initial conditions subject to a force $F(t)$ such that its ODE is given by

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = F(t) \tag{5.264}$$

The solution is

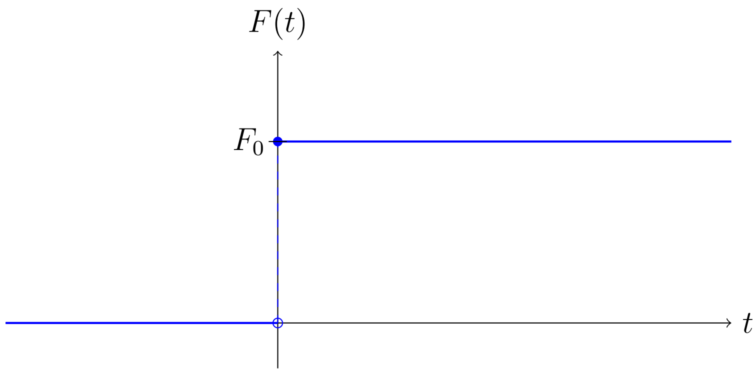
$$x(t) = \int_{-\infty}^{t} \frac{F(\tau)}{m\omega_d} e^{-\zeta\omega_N(t-\tau)} \sin(\omega_d(t-\tau))d\tau \tag{5.265}$$

Example: Driven Oscillator with Green's Function

Consider an oscillator driven by a constant driving force described by

$$F(t) = F_0 H(t) \tag{5.266}$$

We have an undamped oscillator with $x(0) = \dot{x}(0) = 0$. Since we have an undamped



oscillator, $c = 0$ and so our ODE is

$$\mathscr{L}[x] = m\ddot{x} + kx = F_0 H(t) \tag{5.267}$$

There are two ways we can approach this. We can either use the method of undetermined coefficients or via Green's functions. We will see how both methods lead to the same answer. To simplify our problem, we will work only for $t \geq 0$. In other words, we only look for a solution when our oscillator is subject to a driving force

Strategy I: Ansatz and Undetermined Coefficients

The full solution to our oscillator is given by $x(t) = x_h(t) + x_p(t)$ where $x_h(t)$ is the homogenous solution and $x_p(t)$ is the particular solution. We know the homogenous solution for an undamped oscillator is

$$x_h(t) = A\cos(\omega_N t) + B\sin(\omega_N t) \tag{5.268}$$

We can use the ansatz $x_p(t) = a$ where a is a constant. Thus, $\dot{x}_p = 0 \implies \ddot{x}_p = 0$. Then,

$$ka = F_0 H(t) \tag{5.269}$$

We can solve for a to get,

$$a = \frac{F_0}{k} = \frac{F_0 H(t)}{m\omega_N^2} \tag{5.270}$$

Our solution is then,

$$x(t) = A\cos(\omega_N t) + B\sin(\omega_N t) + \frac{F_0}{m\omega_N^2} \tag{5.271}$$

We now apply our initial conditions to see that

$$x(t) = A + \frac{F_0}{m\omega_N^2} = 0 \tag{5.272}$$

$$\implies A = -\frac{F_0}{m\omega_N^2} \tag{5.273}$$

Thus,

$$x(t) = -\frac{F_0}{m\omega_N^2}\cos(\omega_N t) + B\sin(\omega_N t) + \frac{F_0}{m\omega_N^2} \tag{5.274}$$

Computing $\dot{x}$,

$$\dot{x}(t) = \frac{F_0}{m\omega_N}\sin(\omega_N t) + B\omega_N\cos(\omega_N t) + \frac{F_0}{m\omega_N^2} \tag{5.275}$$

Then,

$$\dot{x} = B\omega_N = 0 \implies B = 0 \tag{5.276}$$

Thus,

$$x(t) = \frac{F_0}{m\omega_N^2}\left(1 - \cos(\omega_N t)\right) \tag{5.277}$$

Strategy II: Green's Function

We can write our solution as convolution between our force and our Green's Function

$$x(t) = F(t) * G(t) = \int_{-\infty}^{\infty} F(\tau)G(t-\tau)d\tau \tag{5.278}$$

where the Green's function is

$$(t-\tau) = \frac{1}{m\omega_d}e^{-\zeta\omega_N(t-\tau)}\sin(\omega_d(t-\tau))H(t-\tau) \tag{5.279}$$

Since $c = 0$, we have $\zeta = 0$ and $\omega_d = \omega_N$. Thus, our Green's function simplifies to

$$G(t-\tau) = \frac{H(t-\tau)}{m\omega_N}\sin(\omega_N(t-\tau)) \tag{5.280}$$

Thus,

$$x(t) = \int_{-\infty}^{\infty} F_0 H(\tau)\frac{H(t-\tau)}{m\omega_N}\sin(\omega_N(t-\tau))d\tau \tag{5.281}$$

The two step functions are defined as

$$H(\tau) = \begin{cases} 1, & \tau \geq 0 \\ 0, & \tau < 0 \end{cases} \tag{5.282}$$

$$H(t-\tau) = \begin{cases} 1, & t-\tau \geq 0 \implies t \geq \tau \\ 0, & t-\tau < 0 \implies t < \tau \end{cases} \tag{5.283}$$

Thus, our integral become

$$x(t) = \int_0^t \frac{F_0}{m\omega_N} \sin(\omega_N t - \omega_N \tau) d\tau \tag{5.284}$$

$$= \frac{F_0}{m\omega_N^2} \cos(\omega_N(t-\tau))|_0^\tau \tag{5.285}$$

$$= \frac{F_0}{m\omega_N^2} (1 - \cos(\omega_N t)) \tag{5.286}$$

We get the same answer.

5.5 Fourier Transforms

The Fourier transform can be thought of as the integral version of the Fourier series. It is used to convert a function of time (in the time domain) to a function of frequency (in the frequency domain). Fourier transforms are useful to analyse signals, but they are particularly useful as they convert differential equations in the time domain to algebraic equations in the frequency domain. We can then solve this algebraic equation for our desired quantity and convert back into the time domain. Fourier Transforms describe the distribution of the frequencies of oscillations.

The fourier transform operator $\mathcal{F}$ converts a time function $f(t)$ into a frequency function $\tilde{f}(\omega)$ as

$$\mathcal{F}[f(t)] = \tilde{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \tag{5.287}$$

The inverse fourier transform is given by

$$\mathcal{F}^{-1}[f(t)] = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega t} d\omega \tag{5.288}$$

The Fourier transform gives us a useful representation of the Dirac delta distribution. Consider,

$$f(t) = \delta(t) \tag{5.289}$$

Then,

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} e^{-i\omega t}\delta(t)dt \tag{5.290}$$
$$= e^0 \tag{5.291}$$
$$= 1 \tag{5.292}$$

Then, it should hold that

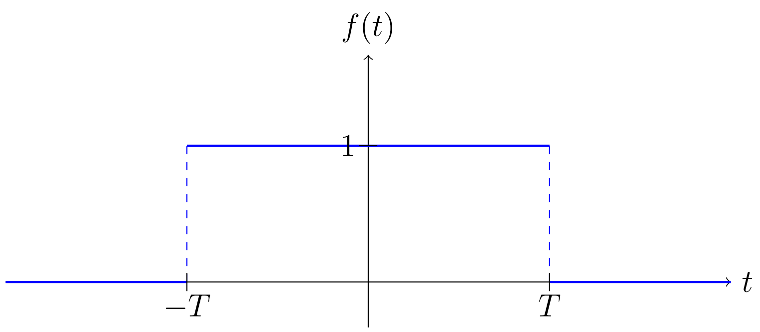
$$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega t}dt \tag{5.293}$$

Example: Square Pulse

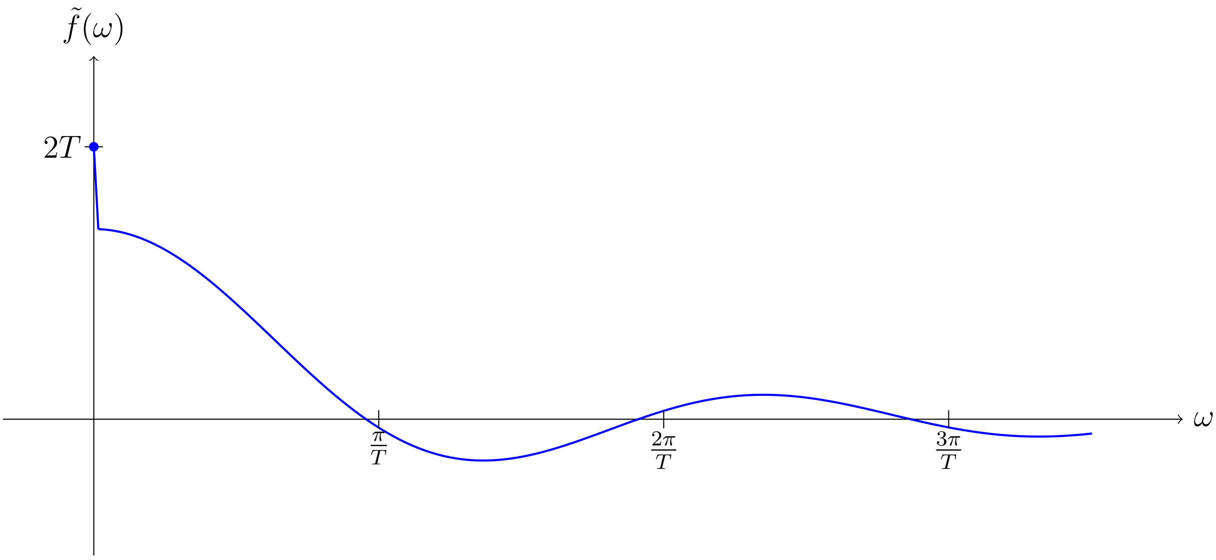
Consider a square pulse defined by

$$f(t) = H(t+T)H(T-t) \tag{5.294}$$

Then,



$$f(\omega) = \int_{-\infty}^{\infty} H(t+T)H(T-t)e^{-\omega t}dt \tag{5.295}$$
$$= \int_{-T}^{T} e^{-i\omega t}dt \tag{5.296}$$
$$= \frac{1}{i\omega}\left(e^{i\omega T} - e^{-i\omega t}\right) \tag{5.297}$$
$$= \frac{2}{\omega}\sin(\omega T) \tag{5.298}$$



We now want to apply the Fourier transform to our harmonic oscillator. We will see how it turns our differential equation in time to an algebraic equation in frequency. The ODE in the time domain is

$$\mathscr{L}[x] = m\ddot{x} + c\dot{x} + kx = F(t) \tag{5.299}$$

We define $\tilde{F}(\omega)$ and $\tilde{X}(\omega)$ as

$$\tilde{F}(\omega) = \mathcal{F}[F] = \int_{-\infty}^{\infty} F(t)e^{-i\omega t}dt \tag{5.300}$$

$$\tilde{X}(\omega) = \mathcal{F}[x] = \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt \tag{5.301}$$

We now need the fourier transforms of $\dot{x}$ and $\ddot{x}$. For a fourier transform to exist, we need the following boundary conditions

$$\lim_{t\to\pm\infty} x(t) = 0 \tag{5.302}$$

With this in mind,

$$\mathcal{F}[\dot{x}] = \int_{-\infty}^{\infty} \dot{x}(t) e^{-i\omega t} dt \tag{5.303}$$

$$= \cancelto{0}{\left[x e^{-i\omega t}\right]_{-\infty}^{\infty}} + \int_{-\infty}^{\infty} i\omega x e^{-i\omega t} dt \tag{5.304}$$

$$= i\omega \tilde{X}(\omega) \tag{5.305}$$

Similarly,

$$\mathcal{F}[\ddot{x}] = -\omega^2 \tilde{X}(\omega) \tag{5.306}$$

Thus, applying the Fourier transform to each term in $\mathscr{L}[x]$ gives

$$-m\omega^2 \tilde{X}(\omega) + ic\omega \tilde{X}(\omega) + k\tilde{X}(\omega) = \tilde{F}(\omega) \tag{5.307}$$

We can rearrange for $\tilde{X}(\omega)$ to get

$$\tilde{X}(\omega) = \frac{\tilde{F}(\omega)}{-m\omega^2 + ic\omega + k} \tag{5.308}$$

Recall from eq. (5.158), that the denominator is the complex frequency response function. Thus,

$$\tilde{X}(\omega) = \tilde{F}(\omega)\tilde{G}(\omega) \tag{5.309}$$

where

$$\tilde{G}(\omega) = \frac{1}{k}\left(1 + 2i\zeta\frac{\omega}{\omega_N} - \frac{\omega^2}{\omega_N^2}\right)^{-1} \tag{5.310}$$

We can then use the inverse Fourier Transform to get $x(t)$

$$x(t) = \mathcal{F}^{-1}\left[\tilde{X}(\omega)\right] \tag{5.311}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\tilde{F}(\omega)}{-m\omega^2 + ic\omega + k} e^{i\omega t} d\omega \tag{5.312}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{F}(\omega)\tilde{G}(\omega) d\omega \tag{5.313}$$

Recall that the Green's function $G(t)$ is the solution to the damped harmonic oscillator subject to a delta force.

$$\mathscr{L}[G(t)] = m\ddot{G}(t) + c\dot{G}(t) + kG(t) = \delta(t) \tag{5.314}$$

We define $\tilde{K}(\omega)$ to be the Fourier Transform of our Green's Function

$$\tilde{K}(\omega) = \int_{-\infty}^{\infty} G(t)e^{-i\omega t}dt \tag{5.315}$$

Further recall that $\mathcal{F}[\delta(t)] = 1$ Then,

$$-m\omega^2\tilde{K}(\omega) + ic\omega\tilde{K}(\omega) + k\tilde{K}(\omega) = 1 \tag{5.316}$$

Thus,

$$\tilde{K}(\omega) = \frac{1}{-m\omega^2 + ic\omega + k} \tag{5.317}$$

But, this is exactly the complex frequency response function $\tilde{G}(\omega)$! Thus, the Green's function and the complex frequency response function are related by a Fourier Transform. Namely,

$$\tilde{G}(\omega) = \mathcal{F}[G(t)] = \int_{-\infty}^{\infty} G(t)e^{-i\omega t}dt \tag{5.318}$$

6 Non-Inertial Reference Frames

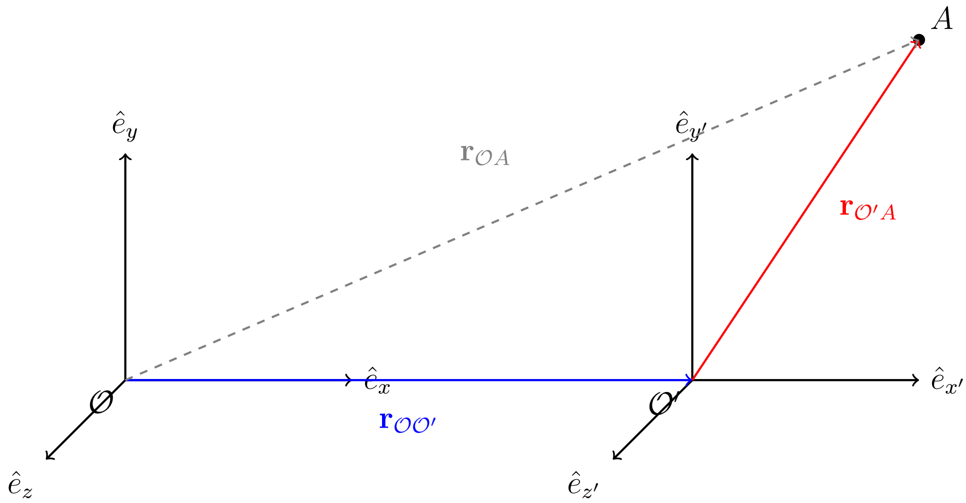
An inertial reference frame (RF) is a frame that is not accelerating. Thus, a non-inertial RF is a frame that is not an inertial RF! Namely, a non-inertial RF is a frame that is accelerating.

Newton's second law in its standard form is $\mathbf{F} = m\ddot{\mathbf{r}}$ where $\mathbf{F}$ is the sum of all real external forces. The systems we have seen thus far are inertial RFs. However, not every system is inertial. For example, the Earth is rotating and thus if we truly want an accurate description of the Earth, we need to take this into account. However, N2L in its standard form only holds true in inertial RFs. In non-inertial RFs, N2L will include additional terms which appear as fictitious forces.

6.1 Kinematics in Non-Inertial Reference Frames

6.1.1 Transformations in Linearly Accelerated Frames

Consider two reference frames accelerated with respect to each other. We have a rest frame $\mathcal{O}$ with basis vectors $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ and an accelerated frame $\mathcal{O}'$ with basis vectors $\mathbf{e}'_x, \mathbf{e}'_y, \mathbf{e}'_z$. We consider $\mathcal{O}$ to be fixed, that is $\dot{\mathbf{e}_x} = \dot{\mathbf{e}_y} = \dot{\mathbf{e}_z} = 0$. Both reference frames are related to each other by a position vector $\mathbf{r}_{\mathcal{O}\mathcal{O}'}$. Another point A, with respect to the rest frame $\mathcal{O}$ and $\mathcal{O}'$ has position vector $\mathbf{r}_{\mathcal{O}A}$ and $\mathbf{r}_{\mathcal{O}'A}$ By the parallelogram law of vector



addition, we see that

$$\mathbf{r}_{\mathcal{O}A} = \mathbf{r}_{\mathcal{O}\mathcal{O}'} + \mathbf{r}_{\mathcal{O}'A} \tag{6.1}$$

The velocity of point A in the rest frame is then

$$\mathbf{v}_{\mathcal{O}A} = \dot{\mathbf{r}}_{\mathcal{O}A} \tag{6.2}$$
$$= \dot{\mathbf{r}}_{\mathcal{O}\mathcal{O}'} + \dot{\mathbf{r}}_{\mathcal{O}'A} \tag{6.3}$$
$$= \mathbf{v}_{\mathcal{O}\mathcal{O}'} + \mathbf{v}_{\mathcal{O}'A} \tag{6.4}$$

where $\mathbf{v}_{\mathcal{O}\mathcal{O}'}$ is the relative velocity between the two frames and $\mathbf{v}_{\mathcal{O}'A}$ is the velocity of A with respect to $\mathcal{O}'$. The acceleration is then

$$\mathbf{a}_{\mathcal{O}A} = \dot{\mathbf{v}}_{\mathcal{O}A} \tag{6.5}$$
$$= \dot{\mathbf{v}}_{\mathcal{O}\mathcal{O}'} + \dot{\mathbf{v}}_{\mathcal{O}'A} \tag{6.6}$$
$$= \mathbf{a}_{\mathcal{O}\mathcal{O}'} + \mathbf{a}_{\mathcal{O}'A} \tag{6.7}$$

where $\mathbf{a}_{\mathcal{O}\mathcal{O}'}$ is the relative acceleration between the two frames and $\mathbf{a}_{\mathcal{O}'A}$ is the acceleration of A with respect to $\mathcal{O}'$. In the rest frame, the force at A is given by

$$\mathbf{F} = m\mathbf{a}_{\mathcal{O}A} = m(\mathbf{a}_{\mathcal{O}\mathcal{O}'} + \mathbf{a}_{\mathcal{O}'A}) \tag{6.8}$$

We can rearrange this to get

$$m\mathbf{a}_{\mathcal{O}'A} = \mathbf{F} - m\mathbf{a}_{\mathcal{O}\mathcal{O}'} \tag{6.9}$$

Since $m\mathbf{a}_{\mathcal{O}\mathcal{O}'}$ is an apparent force that arises from the relative acceleration between the two frames, and is not an actual force in our system. We call it a “fictitious” force.

Thus N2L, in the accelerated frame is

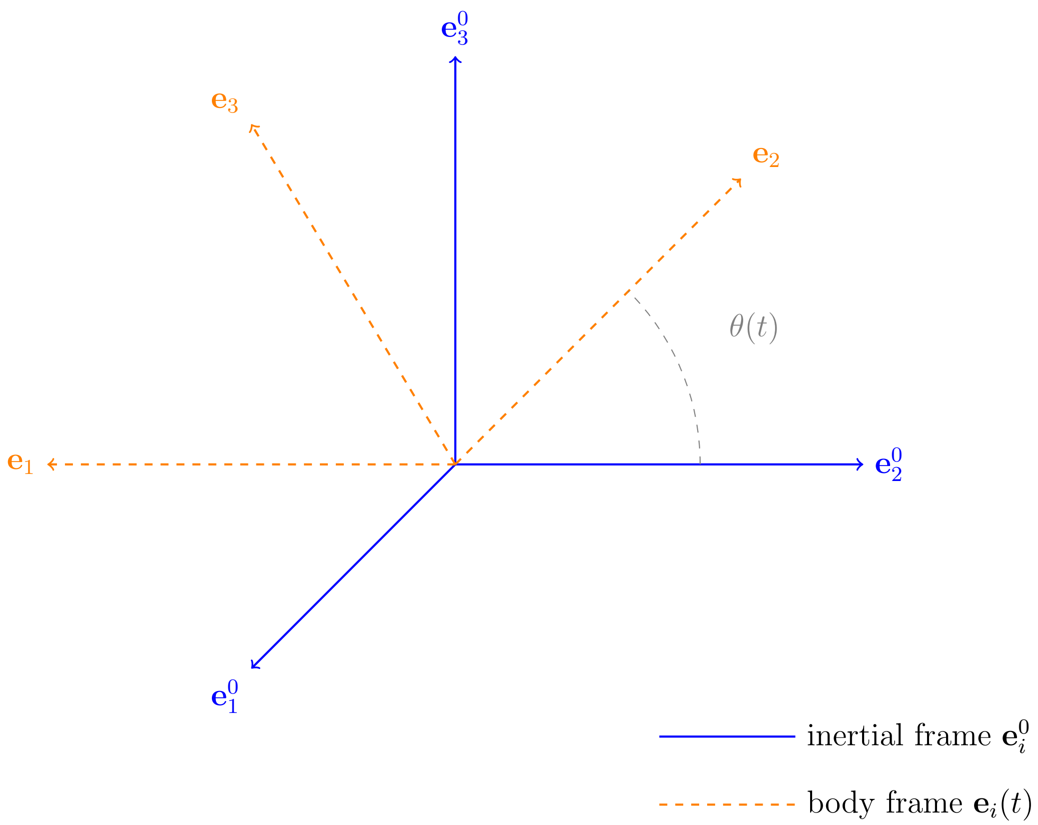
$$\mathbf{F}_{\text{tot}} = m\mathbf{a}_{\mathcal{O}'A} = \mathbf{F}_{\text{real}} + \mathbf{F}_{\text{fictitious}} \tag{6.10}$$

6.1.2 Transformations in Rotating Frames

Now, we consider the case where we do have a rotating frame. Consider two sets of orthonormal basis vectors, $\mathbf{e}_i^0$ and $\mathbf{e}_i$ where $i \in \{1, 2, 3\}$. $\mathbf{e}_i^0$ is the basis vector of the fixed inertial reference frame. Since this is fixed, i.e. it doesn't move or rotate, we have

$$\frac{d\mathbf{e}_i^0}{dt} = 0 \tag{6.11}$$

$\mathbf{e}_i$ is the basis of the rotating frame. We tend to call the rotating frame the body frame. Since this changes with time, we have $\mathbf{e} = \mathbf{e}(t)$



An arbitrary vector u can be expressed in both bases as

$$\mathbf{u} = \sum_{i=1}^{3} u_i^0 \mathbf{e}_i^0 \tag{6.12}$$

$$\mathbf{u} = \sum_{i=1}^{3} u_i \mathbf{e}_i \tag{6.13}$$

where u_i^0 are the components of u in the fixed frame and u_i are the components of u in the body frame. **u** is the same physical vector, only the component representation in either basis differs. We want to find a way to relate **u** in the rest frame and the body frame. The velocity of **u** in the inertial frame is given by

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{inertial}} = \sum_{i=1}^{3} \frac{du_i^0}{dt}\mathbf{e}_i^0 \tag{6.14}$$

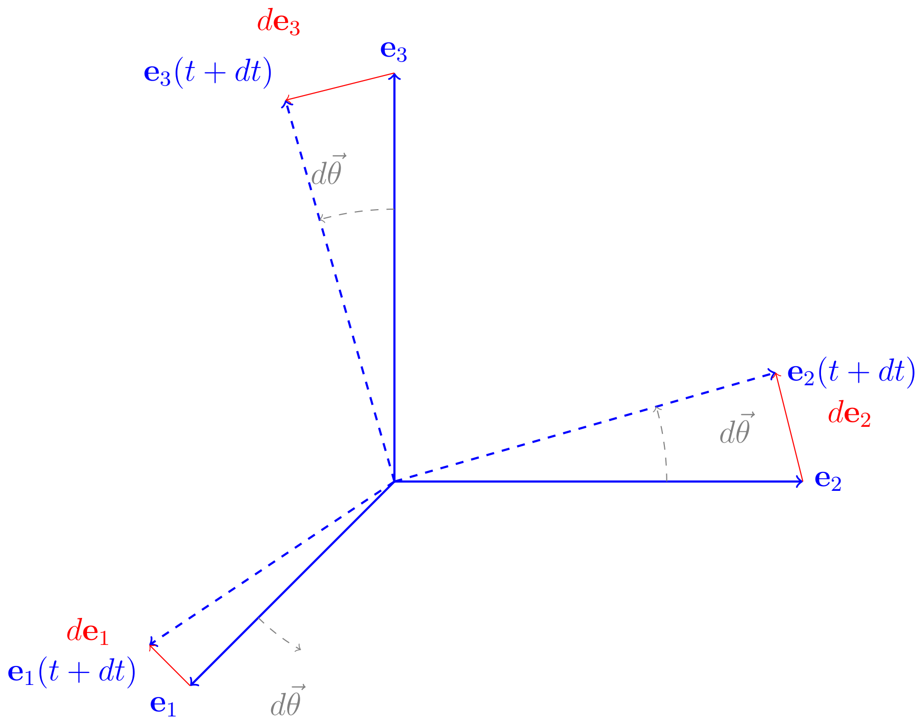
The product rule would give a second term $u_i^0 \frac{d\mathbf{e}_i^0}{dt}$ but since the inertial frame is fixed, the derivatives vanish. In the rotating frame,

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{inertial}} = \sum_{i=1}^{3} \left(\frac{du_i}{dt}\mathbf{e}_i + u_i\frac{d\mathbf{e}_i}{dt}\right) \tag{6.15}$$

Now we need to evaluate the relate $\dot{\mathbf{e}}_i$ to the rotation of the frame. Consider each basis vector $\mathbf{e}_i$ at time t and an infinitesimal time later $t + dt$.

$$\mathbf{e}_i(t + dt) = \mathbf{e}_i(t) + d\mathbf{e_i} \tag{6.16}$$

The infinitesimal change $d\mathbf{e}_i$ is perpendicular to $\mathbf{e}_i$. For an infinitesimal rotation in angle



$d\tilde{\theta}$, one can show geometrically that

$$d\mathbf{e}_i = d\vec{\theta} \times \mathbf{e}_i \tag{6.17}$$

We divide both sides by dt to get

$$\frac{d\mathbf{e}_i}{dt} = \frac{d\vec{\theta}}{dt} \times \mathbf{e}_i \tag{6.18}$$

We define the angular velocity vector $\vec{\omega}$ as

$$\vec{\omega} = \frac{d\vec{\theta}}{dt} \tag{6.19}$$

Then,

$$\frac{d\mathbf{e}_i}{dt} = \vec{\omega} \times \mathbf{e}_i \tag{6.20}$$

Then,

$$\sum_{i=1}^{3} u_i \frac{d\mathbf{e}_i}{dt} = \sum_{i=1}^{3} u_i(\vec{\omega} \times \mathbf{e}_i) \tag{6.21}$$

$$= \vec{\omega} \times \sum_{i=1}^{3} u_i \mathbf{e}_i \tag{6.22}$$

$$= \vec{\omega} \times \mathbf{u} \tag{6.23}$$

Thus, the full rotating frame derivative is

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{rotating}} = \sum_{i=1}^{3} \frac{du_i}{dt}\mathbf{e}_i + \vec{\omega} \times \mathbf{u} \tag{6.24}$$

The first sum is what an observer in the rotating frame would compute. They would only see the components u_i changing, unaware of the frame's rotation. Thus, we call this the body frame time derivative. Then,

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{rotating}} = \left(\frac{d\mathbf{u}}{dt}\right)_{\text{body}} + \vec{\omega} \times \mathbf{u} \tag{6.25}$$

Since $\mathbf{u}$ is the same vector expressed in either frame, we have

$$\left(\frac{d\mathbf{u}}{dt}\right)_{\text{inertial}} = \left(\frac{d\mathbf{u}}{dt}\right)_{\text{body}} + \vec{\omega} \times \mathbf{u} \tag{6.26}$$

We can define the inertial operator as

$$\left(\frac{d}{dt}\right)_{\text{inertial}} = \left(\frac{d}{dt}\right)_{\text{body}} + \vec{\omega}\times \tag{6.27}$$

We can now use this operator to get the velocity and acceleration in a non-inertial frame. Let $\mathbf{r}$ be a position vector in the inertial frame. Then, the velocity in the inertial frame is given by

$$\left(\frac{d\mathbf{r}}{dt}\right)_{\text{inertial}} = \left(\frac{d\mathbf{r}}{dt}\right)_{\text{body}} + \vec{\omega} \times \mathbf{r} \tag{6.28}$$

Thus,

$$\mathbf{v}_{\text{inertial}} = \mathbf{v}_{\text{body}} + \vec{\omega} \times \mathbf{r} \tag{6.29}$$

The acceleration in the inertial frame is then

$$\mathbf{a}_{\text{inertial}} = \left(\frac{d\mathbf{v}_{\text{inertial}}}{dt}\right)_{\text{inertial}} \tag{6.30}$$

Substituting $\mathbf{v}_{\text{inertial}} = \mathbf{v}_{\text{body}} + \vec{\omega} \times \mathbf{r}$,

$$\mathbf{a}_{\text{inertial}} = \left[\left(\frac{d}{dt}\right)_{\text{body}} + \vec{\omega}\times\right] (\mathbf{v}_{\text{body}} + \vec{\omega} \times \mathbf{r}) \tag{6.31}$$

Expanding the operator,

$$\mathbf{a}_{\text{inertial}} = \left(\frac{d}{dt}\right)_{\text{body}} \mathbf{v}_{\text{body}} + \left(\frac{d}{dt}\right)_{\text{body}} (\vec{\omega} \times \mathbf{r}) + \vec{\omega} \times \mathbf{v}_{\text{body}} + \vec{\omega} \times (\vec{\omega} \times \mathbf{r}) \tag{6.32}$$

Applying the product rule to the second term,

$$\left(\frac{d}{dt}\right)_{\text{body}} (\vec{\omega} \times \mathbf{r}) = \frac{d\vec{\omega}}{dt} \times \mathbf{r} + \vec{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_{\text{body}} \tag{6.33}$$

$$= \dot{\vec{\omega}} \times \mathbf{r} + \vec{\omega} \times \mathbf{v}_{\text{body}} \tag{6.34}$$

Substituting back,

$$\mathbf{a}_{\text{inertial}} = \mathbf{a}_{\text{body}} + \dot{\vec{\omega}} \times \mathbf{r} + \vec{\omega} \times \mathbf{v}_{\text{body}} + \vec{\omega} \times \mathbf{v}_{\text{body}} + \vec{\omega} \times (\vec{\omega} \times \mathbf{r}) \tag{6.35}$$

Collecting the two $\vec{\omega} \times \mathbf{v}_{\text{body}}$ terms,

$$\mathbf{a}_{\text{inertial}} = \mathbf{a}_{\text{body}} + \dot{\vec{\omega}} \times \mathbf{r} + 2\vec{\omega} \times \mathbf{v}_{\text{body}} + \vec{\omega} \times (\vec{\omega} \times \mathbf{r}) \tag{6.36}$$

Thus, N2L for a rotating reference frame is given by

$$m\mathbf{a}_{\text{inertial}} = m\mathbf{a}_{\text{body}} + m(\dot{\vec{\omega}} \times \mathbf{r}) + m(2\vec{\omega} \times \mathbf{v}_{\text{body}}) + m\left(\vec{\omega} \times (\vec{\omega} \times \mathbf{r})\right) \tag{6.37}$$

7 Lagrangian Mechanics

Up until now, we have worked with Newtonian mechanics where we have analysed systems by the forces acting on it, and applied N2L to obtain the equation of motion of the system. Lagrangian mechanics is a different formulation of classical mechanics where we now consider the “Lagrangian” of a system. Unlike Newtonian mechanics where we consider forces which are vectors, Lagrangian mechanics works with scalars which tend to make our calculations easier. However, we must first build up the mathematical framework behind the Lagrangian formulation, which is the Calculus of Variations.

7.1 Calculus of Variations

Opening Example: Snell’s Law and Fermat’s Principle

Consider two media with indices of refraction n_1 and n_2 The index of refraction gives the

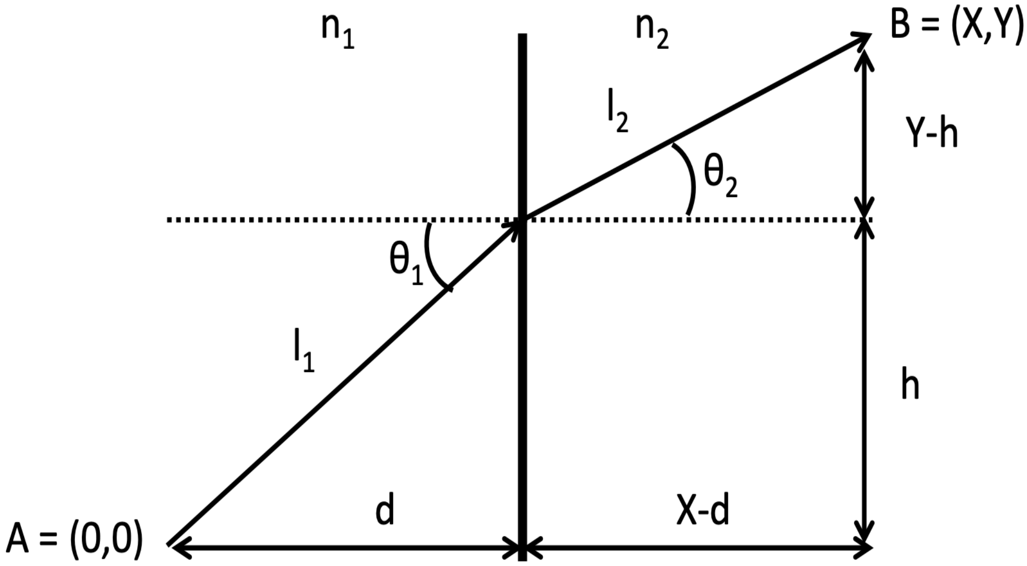


Figure 41: The path light takes

speed of light in that media, namely

$$v = \frac{c}{n} \tag{7.1}$$

Fermat's principle says that light takes the path that minimises the time it takes to get between two points. The time taken to go from point A to point B is

$$
\begin{aligned}
t &= \frac{l_1}{v_1} + \frac{l_2}{v_2} && (7.2)\\
&= \frac{l_1 n_1}{c} + \frac{l_2 n_2}{c} && (7.3)
\end{aligned}
$$

where

$$
\begin{aligned}
l_1 &= \sqrt{d^2 + h^2} && (7.4)\\
l_2 &= \sqrt{(X-d)^2 + (Y-h)^2} && (7.5)
\end{aligned}
$$

Then,

$$
t = \frac{n_1\sqrt{d^2+h^2}}{c} + \frac{n_2\sqrt{(X-d)^2+(Y-h)^2}}{c} \qquad (7.6)
$$

Since light takes the shortest time, we need to find the h that minimises t. Namely,

$$
\begin{aligned}
\frac{dt}{dh} &= \frac{n_1 h}{c\sqrt{d^2+h^2}} - \frac{n_2(Y-h)}{c\sqrt{(X-d)^2+(Y-h)^2}} && (7.7)\\
&= \frac{n_1 h}{c l_1} - \frac{n_2(Y-h)}{c l_2} && (7.8)\\
&= 0 && (7.9)
\end{aligned}
$$

Note that,

$$
\begin{aligned}
\sin\theta_1 &= \frac{h}{l_1} && (7.10)\\
\sin\theta_2 &= \frac{Y-h}{l_2} && (7.11)
\end{aligned}
$$

Thus,

$$
n_1 \sin\theta_1 = n_2 \sin\theta_2 \qquad (7.12)
$$

This is known as Snell's law. It is a relation between n_1, θ_1, n_2, and θ_2 that is obeyed when light takes the path between two points such that it minimises its time of travel.

7.1.1 Euler-Lagrange Equation

The main idea behind the Calculus of Variations is given a set of generalised coordinates $\{q_i(t)\} = \{q_1(t), \ldots, q_n(t)\}$ and generalised velocities $\{\dot{q}_i(t)\} = \{\dot{q}_1(t), \ldots, \dot{q}_n(t)\}$, we want to find a path labelled $q(t)$ such that the quantity S is extremised. $q_i(t)$ could be cartesian coordinates ($\{q_i(t)\} = \{x(t), y(t), z(t)\}$), spherical coordinates ($\{q_i(t)\} = \{r_i(t), \varphi_i(t), \theta_i(t)\}$)or coordinates in some other coordinate system. Given two fixed points $q(t_1), q(t_2)$, the quantity S is defined as

$$S[q(t)] = \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i, t)dt \tag{7.13}$$

where $\mathcal{L}$ is the "Lagrangian". S is a mathematical object known as a functional. It is a function, whose input is a function, and whose output is a number. Extremising S means finding a path q such that S is either maximised or minimised. Such a path q is called an extremal path. One property of the extremal path is that the action is invariant under small variations in our path δq. In other words, if we have an extremal path q, and we vary it slightly by δq, then our action does not change. Mathematically,

$$\delta q \implies \delta S = 0 \tag{7.14}$$

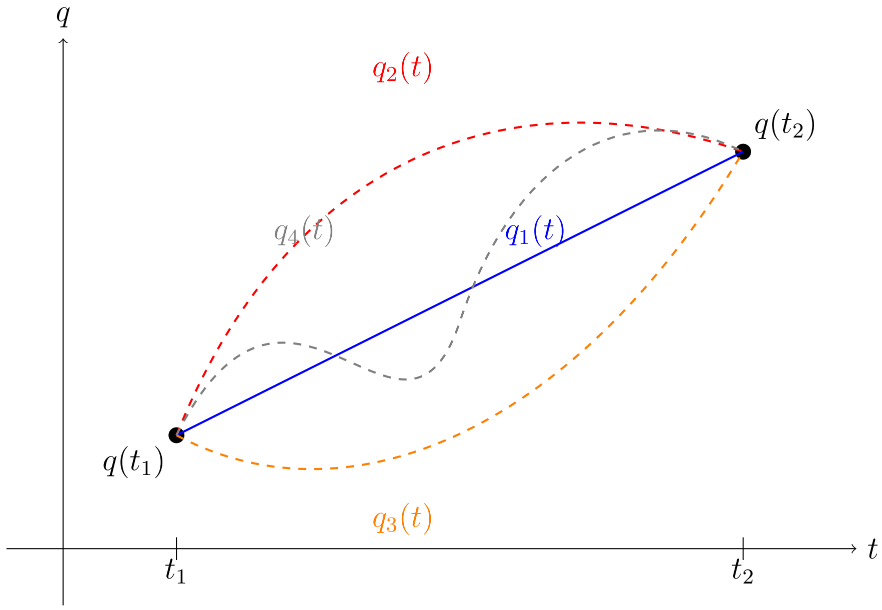
We can use this property to derive what is said to be the most beautiful equation in Classical Mechanics: the Euler-Lagrange equation.

Let $q(t)$ be an extremal path. We consider small variations in our extremal path δq, but we fix endpoints in our path at $q(t_1)$ and $q(t_2)$ such that there is no variation in our endpoints. In other words,

$$\delta q(t_1) = \delta q(t_2) = 0 \tag{7.15}$$

If q is our extremal path, we expect that $\delta S = 0$. To this end, we define the variational derivative of our action as

$$\delta S[q] = S[q + \delta q] - S[q] \tag{7.16}$$



We can then insert the definition of the action to see that

$$\delta S[q] = \int_{t_1}^{t_2} \mathcal{L}(q_i + \delta q_i, \dot{q}_i + \delta \dot{q}_i, t)dt - \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i, t)dt \tag{7.17}$$

Since each δq_i and $\delta \dot{q}_i$ is small as we only consider small variations in our extremal path, we can Taylor expand our Lagrangian to the first order to see that

$$\mathcal{L}(q_i + \delta q_i, \dot{q}_i + \delta \dot{q}_i, t) = \mathcal{L}(q_i, \dot{q}_i, t) + \sum_{i=1}^{n} \frac{\partial}{\partial q_i}\mathcal{L}(q_i, \dot{q}_i, t)\delta q_i + \sum_{i=1}^{n} \frac{\partial}{\partial \dot{q}_i}\mathcal{L}(q_i, \dot{q}_i, t)\delta \dot{q}_i + O(\delta q^2) \tag{7.18}$$

If we use the shorthand $\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i, t)$, then

$$\delta S[q] = \int_{t_1}^{t_2} \mathcal{L} + \sum_{i=1}^{n} \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \sum_{i=1}^{n} \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i dt - \int_{t_1}^{t_2} \mathcal{L} dt \tag{7.19}$$

$$= \int_{t_1}^{t_2} \sum_{i=1}^{n} \left[\frac{\partial \mathcal{L}}{\partial q_i}\delta q_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i \right] dt \tag{7.20}$$

We can use integration by parts on the second term where $u = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$ and $dv = \delta \dot{q}_i$. Note that $\delta \dot{q}_i = \frac{d}{dt}(\delta q_i)$. Thus,

$$\int_{t_1}^{t_2} \sum_{i=1}^{n} \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta \dot{q}_i dt = \cancelto{0}{\sum_{i=1}^{n} \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta q_i \right]_{t_1}^{t_2}} - \int_{t_1}^{t_2} \sum_{i=1}^{n} \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \delta q_i dt \tag{7.21}$$

where the first term vanishes as there are no variations in our endpoints $\delta q(t_1) = \delta q(t_2) = 0$. Thus,

$$\delta S[q] = \int_{t_1}^{t_2} \sum_{i=1}^{n} \left[\frac{\partial \mathcal{L}}{\partial q_i} \delta q_i - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \delta q_i \right] dt \tag{7.22}$$

$$= \int_{t_1}^{t_2} \sum_{i=1}^{n} \left[\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \right] \delta q_i dt \tag{7.23}$$

Now, since q is an extremal path, we must have $\delta S[q] = 0$. Thus,

$$\int_{t_1}^{t_2} \sum_{i=1}^{n} \left[\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \right] \delta q_i dt = 0 \tag{7.24}$$

Since we are considering non-zero variations in our extremal path i.e. $\delta q_i \neq 0$, the sum must be 0. Further, since we consider each generalised coordinate to be independent of each other, we must have

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = 0 \tag{7.25}$$

This is the Euler-Lagrange equation. The Euler-Lagrange equation is the "$\mathbf{F} = m\ddot{\mathbf{r}}$" for the Lagrangian formulation of classical mechanics. It can be used to obtain the equation of motion for different systems. It can also be used to find a path that extremises an action S.

Example: Shortest path between two points in $\mathbb{R}^2$

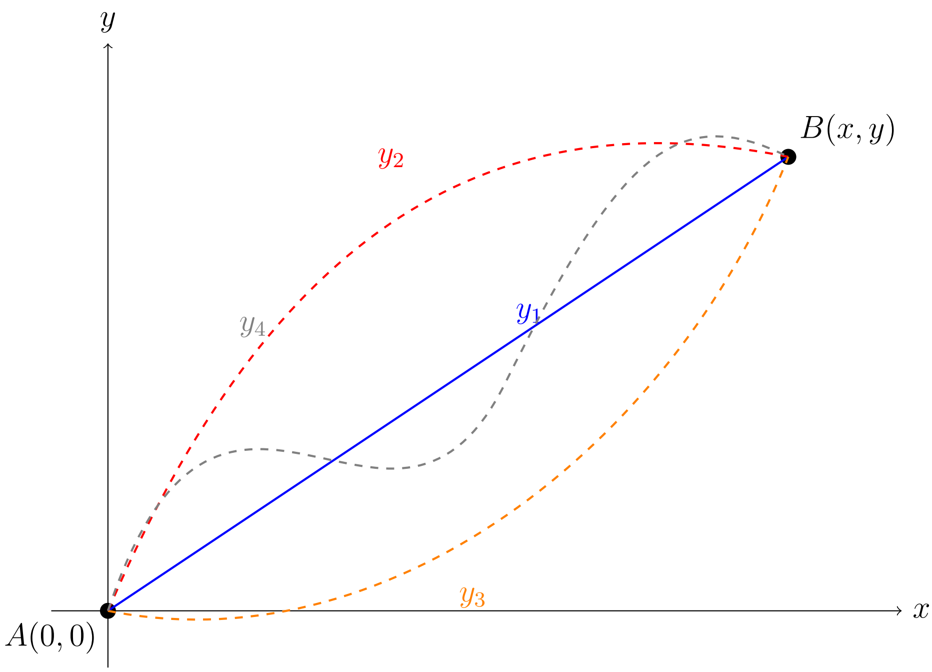
Consider two fixed points $A(0, 0)$ and $B(x, y)$ in $\mathbb{R}^2$. We want to find the path between A and B that minimises the length of the path.

The arclength of any given path between A and B parameterised by $y = y(x)$ is given by

$$L = \int_A^B \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \tag{7.26}$$

Here, our action is the arclength L. If we write out the definition of the action, we have

$$\int_A^B \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx = \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i, t) dt \tag{7.27}$$



Thus, we can map our coordinates as

$$q \to y \tag{7.28}$$

$$\dot{q} \to \frac{dy}{dx} = y' \tag{7.29}$$

$$t \to x \tag{7.30}$$

Thus, our Lagrangian can be re-written as

$$\mathcal{L}(y, y', x) = \sqrt{1 + y'^2} \tag{7.31}$$

and our Euler-Lagrange equation is

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial y'}\right) = 0 \tag{7.32}$$

7.2 Principle of Least Action and the Lagrangian

We now want to be able to describe dynamical systems using our new formulation of classical mechanics. In Newtonian mechanics, we sum all the forces acting on a particle and set it equal to $\mathbf{F} = m\ddot{\mathbf{r}}$ and solve for the one and only path of the particle. In

Lagrangian mechanics, we consider all possible paths that a particle can take. A valid path must be continuous and differentiable everywhere and must have beginning and endpoints. The actual path a particle takes is given by the principle of least action.

Theorem 7.1 (Principle of Least Action)**.** Consider a path described by generalised coordinates $\{q_i\}$. The path taken by a particle between two fixed points $q(t_1)$ and $q(t_2)$ is such that it minimises the action

$$S[q(t)] = \int_{t_1}^{t_2} \mathcal{L}(q_t, \dot{q}_i, t)dt \tag{7.33}$$

The path that minimises the action satisfies the Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = 0 \tag{7.34}$$

The Lagrangian in Classical Mechanics was derived to be the difference in a particle's kinetic and potential energy

$$\mathcal{L} = T - U \tag{7.35}$$

This is a generalised form of the Lagrangian, namely it takes the same form in every conserved system (systems where we have an associated potential energy). We can recover N2L from the Lagrangian.

Consider a particle of mass m moving in a potential $U(\mathbf{r})$. We want to find the equation of motion of the particle using Newtonian and Lagrangian mechanics. We have that the acceleration of the system is given by

$$\mathbf{a} = \ddot{\mathbf{r}} \tag{7.36}$$

Further, the force acting on the particle is given by

$$\mathbf{F} = -\nabla U(\mathbf{r}) \tag{7.37}$$

Then by N2L,

$$m\ddot{\mathbf{r}} = -\nabla U(\mathbf{r}) \tag{7.38}$$

Now using the Lagrangian,

$$\begin{align} \mathcal{L} &= T - U \tag{7.39} \\ &= \frac{1}{2}m|\dot{\mathbf{r}}|^2 - U(\mathbf{r}) \tag{7.40} \end{align}$$

The Euler-Lagrange equation is given by

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) = 0 \tag{7.41}$$

From each term

$$\begin{align} \frac{\partial \mathcal{L}}{\partial \mathbf{r}} &= -\frac{\partial U}{\partial \mathbf{r}} \tag{7.42} \\ &= -\nabla U(\mathbf{r}) \tag{7.43} \\ \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} &= m\dot{\mathbf{r}} \tag{7.44} \\ \implies \frac{d}{dt}(m\dot{\mathbf{r}}) &= m\ddot{\mathbf{r}} \tag{7.45} \end{align}$$

Thus,

$$\begin{align} -\nabla U(\mathbf{r}) - m\ddot{\mathbf{r}} = 0 \tag{7.46} \\ \implies m\ddot{\mathbf{r}} = -\nabla U(\mathbf{r}) \tag{7.47} \end{align}$$

Thus, we get the exact same equation of motion using the Lagrangian as we would if we used the Newtonian formulation. The Lagrangian formulation is said to be a more generalised approach to Newtonian mechanics. Notice how

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} = -\frac{\partial U}{\partial \mathbf{r}} = -\nabla U(\mathbf{r}) \tag{7.48}$$

looks like a force. Further, notice how

$$\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} = m\dot{\mathbf{r}} \tag{7.49}$$

looks like momentum ($\mathbf{p} = m\dot{\mathbf{r}}$). We can extend this. Generally, $\frac{\partial \mathcal{L}}{\partial q}$ will generally not have dimensions of force. Yet, it plays a similar role to a force in most systems. Thus, we call it the generalised force. In particular, the i^{th} component of the generalised force is defined as

$$Q_i = \frac{\partial \mathcal{L}}{\partial q_i} \tag{7.50}$$

Further, $\frac{\partial \mathcal{L}}{\partial \dot{q}_i}$ will generally not have dimensions of momentum. Yet, it plays a similar role to a momentum in most systems. Thus, we call it the generalised momentum. In particular, the i^{th} component of of the generalised momentum is defined as

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \tag{7.51}$$

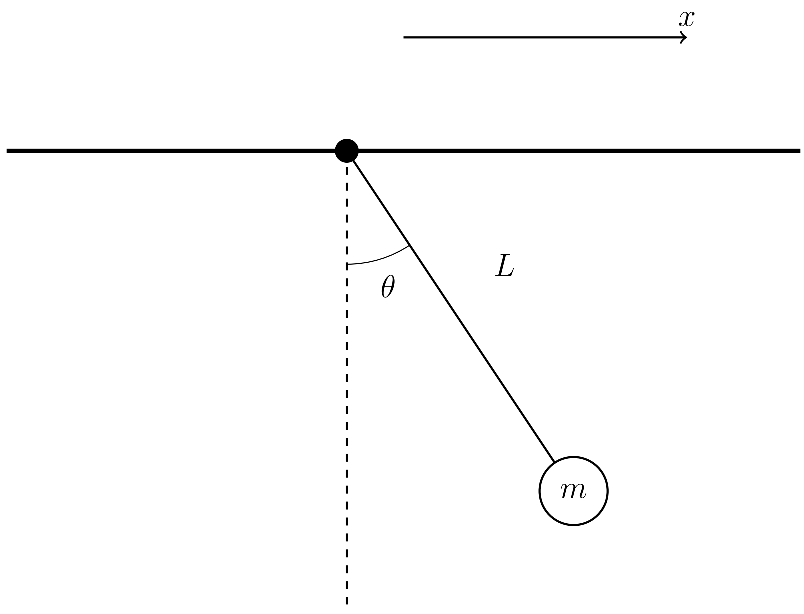
Then, the Euler Lagrange equation becomes

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \frac{dp_i}{dt} \tag{7.52}$$

Thus, the generalised force is equal to the rate of change of generalised momentum. This is a more *general* version of Newton's second law.

Pendulum with translational motion

Consider a simple pendulum of length L with a bob of mass m whose pivot is moving to the right.



The generalised coordinates and generalised velocities are given by

$$\{q_i\} = \{q_1, q_2\} = \{x, \theta\} \tag{7.53}$$

$$\{\dot{q}_i\} = \{\dot{q}_1, \dot{q}_2\} = \{\dot{x}, \dot{\theta}\} \tag{7.54}$$

Thus,

$$\mathcal{L} = T - U \tag{7.55}$$

$$= \frac{1}{2}m(\mathbf{v} \cdot \mathbf{v}) - U \tag{7.56}$$

$$= \frac{1}{2}m\left(\dot{X}^2 + \dot{Y}^2\right) - U \tag{7.57}$$

where

$$X = L\sin\theta + x \tag{7.58}$$

$$\implies \dot{X} = L\dot{\theta}\cos\theta + \dot{x} \tag{7.59}$$

$$Y = L(1 - \cos\theta) \tag{7.60}$$

$$\implies \dot{Y} = \dot{\theta}L\sin\theta \tag{7.61}$$

Thus,

$$T = \frac{1}{2}m\left(\left(L\dot{\theta}\cos\theta + \dot{x}\right)^2 + \left(\dot{\theta}L\sin\theta\right)^2\right) \tag{7.62}$$

$$= \frac{1}{2}m\left(\dot{x}^2 + 2\dot{x}\dot{\theta}L\cos\theta + \dot{\theta}^2L^2\right) \tag{7.63}$$

The potential energy is given by

$$U = mgL(1 - \cos\theta) = mgL - mgL\cos\theta \tag{7.64}$$

The Lagrangian is then given by

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 + m\dot{x}\dot{\theta}L\cos\theta + \frac{1}{2}m\dot{\theta}^2L^2 + mgL - mgL\cos\theta \tag{7.65}$$

Since we have two generalised coordinates, we have two Euler-Lagrange equations. These are

$$\frac{\partial\mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}}\right) = 0 \tag{7.66}$$

$$\frac{\partial\mathcal{L}}{\partial\theta} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = 0 \tag{7.67}$$

We have

$$\frac{\partial\mathcal{L}}{\partial x} = 0 \tag{7.68}$$

$$\frac{\partial\mathcal{L}}{\partial\dot{x}} = m\dot{x} + m\dot{\theta}L\cos\theta \tag{7.69}$$

$$\implies \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}}\right) = m\ddot{x} + m\ddot{\theta}L\cos\theta - m\dot{\theta}^2L\sin\theta \tag{7.70}$$

$$\frac{\partial\mathcal{L}}{\partial\theta} = -m\dot{x}\dot{\theta}L\sin\theta + mgL\sin\theta \tag{7.71}$$

$$\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = m\dot{x}L\cos\theta + m\dot{\theta}L^2 \tag{7.72}$$

$$\implies \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = m\ddot{x}L\cos\theta - m\dot{x}L\dot{\theta}\sin\theta + m\ddot{\theta}L^2 \tag{7.73}$$

Thus,

$$m\ddot{x} + mL\cos(\theta)\ddot{\theta} - mL\sin(\theta)\dot{\theta}^2 = 0 \tag{7.74}$$

$$-m\dot{x}\dot{\theta}L\sin\theta + mgL\sin\theta - m\ddot{x}L\cos\theta + m\dot{x}L\dot{\theta}\sin\theta - m\ddot{\theta}L^2 = 0 \tag{7.75}$$

Particle Motion in Polar Coordinates

We now want to find the Lagrangian and Euler-Lagrange equations that describe the motion of a particle of mass m in a potential $U(r,\varphi)$ in polar coordinates. The generalised coordinates and generalised velocities are

$$\{q_i\} = \{q_1, q_2\} = \{r, \varphi\} \tag{7.76}$$

$$\{\dot{q}_i\} = \{\dot{q}_1, \dot{q}_2\} = \{\dot{r}, \dot{\varphi}\} \tag{7.77}$$

The velocity in polar coordinates is given by,

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi \tag{7.78}$$

The Lagrangian is then given by

$$\mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - U(r,\varphi) \tag{7.79}$$

The Euler-Lagrange equations are

$$\frac{\partial \mathcal{L}}{\partial r} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = 0 \tag{7.80}$$

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\varphi}}\right) = 0 \tag{7.81}$$

Then,

$$\frac{\partial \mathcal{L}}{\partial r} = mr\dot{\varphi}^2 - \frac{\partial U}{\partial r} \tag{7.82}$$

$-\frac{\partial U}{\partial r}$ is the component of the force in the radian direction F_r. Thus,

$$\frac{\partial \mathcal{L}}{\partial r} = mr\dot{\varphi}^2 + F_r \tag{7.83}$$

Further,

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r} \tag{7.84}$$

$$\implies \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = m\ddot{r} \tag{7.85}$$

Then,

$$mr\dot{\varphi}^2 + F_r - m\ddot{r} = 0 \tag{7.86}$$

$$\implies F_r = m(\ddot{r} - r\dot{\varphi}^2) \tag{7.87}$$

$$= ma_r \tag{7.88}$$

Thus, the radial component of the acceleration vector in spherical coordinates falls out naturally. We also have,

$$\frac{\partial \mathcal{L}}{\partial \varphi} = -\frac{\partial U}{\partial \varphi} \tag{7.89}$$

$$= rF_\varphi \tag{7.90}$$

The second equality comes from the fact that

$$\mathbf{F} = -\nabla U \tag{7.91}$$

$$= -\frac{\partial U}{\partial r}\mathbf{e}_r - \frac{1}{r}\frac{\partial U}{\partial \varphi}\mathbf{e}_\varphi \tag{7.92}$$

$$\implies -\frac{\partial U}{\partial \varphi} = rF_\varphi \tag{7.93}$$

Note that this looks like a torque $\vec{\tau} = \mathbf{r} \times \mathbf{F}$. Further,

$$\frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = mr^2\dot{\varphi} \tag{7.94}$$

Note that this looks like an angular momentum. Thus,

$$rF_\varphi = \frac{d}{dt}\left(mr^2\dot{\varphi}\right) \tag{7.95}$$

This gives the result that

$$\vec{\tau} = \frac{d\mathbf{L}}{dt} \tag{7.96}$$

where our generalised force is now a torque and our generalised momentum is angular momentum.

Now imagine we had a potential that depends only on r, i.e. our potential is of the form $U(r)$. Then, the force is

$$\mathbf{F} == -\nabla U(r) = F_r\mathbf{e}_r \tag{7.97}$$

Notice that we have a central force. Further, since $F_\varphi = 0$, we have $rF_\varphi = 0$. Then,

$$\frac{d}{dt}\left(mr^2\dot{\varphi}\right) = 0 \tag{7.98}$$

Notice that this is exactly the conservation of angular momentum we saw in the section on central forces! Many expressions and conservation laws fall cleanly out of the Lagrangian. If U did not depend on φ, our Lagrangian wouldn't either. We can then express our Lagrangian as $\mathcal{L} = \mathcal{L}(r, \dot{r}, \dot{\varphi}, t)$. Our Lagrangian being independent of φ gave us conservation of angular momentum. We can generalise this idea.

In general, if the Lagrangian is independent of a coordinate (i.e. doesn't depend on q_i), then the generalised force vanishes as $\frac{\partial\mathcal{L}}{\partial q_i} = 0$ and the generalised momentum is conserved as

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_i}\right) = 0 \tag{7.99}$$

Thus, we say that the generalised momentum is a constant of motion. If $\mathcal{L}$ is independent of q_i, we say that $\mathcal{L}$ is invariant under variations of q_i and we say q_i is a cyclic coordinate. We will soon see in Noether's theorem that a cyclic coordinate implies a symmetry, which in turn implies a symmetry law. Namely,

$$\text{Invariance/Cyclicity} \iff \text{Symmetry} \iff \text{Conservation Law} \tag{7.100}$$

In the meantime, we return to a previous worked example from section 3.

Example Revisited: Bead on a Spinning Hoop

We return to this example to see how the Lagrangian greatly simplifies finding the equation of motion of the system.

Consider a hoop of fixed radius R that spins around its vertical axis at a fixed rate Ω. A bead of mass m is free to move along the hoop with no friction. The setup is in a gravitational field.

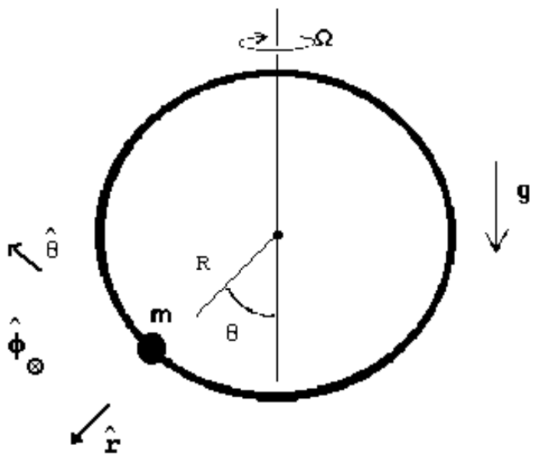


Figure 42: Spinning hoop

The generalised coordinates and generalised velocities are given by

$$\{q_i\} = \{q_1, q_2, q_3\} = \{R, \varphi, \theta\} \tag{7.101}$$

$$\{\dot{q}_i\} = \{\dot{q}_1, \dot{q}_2, \dot{q}_3\} = \{0, \Omega, \dot{\theta}\} \tag{7.102}$$

Thus,

$$\mathbf{v} = R\dot{\theta}\mathbf{e}_\theta + R\Omega\sin\theta\mathbf{e}_\varphi \tag{7.103}$$

$$\mathbf{v}\cdot\mathbf{v} = R^2\dot{\theta}^2 + R^2\Omega^2\sin^2\theta \tag{7.104}$$

The Lagrangian is then given by,

$$\mathcal{L} = \frac{1}{2}m\left(R^2\dot{\theta}^2 + R^2\Omega^2\sin^2\theta\right) - mgR(1-\cos\varphi) \tag{7.105}$$

$$= \frac{1}{2}mR^2\dot{\theta}^2 + \frac{1}{2}mR^2\Omega^2\sin^2\theta - mgR + mgR\cos\varphi \tag{7.106}$$

The Euler-Lagrange equation in r gives,

$$\frac{\partial\mathcal{L}}{\partial r} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{r}}\right) = 0 \tag{7.107}$$

$$\implies mR\Omega^2\sin^2\theta - mg + mg\cos\varphi = 0 \tag{7.108}$$

The Euler-Lagrange equation in φ gives,

$$\frac{\partial\mathcal{L}}{\partial\varphi} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\varphi}}\right) = 0 \tag{7.109}$$

The Euler-Lagrange equation in θ gives,

$$\frac{\partial\mathcal{L}}{\partial\theta} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = 0 \tag{7.110}$$

$$\implies \left(mR^2\Omega^2\sin\theta\cos\theta - mgR\sin\theta\right) - \frac{d}{dt}\left(mR^2\dot{\theta}\right) = 0 \tag{7.111}$$

$$\implies mR^2\Omega^2\sin\theta\cos\theta - mgR\sin\theta = mR^2\ddot{\theta} \tag{7.112}$$

$$\implies \ddot{\theta} = \Omega^2\sin\theta\cos\theta - \frac{g}{R}\sin\theta \tag{7.113}$$

This is exactly what we derived using N2L but in a much simpler method.

In this problem, we had a constraint: the particle could only move along the hoop and so the particle was always constrained such that $r = R$. We can formally introduce constraints into the Lagrangian using Lagrange multipliers.

7.3 Constraints and Lagrange Multipliers

7.3.1 Global Constraints

The aim of using Lagrange multipliers is to extremise a functional S_1 while keeping another function S_2 (which represents our constraint) fixed. That is, we want to find a path q such that

$$S_1[q(t)] = \int_{t_1}^{t_2} \mathcal{L}_1(q_i, \dot{q}_i, t)dt \tag{7.114}$$

gives $\delta S_1 = 0$ while

$$S_2[q(t)] = \int_{t_1}^{t_2} \mathcal{L}_2(q_i, \dot{q}_i, t)dt = \text{constant} \tag{7.115}$$

We achieve this by using a Lagrange multiplier λ and constructing a compound functional C given by

$$C = S_1[q(t)] - \lambda S_2[q(t)] \tag{7.116}$$

The variation of C is given by

$$\delta C = \delta S_1 - \lambda \delta S_2 \tag{7.117}$$

Since we want to extremise S_1, we must have $\delta S_1 = 0$. Further, since S_2 is fixed, then there cannot be any variation in S_2. Thus, $\delta S_2 = 0$. Then, it follows that

$$\delta C = 0 \tag{7.118}$$

Thus, we can write down an Euler-Lagrange equation for the associated compound functional,

$$\frac{\partial \mathcal{L}_C}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}_C}{\partial \dot{q}_i}\right) = 0 \tag{7.119}$$

where $\mathcal{L}_C$ is the compound Lagrangian. From the definition of C,

$$C[q(t)] = S_1[q(t)] - \lambda S_2[q(t)] \tag{7.120}$$

$$= \int_{t_1}^{t_2} \mathcal{L}_1(q_i, \dot{q}_i, t) - \lambda \mathcal{L}_2(q_i, \dot{q}_i, t) dt \tag{7.121}$$

Thus, the compound Lagrangian is given by,

$$\mathcal{L}_C = \mathcal{L}_1 - \lambda \mathcal{L}_2 \tag{7.122}$$

7.3.2 Local Constraints

Now consider a functional of two or more functions where the two functions are related by a constraint. Taking the example of two functions, the functional takes the form of

$$S[x(t), y(t)] = \int_{t_1}^{t_2} \mathcal{L}(x(t), \dot{x}(t), y(t), \dot{y}(t), t) dt \tag{7.123}$$

and the constraint is of the form,

$$f(x(t), y(t), t) = 0 \tag{7.124}$$

The key distinction from the global case is that this constraint must hold at *every* instant t, not just as an integrated quantity. We therefore introduce a Lagrange multiplier $\lambda(t)$ which is itself a function of time, rather than a single constant. The compound functional is constructed as

$$C[x(t), y(t)] = S[x(t), y(t)] - \int_{t_1}^{t_2} \lambda(t) f(x(t), y(t), t)\, dt \tag{7.125}$$

which can be combined into a single integral,

$$C[x(t), y(t)] = \int_{t_1}^{t_2} \mathcal{L}(x, \dot{x}, y, \dot{y}, t) - \lambda(t) f(x, y, t) dt \tag{7.126}$$

The compound Lagrangian is therefore

$$\mathcal{L}_C = \mathcal{L} - \lambda(t) f(x, y, t) \tag{7.127}$$

Since the constraint $f = 0$ holds along the true path, the second term vanishes on that path and C agrees with S. Extremising C with respect to independent variations of $x(t)$ and $y(t)$ gives

$$\delta C = \int_{t_1}^{t_2} \left(\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \lambda \frac{\partial f}{\partial x} \right) \delta x + \left(\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} - \lambda \frac{\partial f}{\partial y} \right) \delta y dt = 0 \tag{7.128}$$

Because $x(t)$ and $y(t)$ are no longer independent (they are linked through $f = 0$), we cannot immediately set each bracket to zero. Instead, we choose $\lambda(t)$ to make one of the brackets vanish identically, after which the remaining variation becomes free, and that bracket must also vanish. The result is a system of Euler-Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \lambda(t) \frac{\partial f}{\partial x} \tag{7.129}$$

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = \lambda(t) \frac{\partial f}{\partial y} \tag{7.130}$$

together with the constraint equation itself,

$$f(x(t), y(t), t) = 0 \tag{7.131}$$

This gives three equations for three unknowns: $x(t)$, $y(t)$, and $\lambda(t)$. The right-hand sides $\lambda \partial f / \partial q_i$ are interpreted physically as the generalised constraint forces i.e. the forces that the constraint exerts on the system to keep it on the surface $f = 0$. This is a major advantage over simply eliminating one coordinate using $f = 0$: that approach removes the constraint from the problem entirely, but the Lagrange multiplier method retains $\lambda(t)$ and thus gives direct access to the constraint forces.

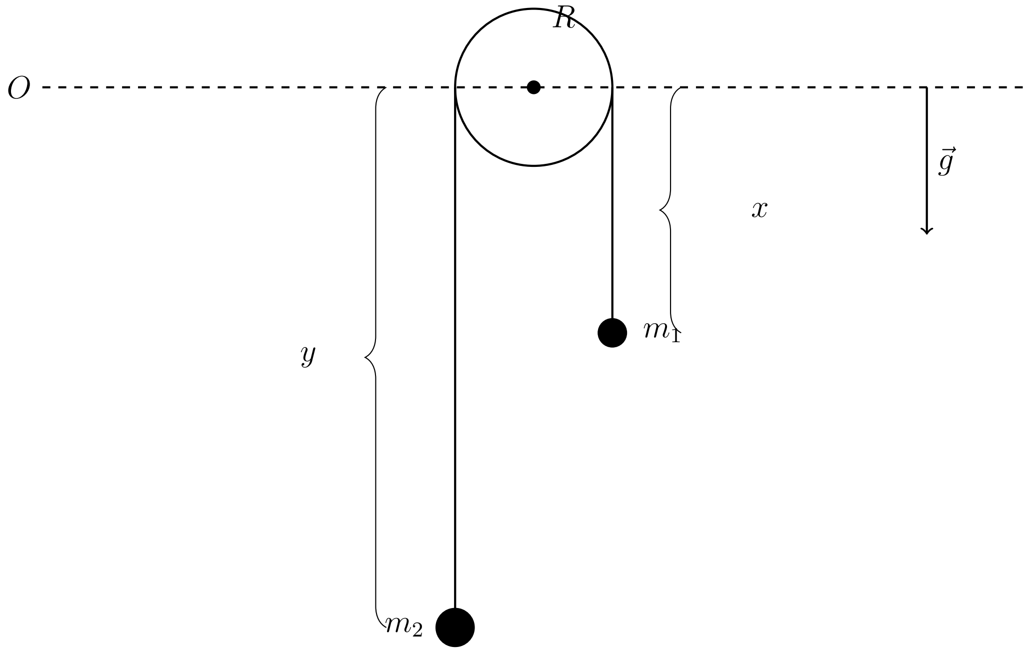
The generalisation to n coordinates $q_1, \ldots, q_n$ subject to k constraints $f_\alpha(q_i, t) = 0$ for $\alpha = 1, \ldots, k$ is straightforward. One Lagrange multiplier $\lambda_\alpha(t)$ is introduced for each constraint, and the Euler-Lagrange equations become

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \sum_{\alpha=1}^{k} \lambda_\alpha(t) \frac{\partial f_\alpha}{\partial q_i} \tag{7.132}$$

supplemented by the k constraint equations $f_\alpha = 0$.

Example: Atwood Machine

Consider an Atwood machine (in a gravitational field) created with two masses: m_1 that is suspended a distance x from the center of a frictionless pulley and a mass m_2 suspended a distance y from the center of the pulley In this problem, the length of rope is fixed and



is given by

$$l = x + y + \pi R \tag{7.133}$$

where the πR term is the portion of the rope over the pulley. The generalised coordinates and velocities are given by

$$\{q_i\} = \{q_1, q_2\} = \{x, y\} \tag{7.134}$$

$$\{\dot{q}_i\} = \{\dot{q}_1, \dot{q}_2\} = \{\dot{x}, \dot{y}\} \tag{7.135}$$

Further,

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 \tag{7.136}$$

$$U = -m_1 gx - m_2 gy \tag{7.137}$$

The Lagrangian of our system is then given by

$$\mathcal{L} = T - U \tag{7.138}$$

$$= \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{y}^2 + m_1 gx + m_2 gy \tag{7.139}$$

Since the length of the rope is fixed, we have a local constraint given by

$$f(x, y) = l - (x + y + \pi R) = 0 \tag{7.140}$$

The compound Lagrangian is then,

$$\mathcal{L}_C = \mathcal{L} - \lambda f \tag{7.141}$$

$$= \mathcal{L} - \lambda(l - (x + y + \pi R)) \tag{7.142}$$

$$= \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{y}^2 + m_1 gx + m_2 gy - \lambda l + \lambda x + \lambda y + \lambda\pi R \tag{7.143}$$

There are two associated Euler-Lagrange equations, one for each generalised coordinate

$$\frac{\partial\mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}}\right) = 0 \tag{7.144}$$

$$\frac{\partial\mathcal{L}}{\partial y} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{y}}\right) = 0 \tag{7.145}$$

Thus,

$$m_1 g + \lambda - m_1\ddot{x} = 0 \tag{7.146}$$

$$m_2 g + \lambda - m_2\ddot{y} = 0 \tag{7.147}$$

Thus,

$$m_1\ddot{x} = m_1 g + \lambda \tag{7.148}$$

$$m_2\ddot{y} = m_2 g + \lambda \tag{7.149}$$

We can subtract the first equation from the second to get rid of λ,

$$m_1\ddot{x} - m_2\ddot{y} = (m_1 - m_2)g \tag{7.150}$$

We can now use the constraint to find a relationship between $\ddot{x}$ and $\ddot{y}$. Recall that,

$$l - (x + y + \pi R) = 0 \tag{7.151}$$

Taking second time derivatives on both sides give,

$$-\ddot{x} - \ddot{y} = 0 \tag{7.152}$$

$$\ddot{x} = -\ddot{y} \tag{7.153}$$

Substituting gives,

$$\ddot{x} = \frac{(m_1 - m_2)g}{m_1 + m_2} \tag{7.154}$$

$$\ddot{y} = \frac{(m_2 - m_1)g}{m_1 + m_2} \tag{7.155}$$

These are the exact equations we saw in PHYS 211. To interpret the Lagrangian multiplier λ, we see that

$$m_1\ddot{x} = m_1 g + \lambda \tag{7.156}$$

$$m_2\ddot{y} = m_2 g + \lambda \tag{7.157}$$

By N2L,

$$m_1\ddot{x} = F_g + F_T \tag{7.158}$$

$$m_2\ddot{y} = F_g + F_T \tag{7.159}$$

Thus, the lagrange multiplier is related to the tension force for this problem.

7.4 Noether's Theorem and Conservation Laws

We have seen that certain symmetries in our problem give us conserved quantities. For example, we were able to rotate the plane by $\theta = \frac{\pi}{2}$ in a central force field and use conservation of angular momentum. We also saw that cyclic coordinates give us conservation of generalised momentum. We can formally ask when we get conservation laws and how they are related to symmetries. We will see that there is a connection between

$$\text{Invariance of the Lagrangian under Cyclic Coordinates} \iff \text{Symmetries} \iff \text{Conservation Laws} \tag{7.160}$$

First, we need to define conservation. We say a function $f(q, \dot{q}, t)$ is conserved if

$$\frac{d}{dt}\left(f(q, \dot{q}, t)\right) = 0 \tag{7.161}$$

We can expand this using partial derivatives to see that

$$\frac{d}{dt}f(q, \dot{q}, t) = \sum_i \left[\frac{\partial f}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial f}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\right] + \frac{\partial f}{\partial t} \tag{7.162}$$

We will apply this to the Lagrangian to see when our Lagrangian implies conservation of energy.

7.4.1 Conservation of Energy from the Lagrangian

We apply the expansion above to the Lagrangian $\mathcal{L} = \mathcal{L}(q, \dot{q}, t)$ to see,

$$\begin{aligned} \frac{d\mathcal{L}}{dt} &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} \right] + \frac{\partial \mathcal{L}}{\partial t} && (7.163) \\ &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} \right] + \frac{\partial \mathcal{L}}{\partial t} && (7.164) \end{aligned}$$

We can substitute the Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \tag{7.165}$$

Thus,

$$\begin{aligned} \frac{d\mathcal{L}}{dt} &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} \right] + \frac{\partial \mathcal{L}}{\partial t} && (7.166) \\ &= \sum_i \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} \right] + \frac{\partial \mathcal{L}}{\partial t} && (7.167) \\ & && (7.168) \end{aligned}$$

We can recognise the summand as the total derivative $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i \right)$. Thus,

$$\frac{d\mathcal{L}}{dt} = \sum_i \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i \right) \right] + \frac{\partial \mathcal{L}}{\partial t} \tag{7.169}$$

Since we have at most a finite number of generalised coordinates, we can pull the time derivative outside the sum

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt} \left(\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i \right) + \frac{\partial \mathcal{L}}{\partial t} \tag{7.170}$$

Recall that the i$^{\text{th}}$ component of the generalised momentum is $p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$. Then,

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt} \left(\sum_i p_i \dot{q}_i \right) + \frac{\partial \mathcal{L}}{\partial t} \tag{7.171}$$

We can move the time derivative of the Lagrangian of the other side to get

$$\frac{d}{dt}\left(\sum_i p_i \dot{q}_i - \mathcal{L}\right) + \frac{\partial \mathcal{L}}{\partial t} = 0 \tag{7.172}$$

We define the quantity,

$$\mathcal{H} = \sum_i p_i \dot{q}_i - \mathcal{L} \tag{7.173}$$

Then,

$$\frac{d\mathcal{H}}{dt} + \frac{\partial \mathcal{L}}{\partial t} = 0 \tag{7.174}$$

Thus if $\frac{\partial \mathcal{L}}{\partial t} = 0$, then $\mathcal{H}$ is a conserved quantity. We call $\mathcal{H}$ the Hamiltonian. Further,

$$\mathcal{H} = \sum_i p_i \dot{q}_i \tag{7.175}$$

$$= \sum_i m\dot{q}_i \cdot \dot{q}_i \tag{7.176}$$

$$= \sum_i m\dot{q}_i^2 \tag{7.177}$$

$$= 2T \tag{7.178}$$

Thus,

$$\mathcal{H} = 2T - \mathcal{L} \tag{7.179}$$

$$= 2T - (T - U) \tag{7.180}$$

$$= T + U \tag{7.181}$$

$$= E_{tot} \tag{7.182}$$

Thus, if $\frac{\partial}{\partial t} = 0$ we have

$$\frac{d\mathcal{H}}{dt} = \frac{dE_{tot}}{dt} = 0 \tag{7.183}$$

This result is profound. If the Lagrangian does not explicitly depend on time, the Hamiltonian is equal to the total energy, and the total energy is conserved.

$$\mathcal{H} = T + U = E_{tot} = \text{constant} \tag{7.184}$$

7.4.2 Noether's Theorem

Noether's Theorem was written down by Emmy Noether, a German mathematician in the late 1800s. This is widely considered as one of the most important results in Physics.

We start by considering a coordinate transformation

$$q_i(t) \to \bar{q}_i(t, \varepsilon) \tag{7.185}$$

where ε is a continuous parameter such that

$$\bar{q}_i(t, \varepsilon = 0) = q_i(t) \tag{7.186}$$

For example, if we rotate a coordinate system by a small angle, then ε is the small angle. Then, we can state Noether's theorem

Theorem 7.2 (Noether's Theorem)**.** If a coordinate transform $q_i(t) \to \bar{q}_i(t, \varepsilon)$ is a continuous symmetry, then the quantity

$$Q = \sum_i \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{\partial \bar{q}_i}{\partial \varepsilon} \right) |_{\varepsilon=0} = \text{constant} \tag{7.187}$$

is conserved.

Remark. A continuous symmetry means that the action

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i, t) dt \tag{7.188}$$

is invariant under the transformation $q_i(t) \to \bar{q}_i(t, \varepsilon)$. That is,

$$S[q(t)] = S[\bar{q}(t, \varepsilon)] \tag{7.189}$$

If the action is invariant under the transformation, then so is the Lagrangian. Further, Noether's theorem is constructive. It tells us exactly how to compute the quantity Q that is conserved if the coordinate transformation is a continuous symmetry.

Proof. Let $\mathcal{L}(q_i, \dot{q}_i, t)$ be the Lagrangian and suppose $q_i(t) \to \bar{q}_i(t, \varepsilon)$ is a continuous symmetry. Then, the Lagrangian must be invariant under the transformation. That is,

$$\mathcal{L}(q_i, \dot{q}_i, t) = \mathcal{L}(\bar{q}_i, \dot{\bar{q}}_i, t)\ \forall\ \varepsilon \tag{7.190}$$

In other words, the Lagrangian viewed as a function of ε through the transformed coordinates is constant. Thus,

$$\frac{d}{d\varepsilon}\mathcal{L}(\bar{q}_i, \dot{\bar{q}}_i, t)|_{\varepsilon=0} = 0 \tag{7.191}$$

We can expand this to get,

$$\frac{d\mathcal{L}}{d\varepsilon}|_{\varepsilon=0} = \sum_i \left(\frac{\partial \mathcal{L}}{\partial q_i}\frac{\partial \bar{q}_i}{\partial \varepsilon} + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{\partial \dot{\bar{q}}_i}{\partial \varepsilon} \right) |_{\varepsilon=0} = 0 \tag{7.192}$$

Since $\frac{\partial}{\partial \varepsilon}$ and $\frac{d}{dt}$ commute,

$$\frac{\partial \dot{\bar{q}}_i}{\partial \varepsilon} = \frac{d}{dt}\left(\frac{\partial \bar{q}_i}{\partial \varepsilon} \right) \tag{7.193}$$

Then, substituting the Euler-Lagrange equation gives us

$$0 = \sum_i \tag{7.194}$$

■

Example: Translation and Linear Momentum

Consider the following Lagrangian of a particle in 1D where the potential only depends on the distance between two points,

$$\mathcal{L} = T - U(|x_a - x_b|) \tag{7.195}$$

Consider a coordinate transformation that describes a translation

$$x \to \bar{x} = x + \varepsilon \tag{7.196}$$

$$\implies \dot{\bar{x}} = \dot{x} \tag{7.197}$$

Further,

$$|\bar{x}_a - \bar{x}_b| = |x_a + \varepsilon - (x_b + \varepsilon)| \tag{7.198}$$

$$= |x_a - x_b| \tag{7.199}$$

Thus, the Lagrangian is,

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - U(|x_a - x_b|) \tag{7.200}$$

$$= \frac{1}{2}m\dot{\bar{x}} - U(|\bar{x}_a - \bar{x}_b|) \tag{7.201}$$

Thus, the Lagrangian is invariant under the coordinate transformation which implies the action is invariant under the coordinate transformation. Thus, we have a continuous symmetry. Then, by Noether's theorem (we don't have a sum as we are only in 1D),

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{x}} \frac{\partial \bar{x}}{\partial \varepsilon}|_{\varepsilon=0} \tag{7.202}$$

$$= m\dot{x}(1) \tag{7.203}$$

$$= m\dot{x} \tag{7.204}$$

$$= p \tag{7.205}$$

Thus, invariance under spatial transformations mean linear momentum is conserved.

Example: Rotations and Angular Momentum

Example: Time and Energy